Chapter 1
Oceans and Life

The Big Questions

1. *What is the importance of ocean relief?*
2. *How does ocean water movement impact life and environment?*
3. *Why are maritime zones important for a country?*
4. *How can we use ocean resources in a sustainable way?*
5. *How do early warning systems help protect people from tsunamis and cyclones?*

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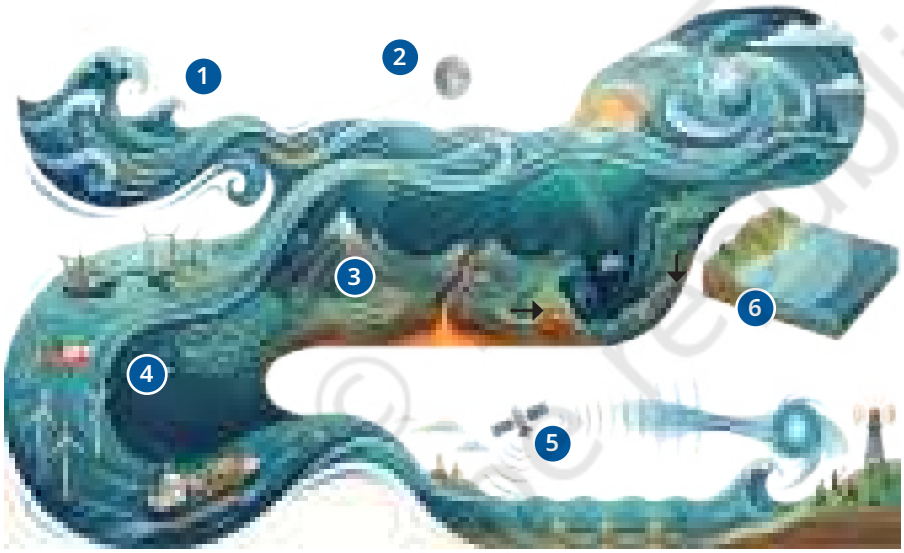

1. Ocean water is always in motion.

Waves, tides, and currents shape our climate, support marine life, and impact travel.

2. The gravitational pull creates tides.

The daily rise and fall of sea level is crucial for navigation and generating tidal energy.

3. The ocean floor is a hidden landscape.

Like land, it has features like mountains, plains, mid-oceanic ridges, trenches, etc.

4. The 'Blue Economy': sustaining ocean resources.

Oceans provide food, energy, and jobs, which must be used sustainably.

5. Early warning systems protect coastal life.

Seismic sensors and satellites detect tsunamis and cyclones, allowing for timely evacuations.

6. Maritime zones create 'rules of the sea'.

International laws like UNCLOS define a country's rights from its coast to the high seas.

Fig. 1.1. The dynamic ocean system

In earlier grades, you learnt about continents and oceans. In Indian culture, the ocean—commonly called *Samudra*—is considered powerful and sacred, and features prominently in stories and rituals.

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Oceans cover most of our planet and conceal many secrets beneath their waters. Have you ever wondered what the land under the oceans looks like, why ocean water is always moving, or what wealth and resources lie beneath the sea? How do areas like maritime zones help countries use the oceans, and how can we protect ourselves from ocean hazards, such as cyclones and tsunamis?

In this chapter, we will explore these human–ocean connections and discover how oceans shape both our daily lives and the world around us.

Ocean Relief

As we explore human–ocean connections, let us begin with the ocean floor and its features. Just as land has mountains, valleys, and plains, the ocean floor, too, has several landforms, together known as **submarine relief** or **ocean relief.** These include underwater mountains, valleys, plateaus, and trenches. The Mariana Trench in the Pacific Ocean is the deepest place on Earth, while the Mid-Ocean Ridge is the world's longest mountain range, hidden beneath the ocean. Some underwater mountains like Mauna Kea (Hawai'i) are even taller than Mount Everest.

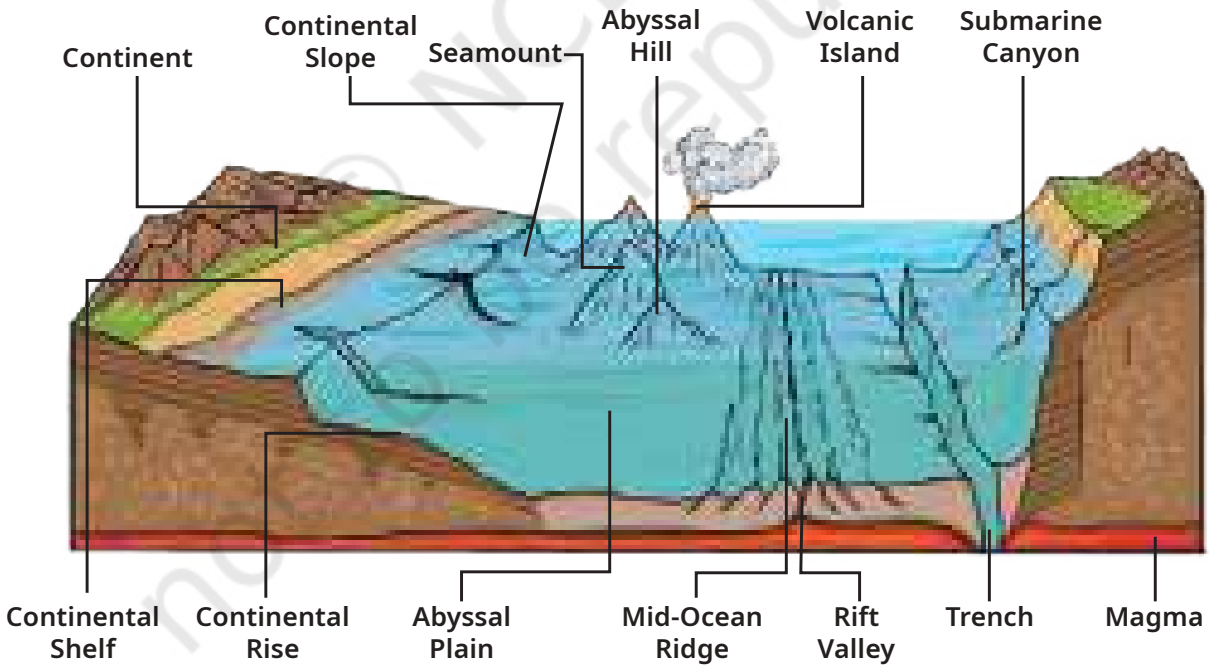

Fig. 1.2. Submarine relief

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Looking at Fig. 1.2

→ How do you think these oceanic features would have been formed?

→ Which part of the ocean floor appears deep and shallow?

→ How do you think these underwater features would have an effect on marine life and ocean movements?

Important submarine relief features

Continental Shelf

The submerged, shallow edge of a continent; extends under the ocean. Rich in marine life and resources like oil and gas.

Continental Slope

The steep slope marking the true edge of a continent.

Abyssal Plain

The vast, flat, and deep ocean floor.

Mid-Ocean Ridge

An immense underwater mountain range formed by volcanic activity.

Ocean Trenches

The deepest parts of the ocean.

Seamounts and Guyots

Seamounts are often volcanic, with some flattened by wave action over millennia to form **guyots**.

Seamount:
An underwater mountain, usually of volcanic origin, that rises from the ocean floor but does not reach the sea surface.

Guyot:
A flat-topped underwater mountain, usually a former seamount whose summit was eroded and later submerged.

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Movement of Ocean Water—Waves, Tides, and Currents

The water in the oceans is constantly in motion. Waves, tides, and currents are not merely movements of ocean water; they help shape the Earth's surface, support marine life, aid fishing and travel, and even influence climate. Let us learn more about these oceanic movements.

Waves

When the surface of the ocean rises and falls alternately, it is called a **wave**. Waves are primarily caused by winds blowing across the water. They carry energy across the oceans over long distances. Observe Fig. 1.3 and note the crests (high peaks) and troughs (lower valleys).

Wavelength: The horizontal distance between two consecutive wave crests or troughs.

Waves are often called coastal architects as they shape the shoreline through erosion and sediment transport. Observe the various landforms formed by waves in Fig. 1.4 and note how wave action creates erosional features, such as sea cracks, caves, arches, sea stacks, and stumps. Also notice the original headland shown in the figure.

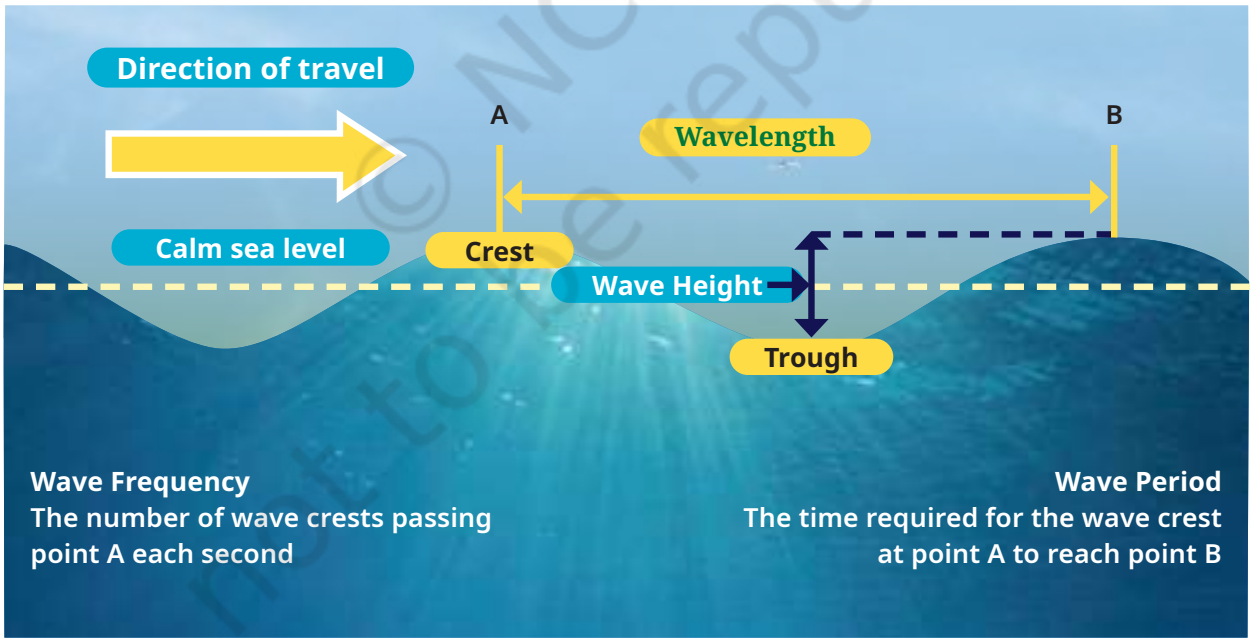

Fig. 1.3. Anatomy of waves

In addition to erosional landforms, waves also form depositional features such as beaches.

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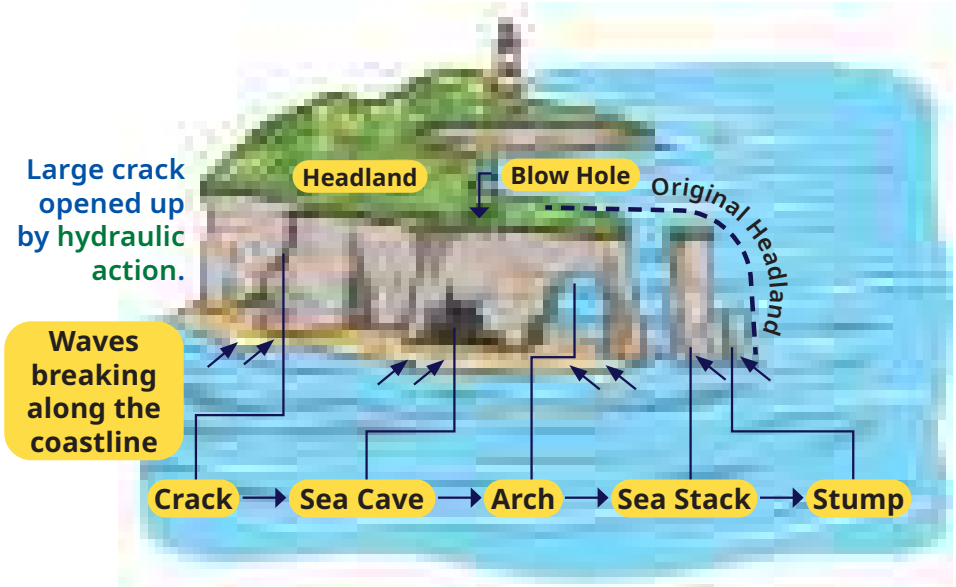

Fig. 1.4. Landform features formed by waves

LET'S EXPLORE

Find out the names of some famous sea beaches in our country and locate them on the map of India.

During storms, strong winds can generate huge waves that may cause significant damage along coastlines. The giant waves majorly caused by underwater earthquakes are called **tsunamis**, which can be extremely destructive. On reaching the shore, tsunamis may attain a maximum vertical height above sea level, known as the run-up height, of up to 30 metres, or even more in extreme cases.

Hydraulic action: The erosion of rock and soil caused by the force of moving water, which loosens and removes material.

Tides

Tides are the rhythmic rise and fall of ocean water that occur twice a day in most parts of the world. When the water rises to its highest level and covers much of the shore, it is called high tide (flood tide). When the water falls to its lowest level and moves away from the shore, it is called low tide (ebb tide). Observe Fig. 1.5 and note the change in water level during high and low tides. Do you notice the oval layer around the Earth indicating water level? Can you also see how the water level rises during high tide and falls during low tide?

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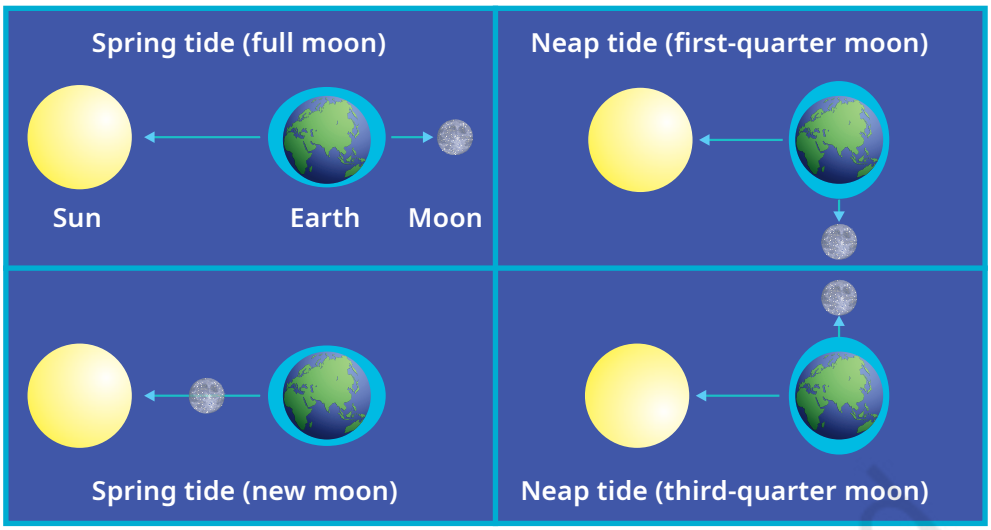

(a) *(b)*

Fig. 1.5 (a) Spring tide (b) Neap tide (Schematic diagram of tidal forcing showing idealised bulges)

Spring Tides and Neap Tides

You may wonder what causes this regular rise and fall of ocean water. To understand this, let us explore how tides are formed.

- **Spring tides:** Observe Fig. 1.5(a), where the Sun, the Earth, and the Moon are aligned in a straight line. Spring tides occur during this alignment, which happens on full moon and new moon days. At this time, the combined gravitational pull of the Sun and the Moon causes high tides of unusual heights and low tides that are unusually lower. The figure also shows the bulging of water on opposite sides of the Earth due to this strong gravitational pull, leading to the formation of spring tides.
- **Neap tides:** When the Moon is in its first- and third-quarter phases, as shown in Fig. 1.5(b), the gravitational pull of the

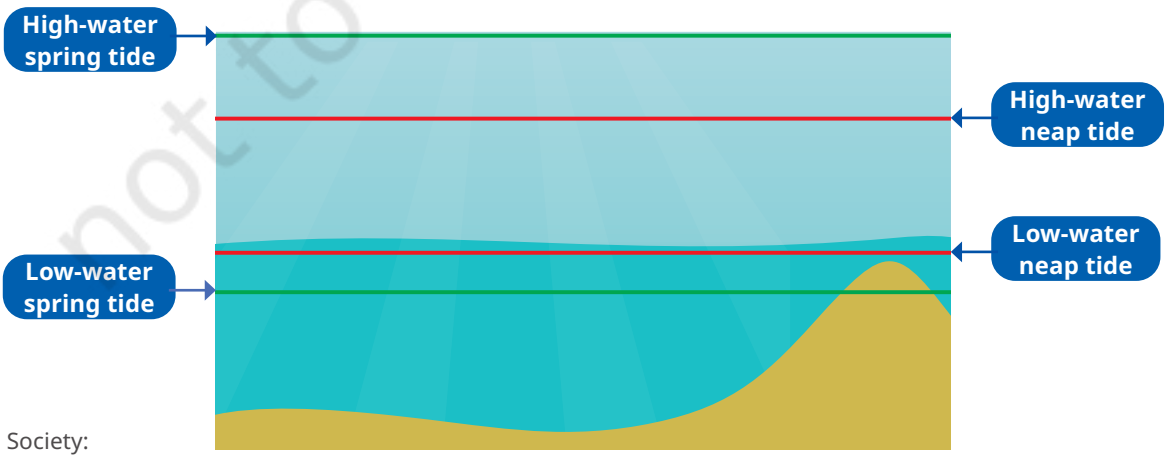

Fig. 1.6. Levels of spring and neap tides

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Sun and the Moon acts in different directions. As a result, the difference between high tide and low tide becomes smaller, producing neap tides. Fig. 1.6 shows the levels of high and low water during both spring and neap tides, along with the average low-water levels for each.

High tides are important for navigation because they raise the water level near the shore, allowing ships to enter harbours more easily. In ports and coastal areas, tidal information is essential, as tidal ranges in some ports can reach 4–5 metres. The Gulf of Khambhat has one of the highest tidal ranges, at about 12 metres. Before a ship berths at a port, it must check both the water depth and the tidal range to ensure safe docking, especially during low tide.

Some sea channels, such as the Singapore Strait, contain shallow areas. Hence, ships such as Very Large Crude Carriers (VLCCs) passing through these regions often wait for hours to transit only during high tide to ensure sufficient water depth and avoid running aground.

Tidal changes also generate strong currents because large volumes of water flow in and out of coastal areas. These tidal currents can reach speeds of 5–6 knots (about 9–11 km/h).

Fig. 1.7. Observe how high and low tides help in fishing activities

High tides also support fishing activities (Fig. 1.7). During high tide, many fish move closer to the shore in search of food and shelter, enabling fishing communities to obtain a better catch.

The rise and fall of tidal water are also used to generate electricity in some places (Fig. 1.8). India has a vast coastline of 11,098.81 kilometres, and although tidal energy is not yet widely used on a large scale, certain areas have been identified as suitable for its development. The Gulf of Kutch, the Gulf of Khambhat, and the Sundarbans are among the major tidal energy sites in India.

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Fig. 1.8. Tides at the Bhadkeshwar Mahadev Temple, Dwarka, Gujarat

The Bhadkeshwar Mahadev Temple in Dwarka is situated on a small hillock in the Arabian Sea. Visitors can reach the temple only during low tide, as the stone pathway leading to it is completely submerged during high tide.

Ocean Currents

Coriolis force: A force caused by Earth's rotation that deflects any free moving object from its straight path to the right in the Northern Hemisphere and to the left in the Southern Hemisphere.

Ocean currents are continuous streams of water that flow across the ocean surface in definite directions (Fig. 1.9). They are caused by factors, such as winds, the Earth's rotation (**Coriolis force**), differences in temperature and salinity, and the shape of continents and ocean basins. Ocean currents may be warm or cold depending on the direction from which the water flows. Currents flowing from warmer regions are called warm currents, while those flowing from colder regions are called cold currents. On maps, warm currents are usually shown in red and cold currents in blue.

THINK ABOUT IT

Observe the map carefully and note the movement patterns of cold and warm currents. Do you think these currents affect the temperature of nearby coastal regions?

The Gulf Stream, a warm current originating in the Gulf of Mexico in the Atlantic Ocean, flows towards Norway and keeps its waters relatively warm. As a result, sea ice does not form easily in the Norwegian region despite its high latitude (60°–70°). In contrast, the Labrador Current, a cold current flowing along the east coast of Canada, carries freezing water and supports the formation of sea ice and icebergs even at lower latitudes.

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Observe Fig. 1.9 and take a note on ocean currents along the coastline of India. What is the nature of ocean currents?

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Fig. 1.9. Major ocean currents

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The regions where warm and cold currents meet are known as fishing grounds. These waters are rich in nutrients, supporting a large variety of fish and making them important fishing grounds.

Cyclones and Tsunamis—Early Warning Systems

Oceans and seas can give rise to sudden natural hazards such as cyclones, which are large rotating storms, and tsunamis, which are giant waves caused by earthquakes. To protect people from such dangers, scientists have developed Early Warning Systems that work like giant alarm systems for coastal areas. To reduce risks, people, communities, and governments must act in advance. Early Warning Systems help detect cyclones and tsunamis, predict disasters, and issue timely warnings so that people can evacuate and stay safe.

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Most deaths during the 2004 Indian Ocean tsunami occurred in Indonesia. However, the Island of Simeulue was largely spared from the disaster. Researchers suggest that this was partly because

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of local folklore, passed down through generations, that helped residents recognise the warning signs of a tsunami and respond quickly. Talk to your elders and find out whether similar folk traditions or stories related to natural disasters exist in your culture. Share these with your classmates and discuss how traditional knowledge can help communities stay safe.

India has several Cyclone Warning Services and Marine Weather Services that monitor storms along both the eastern and western coasts. For early warning systems to be effective, they must function continuously. India uses advanced technologies to predict cyclones at an early stage, which has significantly reduced the loss of human and animal lives.

The Government of India has launched the National Cyclone Risk Mitigation Project (NCRMP) to build cyclone shelters, strengthen warning systems, and improve emergency response mechanisms. The project focuses on cyclone-prone States and Union Territories and includes a Decision Support System (DSS) for disaster management authorities. The Indian National Centre for Ocean Information Services (INCOIS), in collaboration with the India Meteorological Department (IMD), has also established the Storm Surge Early Warning System (SSEWS) for the Indian coasts. Its main objective is to protect coastal communities by forecasting cyclone-induced storm surges and the extent of coastal inundation.

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The **GAGAN-Enabled Mariner's Instrument for Navigation and Information (GEMINI)** is a handheld device designed for fishermen that plays an important role in ensuring the safety of fishing communities at sea (Fig. 1.10). The device receives satellite information and transmits it to mobile phones through Bluetooth. Using its dedicated mobile application, fishermen can access

Fig. 1.10. Local fishermen at work in the Arabian Sea; Alappuzha, Keralam

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important information in their local language, including alerts about high waves and strong winds, cyclone and tsunami warnings, potential fishing zone advisories, current speeds, **bathymetry** data, mapping, and risk management information.

Under the Pradhan Mantri Matsya Sampada Yojana (PMMSY), the government provides financial support for GEMINI devices to improve the safety of coastal communities while they are at sea (Fig. 1.11).

Bathymetry: The study and mapping of the underwater topography of water bodies, such as oceans, rivers, lakes, and streams.

How GEMINI Works

- The GEMINI device receives data from the GAGAN satellite and transfers it to fishermen's mobile phones through Bluetooth.
- The INCOIS mobile application decodes and displays this information in nine regional languages.
- At present, the communication channel can transmit information only up to 50 nautical miles into the sea.
- An upgraded version is expected to provide a two-way communication system on a pilot basis, allowing fishermen to send feedback as well.
- This upgraded system will benefit fishermen travelling up to 300 nautical miles in search of better fishing opportunities.

Fig. 1.11. GEMINI

Focus on Cyclones

Cyclones are caused by atmospheric disturbances over warm ocean waters.

Naming the cyclone: Cyclones in the North Indian Ocean are named sequentially from a list of 169 names submitted by 13 member countries, including India. This system aids clear communication and public awareness. Recent examples include Biparjoy (2023) and Remal (2024). Note: You may track cyclones and their affected areas in India on the School Bhuvan–NCERT Geoportal (Fig. 1.12).

Fig. 1.12. Satellite image of a cyclone

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On 26 December 2004, the world witnessed unprecedented devastation caused by the Indian Ocean Tsunami. The disaster claimed around 2,30,000 lives and caused severe environmental damage across many countries bordering the Indian Ocean. India was among the worst-affected countries, as massive waves swept away people and structures along the coasts. Tamil Nadu, Puducherry, Andhra Pradesh, Keralam, and the Andaman & Nicobar Islands especially suffered extensive damage. The scale of the disaster led many countries to establish effective tsunami early warning systems and stronger disaster-management mechanisms. Under an initiative of the Indian Tsunami Early Warning Centre (ITEWC), the Government of India set up a real-time network of tsunami detection buoys (Fig. 1.14) designed to detect, measure, and monitor tsunamis.

The photograph in Fig. 1.13 shows some areas of Chennai affected by the Indian Ocean tsunami (2004).

Fig. 1.13. Some of the tsunami-affected areas in Chennai (2004)

Fig. 1.14. Tsunami detection buoy

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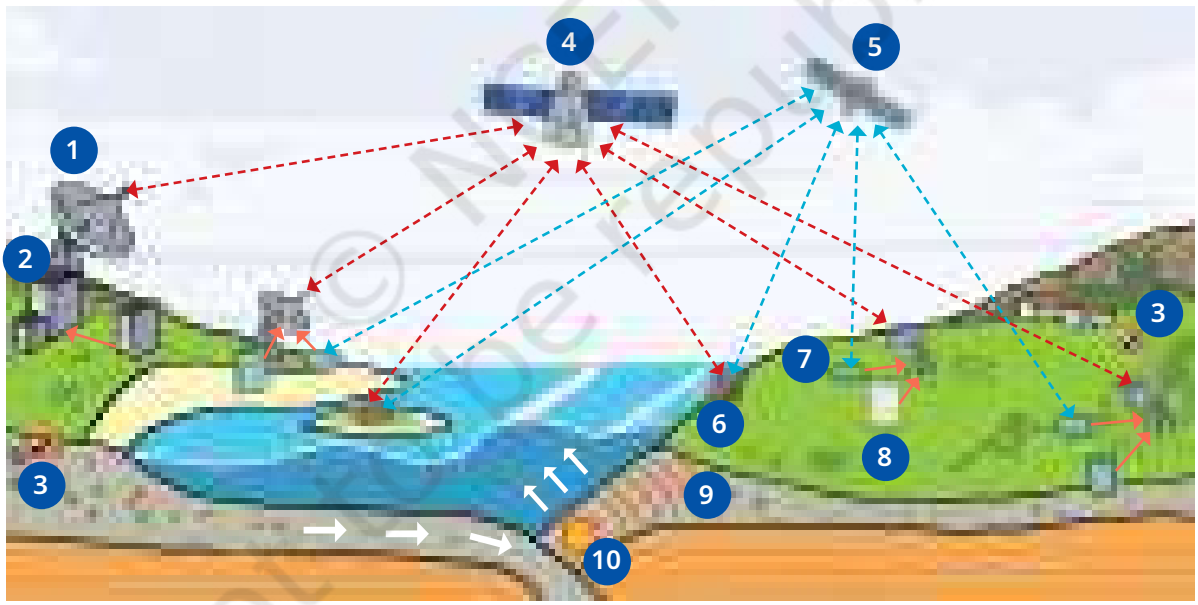
Detection
A network of seismic stations detects undersea earthquakes in real time.

Confirmation
Ocean Bottom Pressure Recorders (OBPRs) placed on the seafloor detect the passage of tsunami waves, confirming the threat.

Tracking
Coastal tide gauges and radars monitor changes in sea level to track the arrival and movement of tsunami waves.

Warning Dissemination
Alerts are transmitted via satellite to government agencies and are also sent directly to officials and the public through SMS and other communication channels.

Fig. 1.15. Functioning of the tsunami early warning system



1 Warning centre	2 Modelling/ simulation	3 Capacity development	4 Communication satellite	5 GPS satellite
6 Tide gauges station	7 GPS station	8 Seismometer station	9 Seismic waves	10 Earthquake

Fig. 1.16. The different elements of the tsunami early warning system

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For an early warning system to save lives, information must reach the public quickly and effectively. The Indian Tsunami Early Warning Centre (ITEWC) is tasked with sending alerts to the Government of India, which then informs state governments and emergency response teams. Reliable communication is ensured through satellite-based communication networks, SMS, emails, and coastal display boards.

Besides technology, public awareness and training programmes are equally important for an effective early warning system. Educating officials and coastal communities on how to interpret warnings helps ensure that when disasters such as cyclones are predicted, people know how to respond appropriately.

THINK ABOUT IT

If you receive a tsunami or cyclone alert on your phone right now, do you know the five steps your family should take to stay safe?

Marine Resources and Their Significance

Do you recall learning about natural resources in the Grade 8 Social Science Part 1 textbook? These resources are obtained from land, water, forests and animals, and other natural sources. Oceans, too, are vast storehouses of resources, and the materials and features found in oceans that have intrinsic or economic value are called marine resources. Marine resources can be broadly classified into two categories—Marine Living Resources and Marine Non-living Resources. They include fish and seafood, biodiversity, oil and natural gas, minerals, sand and gravel, renewable blue energy sources such as tidal and wave energy, tourism potential, and unique ecosystems such as coral reefs. Fig. 1.17 presents a detailed classification of marine resources.

Significance of Marine Resources

Marine resources are vital for food security, economic activities, and maintaining a healthy environment (Fig. 1.17). Countries with long coastlines, such as India, benefit greatly from the sea. Fishing, trade, and tourism provide livelihoods to millions of people, while oceans also support biodiversity and sustain numerous plant and animal species. However, marine resources must be used responsibly without disturbing marine ecosystems.

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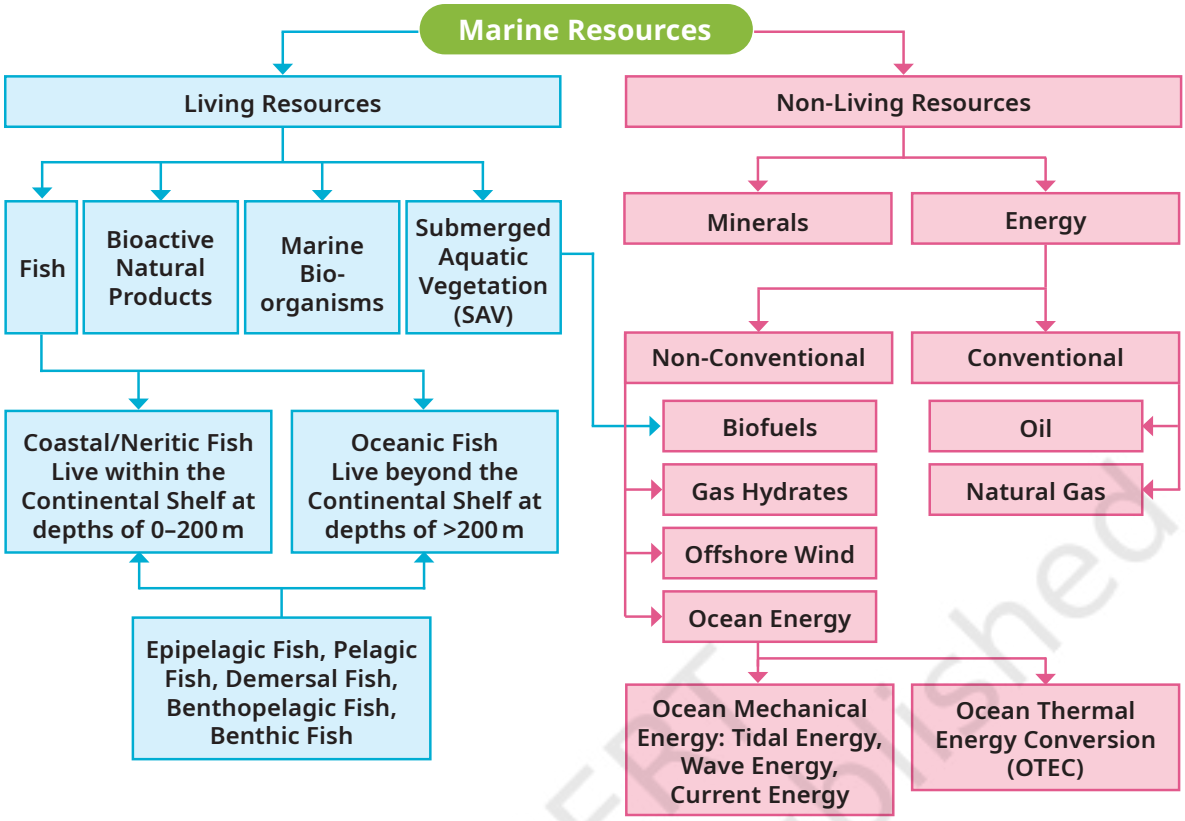

Fig. 1.17. Classification of marine resources

The Ocean is a Vast Storehouse of Resources Vital for Life, Livelihoods, and Culture

Marine resources hold immense value for humanity and for coastal nations like India, they are the bedrock of local economies and traditions.

Food and Livelihood: Fish are a major source of protein, while fishing and aquaculture support important industries and livelihoods.

Minerals and Materials: Sand, gravel, and deep-sea minerals.

Energy: Offshore oil and gas reserves; immense potential for renewable 'blue energy' from tides and waves.

Trade and Tourism: 90% of global trade by volume is carried by sea; coastal tourism is a multi-billion dollar industry.

Cultural Significance:
The sacred *śhaṅkha* (conch) and *kapardikā* (cowrie shells), once used as currency, are regarded as divine treasures from the mythological *Samudra Manthana*, reflecting the close connection among the sea, prosperity, and culture.

Fig. 1.18. Significance of marine resources

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Blue Economy is the Future: Sustainable Growth Powered by Healthy Oceans

The World Bank defines **Blue Economy** as "the sustainable use of ocean resources to promote economic growth, improved livelihoods, and jobs, while preserving the health of the marine ecosystem".

Connecting to Global Goals: This concept is enshrined in the UN's Sustainable Development Goal (SDG) 14: Life Below Water. Its key aims are to:

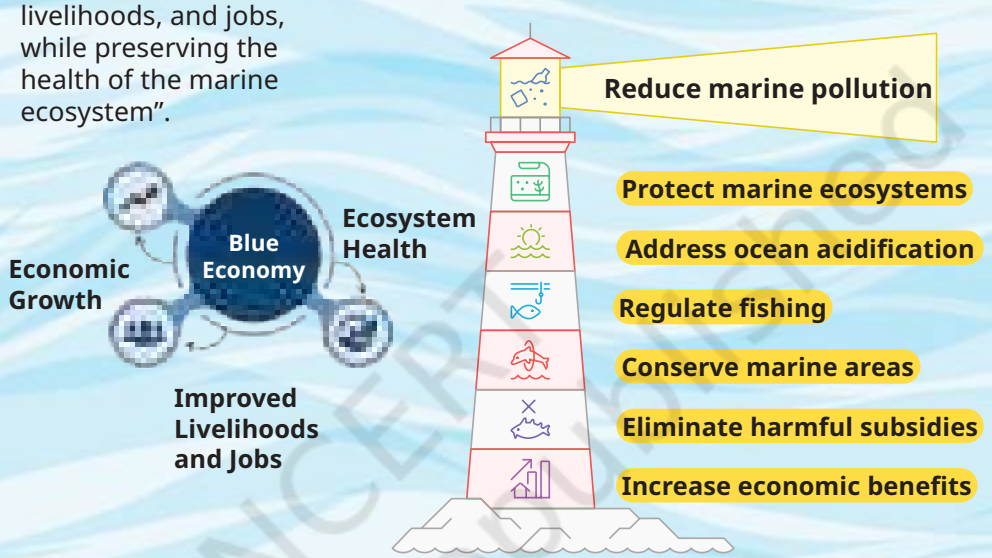

Fig. 1.19. The role of Blue Economy

Maritime Zones

Have you ever wondered who owns the oceans and how countries decide where they can fish or travel? To understand this, we need to learn about maritime zones—different areas of the ocean defined by international law.

The sea area closest to a country is called its territorial waters, over which the country has near-complete control, subject to the right of innocent passage for foreign ships. Just as on land, the government can make laws relating to customs, immigration, and pollution control in this zone.

Observe Fig. 1.20 and note the extent of the territorial sea in **nautical miles**. Beyond this lies the contiguous zone, extending from the baseline up to 24 nautical miles, where the country has limited control over customs, fiscal matters, and illegal immigration.

Nautical mile: A unit of measurement used in maritime and air navigation. It is defined as exactly 1,852 metres (about 6,076 feet). Based on the Earth's circumference, one nautical mile corresponds to one minute of latitude making it a convenient unit for charting and navigation across the world's seas.

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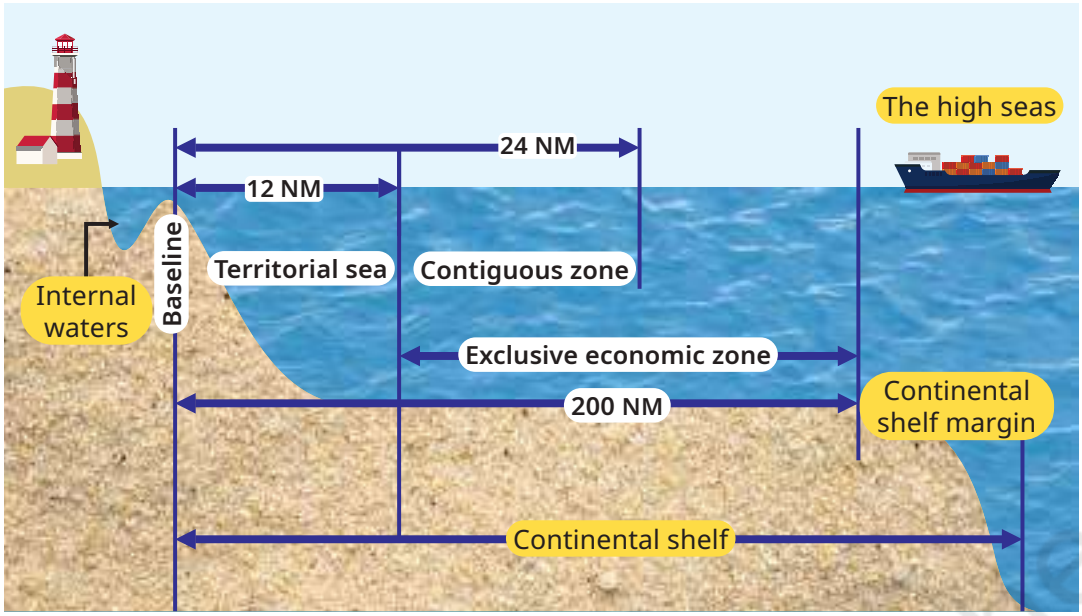

Fig. 1.20. Maritime zones (Indicative only)

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Innocent Passage is a principle that allows foreign ships to pass through the territorial waters of a country as long as they do not threaten its peace, good order, or security. Ships must move continuously without stopping, except in emergencies or cases of breakdowns. This right does not permit activities such as fishing, military operations, or other non-peaceful actions.

Beyond the contiguous zone, lie the Exclusive Economic Zones (EEZs), which extend from the baseline up to 200 nautical miles from the coast, and are defined under the United Nations Convention on the Law of the Sea (UNCLOS) by global agreement. In these zones, a country has special rights to use ocean resources, such as fish, oil, and gas, although ships from other countries may still pass through peacefully. It is important to note, however, that the country does not fully own or control the waters of the EEZ.

Beyond the EEZs of all countries lie the high seas, also known as international waters. Covering about 64 per cent of the world's oceans, these waters do not belong to any single country and are therefore considered part of the global commons. All countries enjoy equal rights on the high seas, including navigation, fishing, flying over them, laying undersea cables, building lawful structures such as artificial lands, and exploring for minerals.

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Although the high seas are essential, they also present many challenges. Since no single country controls them, issues, such as overfishing, illegal fishing, and conflicts over resources often arise. These problems affect millions of people who depend on the sea for food and livelihoods. Ensuring navigation and safety is also more difficult without clear enforcement mechanisms. On the high seas, the laws of the flag state—the country in which a ship is registered—apply.

International Maritime Rules and Regulations

In the earlier sections of this chapter, you learnt that oceans are vital for trade, travel, and communication. To ensure that activities at sea remain safe, fair, and peaceful, countries follow a set of international rules and agreements known as maritime law or maritime regulations. These laws regulate how ships operate and move; protect the marine environment; address issues such as piracy, terrorism, natural disasters, and accidents; safeguard trade routes; and help settle disputes between countries.

LET'S EXPLORE

Maritime disputes are conflicts or disagreements between countries over territory, resources, or rights in naval areas such as oceans, seas, or coastal regions. Can you find out some countries that have conflicts over their territorial sea waters?

Important Maritime Rules and Regulations

Just as there are traffic rules on land, ships must follow "rules of the road at sea" to avoid accidents. These rules, known as COLREGs (Convention on the International Regulations for Preventing Collisions at Sea), include maintaining a proper lookout, deciding which side to pass on, identifying the give-way and stand-on vessels, proceeding at a safe speed, and using lights, sound signals, shapes (for different types of vessels such as fishing boats), along with appropriate actions to prevent collisions. In addition, ships use international signals and radio communication to share information and prevent accidents, especially in crowded or hazardous waters.

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Ships must also meet safety standards and are prohibited from polluting the seas. The MARPOL (International Convention for the Prevention of Pollution from Ships) Convention is an important international agreement aimed at reducing pollution from ships.

Why are Maritime Rules Important?

International maritime rules, regulations, and agreements help keep the oceans safe and ensure the fair and sustainable use of ocean resources. These rules enable countries to cooperate in using marine resources without conflict while protecting marine life, ships, sailors, and coastal communities. Without such regulations, the oceans could become dangerous and chaotic, with increased pollution, accidents, collisions, and disputes over resources. By establishing common standards for all ships, regardless of the country they belong to, maritime rules help keep the oceans safe, clean, and open for trade and travel.

Before we move on...

- Oceans are vital for global trade, travel, communication, and climate regulation.
- Submarine relief refers to the physical features of the ocean, such as the continental shelf, ridges, trenches, and seamounts, which influence marine life, resource availability, and undersea exploration.
- Movements of ocean water—waves, tides, and currents—affect marine ecosystems, global climate, and human activities like fishing and navigation.
- Maritime zones include territorial waters, EEZs (Exclusive Economic Zones), and the high seas.
- Sustainable use of marine resources through cooperation, regulation, and conservation is essential to protect aquatic life and support future generations.
- Maritime rules and regulations exist to resolve disputes over marine boundaries.
- International laws, such as UNCLOS (United Nations Convention on the Law of the Sea), help define maritime zones and settle disputes among countries.

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Questions and activities

1. How do different features of the ocean floor affect human activities such as fishing, transport, and underwater exploration?
2. How do waves, tides, and ocean currents influence climate, marine life, and human activities around the world?
3. Describe the importance of marine resources.
4. Draw a poster that encourages people to protect marine resources. Include a slogan (like Save Our Seas or Fish for the Future) and explain your message in 2–3 sentences.
5. Using a map of India, mark at least five coastal states that might need tsunami warnings.
6. Work in a group and act out what would happen when a cyclone warning is issued. Highlight what role different people, like scientists, officials, and coastal villagers, would play.
7. Suppose a foreign ship is sailing through India's territorial sea. What principle allows the ship to pass through these waters, subject to certain conditions?
8. A country has an Exclusive Economic Zone (EEZ) extending up to 200 nautical miles from its baseline. What rights does the country have over the natural resources in this zone?
9. Create a model of an ocean with ship routes, protected zones, and safety measures. Based on the model, explain how it follows maritime laws and protects the ocean.

Notes

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Chapter 2

Life on Earth

The Big Questions

1. *How do physiography and climate shape life on Earth?*
2. *How are humans and other life forms interconnected?*
3. *What is the importance of preserving various life forms, and how can we contribute to their conservation?*

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Geography helps us understand how climate, landforms, and water shape life on Earth—and how humans adapt their lives to these conditions. From forests and grasslands to deserts and coastal regions, different environments support different forms of life. At the same time, human activities influence these natural systems in significant ways. This chapter explores the close relationships among physical features, climate, and living organisms, and encourages us to reflect on our responsibility towards preserving the diversity of life on Earth.

Imagine our planet as a giant patchwork quilt, where each patch has its own colours, weather, and living beings. These huge patches are called **biomes**. The word 'biome' is made up of the word *bio*, derived from the Greek word *bios*, meaning 'life', and *-ome*, a suffix used in biology to denote a group of a specific type of biological constituents. Biomes are large natural regions of the world where certain types of plants and animals live under similar climatic conditions. For example, icy tundras, hot deserts, green forests, and vast grasslands are all different types of biomes. In simple terms, a biome is like a broad environmental region, while an **ecosystem** is like a neighbourhood within it, where plants, animals, and non-living things interact closely. Each biome is unique, but together they form the living world of our planet, supporting diverse forms of life and helping maintain the balance of nature.

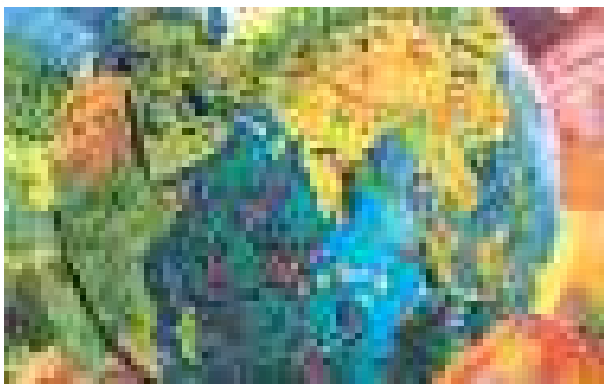

Fig. 2.1. Earth as a giant patchwork quilt

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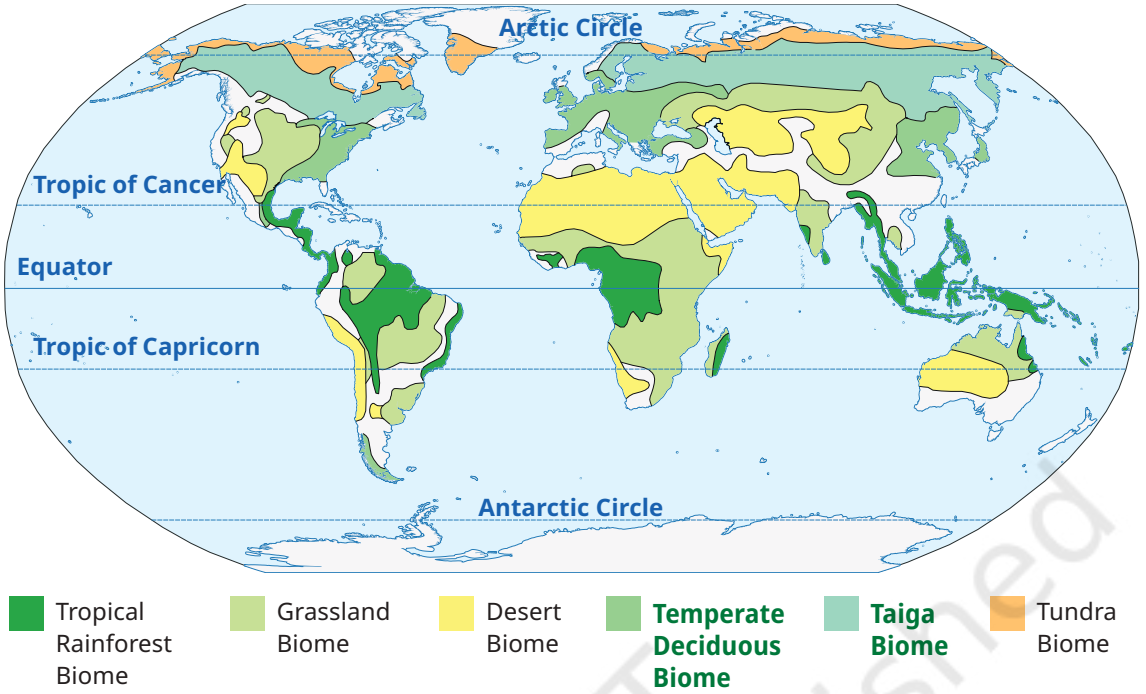

Fig. 2.2. World map showing distribution of biomes (map not to scale)

Temperate deciduous biome: This biome is characterised by forests of broad-leaved trees that shed their leaves seasonally, with moderate temperatures and distinct seasons.

Taiga biome: This biome is characterised by coniferous forests consisting mainly of pines, spruces, and larches.

LET'S MAP

Look at the world map in Fig. 2.2 and answer the following questions.

→ Which type of biome is found in smaller, separate regions, and why?

→ Which continent has the least number of biomes? What could be the reason for this?

→ Why are there areas that do not have any specific biomes?

Types of Biomes

Now that you have understood the meaning and uniqueness of biomes, let us identify the different types of biomes. Different **biomes** support diverse plant and animal life and can be categorised into **terrestrial** (land-based) and **aquatic biomes** (water-based).

There are several factors that determine which types of plants and animals can survive and thrive in a biome. These include temperature, precipitation patterns, dominant plant species, growth forms of vegetation, etc.

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Terrestrial Biomes

Terrestrial biomes are classified into forests, grasslands, deserts, and tundra, based on their climate.

Forest Biome

A forest biome is like nature's giant green umbrella—warm, wet, and bursting with life at every level, from the tallest trees to the damp forest floor. **Tropical rainforests** are mainly found near the Equator, between the Tropic of Cancer and the Tropic of Capricorn (23.5° N to 23.5° S). These regions remain hot and wet throughout the year, which supports the growth of dense forests. The largest tropical rainforests are located in the Amazon Basin in South America, the Congo Basin in Africa, and parts of Southeast Asia, such as Indonesia and Malaysia. Smaller rainforest regions are also found in Central America, as well as in India's Andaman & Nicobar Islands, and the Northeastern states. Together, these areas form the world's richest

Fig. 2.3. (Left to right, top to bottom) A beach, Andaman Islands; Dzukou Valley, Nagaland; Nohkalikai Falls, Sohra; Gurudongmar Lake, Sikkim

storehouse of biodiversity with rich genetic resources. They preserve unique genetic material, supporting conservation and scientific research, and are hence also known as a 'gene bank'. Forests also play an important role in maintaining oxygen levels and balancing the Earth's climate.

LET'S EXPLORE

Look at the layers of vegetation in a rainforest. What do the yellow shapes in the figure indicate?

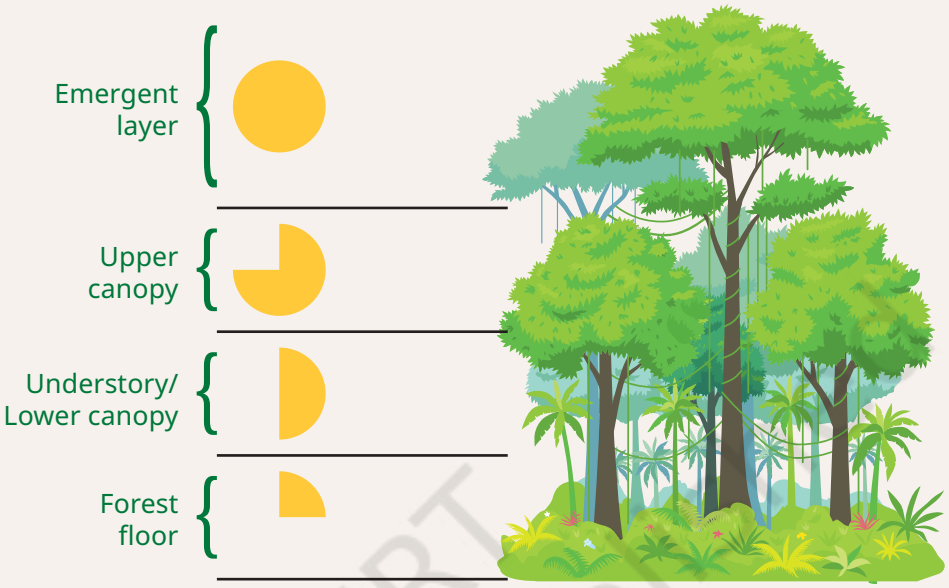

Fig. 2.4. Ideal structure of a rainforest biome

Grassland Biome

Grasslands are large open areas covered mainly with grass and very few trees. These develop in regions that receive moderate rainfall, which is not enough to support dense forests. Temperate grasslands (such as the prairies in North America and the steppes in Asia) have

Fig. 2.5. Typical grassland landscape

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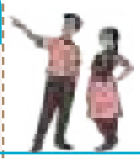
fertile soil and are often used for farming. Tropical grasslands, also known as Savannas (as in Africa), have scattered trees and are home to large animals such as lions, elephants, and giraffes. Grasslands support grazing animals and are important for human activities like agriculture and livestock rearing.

LET'S EXPLORE

Observe Fig. 2.5 and explore which of these animals you will find in the grasslands of India.



Desert Biome

Deserts are the driest biomes, receiving very little rainfall. They can be hot, like the Sahara in Africa, or cold, like the Gobi in Asia. The Rann of Kutch in Gujarat is one of the world's largest salt deserts, turning into a marshy wetland during the monsoon and a vast white salt desert during the dry season. The extreme conditions of a desert make it difficult for life to survive; however, plants like cacti and animals such as camels, lizards, and snakes are specially adapted to these conditions. They can store water and withstand heat and dryness. Deserts often have sandy soil and sparse vegetation. Despite their harshness, they are home to unique ecosystems, and many of them also provide valuable resources such as minerals and oil.

Fig. 2.6. Life in the Thar Desert

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LET'S EXPLORE

→ What types of activities do you think people in desert regions generally undertake to sustain their households and livelihood?

→ With the help of your teacher, explore how do people address issues related to the availability of water.

(a) *(b)*

Fig. 2.7. Camels in deserts

→ Compare the physical features of the Rajasthan camel (Fig. 2.7a) and the Nubra Valley (Ladakh) camel (Fig. 2.7b).

→ Considering the climatic conditions, why do you think the Nubra Valley camel might have a different fur density than the Rajasthan camel?

Tundra Biome

The tundra biome is the coldest of all and is found near the Arctic and Antarctic regions. It has very long, harsh winters and short summers. The soil is frozen almost year-round—a condition called permafrost—which makes it difficult for trees to grow. The vegetation is limited to mosses, lichens, and small shrubs. And animals, such as reindeer, arctic foxes, and polar bears, which are well adapted to survive freezing temperatures, are often found here. The tundra is fragile, and even small temperature changes can affect its ecosystem.

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Fig. 2.8. Tundra (left) and the Himalaya (right)

THINK ABOUT IT

Tundra in polar regions has mosses, lichens, shrubs, etc., while the Himalaya, in the subtropics, is dominated by conifers. What could be the reason for this? What difference do you find in the animals found in these two regions? Which species are found only in the Himalaya? Discuss in class.

Aquatic Biomes

Aquatic biomes cover about three-fourths of the Earth's surface and are divided into freshwater and marine biomes (Fig.2.9). **Freshwater biomes** include rivers, lakes, ponds, and wetlands. These are crucial for agriculture, getting access to drinking water, and supporting species, such as fish, frogs, and insects. **Marine biomes** include oceans, seas, and coral reefs, which are rich in biodiversity and influence the Earth's climate by storing and distributing heat. They are also home to countless species, including whales, dolphins, fish, and plankton, and provide food and livelihood for millions of people.

Fig. 2.9. The diversity of the aquatic biome

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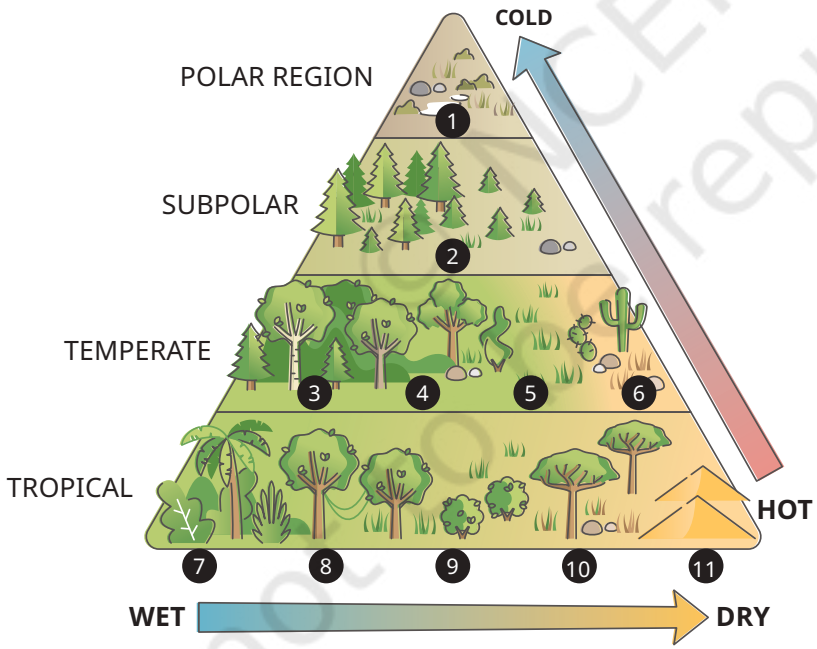
LET'S EXPLORE

Explore your local environment. This could be your village, city, town, or residential area. Observe how people, plants, and animals build structures and interact. Complete the table with your findings.

Component type	Examples observed	Interaction with humans
Trees/Vegetation/Plants		
Insects		
Animals		
Non-living things		

Biome Pyramid

A **biome pyramid** is a graphical representation that illustrates the relationship between different organisms within a biome, usually in terms of energy, number, or biomass. At the base of the pyramid



1 Tundra	**6** Desert
2 Coniferous Forest (Boreal Forest)	**7** Rainforest
3 Deciduous Forest	**8** Scrubland
4 Chaparral	**9** Tropical Seasonal Forest
5 Grassland	**10** Savanna
11 Desert	

Fig. 2.10. Biome pyramid

are the producers, such as plants and algae, which capture solar energy through photosynthesis. The next levels consist of herbivores (primary consumers), carnivores (secondary consumers), and top predators (tertiary consumers). As we move up the pyramid, the

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number of organisms, biomass, and available energy decrease as energy is lost at each trophic level as heat during metabolic processes. Thus, a biome pyramid helps in understanding the structure of ecosystems, the flow of energy, and the balance between different living organisms in a biome.

LET'S MAP

Mark the following regions at appropriate levels on the Fig. 2.10: Andaman & Nicobar Islands, Keralam, Ladakh, Chhattisgarh, Odisha, Meghalaya, and Lakshadweep. Where will you place your home?

There is no arrow on the third side of the triangle. What would happen if we drew an arrow? Would it point upwards or downwards?

Interconnectedness between Biomes and Living Beings

Biomes are closely connected to living beings because they provide a natural environment where plants, animals, and humans can survive and interact. At the same time, living beings also influence biomes—animals help in pollination and seed dispersal, humans modify landscapes for farming or cities, and microorganisms recycle nutrients. In this way, biomes and living beings are interconnected systems that depend on one another to maintain ecological balance and biodiversity. We will discuss some of the aspects of this interconnectedness in the following section.

Maintaining Ecological Balance and Biodiversity

Biodiversity

Biodiversity is the variety of life on Earth, including plants, animals, microorganisms, and the ecosystems they inhabit (Fig. 2.11). It refers not only to the number of different species but also to the genetic differences within species and the variety of habitats and ecological processes that sustain life. In simple terms, biodiversity means the richness of life forms and the natural balance they create, which is essential for maintaining healthy ecosystems and supporting human survival.

Fig. 2.11. Biodiversity in wild animals

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The **geographical aspects of biodiversity conservation** focus on how location, climate, landforms, and ecosystems influence the protection and management of living organisms. Since biodiversity is unevenly distributed across the Earth, geography plays a vital role in conservation. Some key aspects of conservation are—

Hotspots of Biodiversity: Certain regions, such as the Himalaya, the Western Ghats, the North–East India, and the Andaman & Nicobar Islands, are rich in unique species found nowhere else. Identifying and protecting these hotspots is a priority in conservation.

Fig. 2.12. (Left to right, top to bottom) Endangered Nilgiri Tahr Wild Goat; Himalayan Snowcock Bird; Indian Spider, Western Ghats; Big Himalayan Yak; Mangrove Forest, Andaman Islands

Ecosystem-Based Conservation: Different biomes (forests, wetlands, deserts, coasts, and mountains) support distinct life forms. Conservation strategies must suit the geographical setting; for example, mangrove protection along coasts or afforestation in mountains.

Protected Areas: National parks, wildlife sanctuaries, biosphere reserves, and World Heritage Sites are chosen based on geography to preserve critical habitats and prevent habitat fragmentation.

Geographical Barriers and Corridors: Natural features like rivers, mountains, and valleys serve as barriers or corridors that influence animal movement and gene flow. Conservation planning uses this knowledge to create wildlife corridors.

Human–Environment Interaction: Geography also considers how local communities use land, water, and forests. Sustainable practices vary with geography, such as terrace farming in the hills or agroforestry in the plains.

In short, biodiversity conservation is an intrinsic part of geography because the survival of species depends on their location, habitat conditions, and human activity within those regions.

LET'S MAP

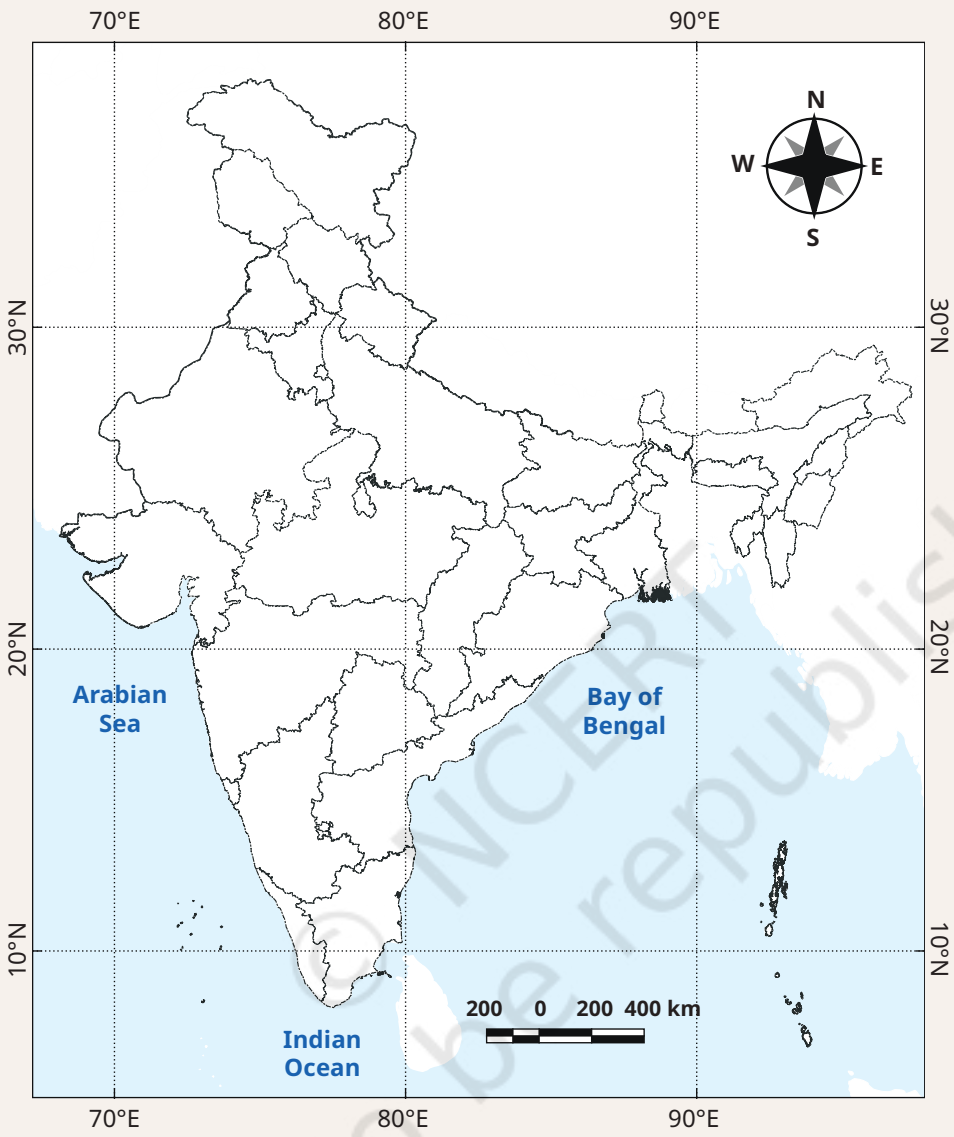

Fig. 2.13. Political Map of India

Mark and name the following regions on the map in Fig. 2.13. Also gather more information about these biodiversity regions from your teacher or other sources and discuss their features with your peers.

1. Namdapha National Park
2. Rann of Kutch
3. Gahirmatha beach
4. Gulf of Mannar
5. Pachmarhi
6. Achanakmar–Amarkantak

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Forest Ecotourism

Ecotourism: Travelling to nature-friendly places without harming the environment, and supporting local people.

Ecotourism in forests refers to responsible travel to forested areas where tourists enjoy nature while contributing to its conservation. Forests are home to a wide variety of plants, animals, rivers, and waterfalls, which attract people who want to experience natural beauty, wildlife, etc. Unlike mass tourism, ecotourism emphasises minimal environmental impact, respect for local communities, and learning about ecosystems. For example, visiting national parks, wildlife sanctuaries, or trekking through rainforests allows visitors to observe biodiversity while supporting conservation programmes and generating income for the local people. Thus, ecotourism in forests helps balance tourism, environmental protection, and sustainable development.

DON'T MISS OUT

The Swadesh Darshan 2.0 Scheme, launched by the Ministry of Tourism, Government of India, aims to develop sustainable tourism destinations with a tourist- and destination-centric approach, and a focus on factors such as sustainable development, conservation, and local community involvement.

LET'S EXPLORE

Thenmala Ecotourism in Keralam

→ Thenmala, located in the Kollam district of Keralam, is India's first planned ecotourism destination, developed jointly by the Tourism Department, the Forest Department, and the Irrigation Department of Keralam. The name *Thenmala* means 'Honey Hill', reflecting the area's natural richness and biodiversity within the Western Ghats—a UNESCO World Heritage Site. The project was launched to promote sustainable tourism while conserving forests and supporting local communities, by employing the inhabitants as guides, eco-guards, and service providers, ensuring that the tourism benefits are shared within the community. Thenmala's success lies in its balanced approach—combining recreation, environmental education, and conservation—making it a model for responsible forest-based ecotourism in India (Fig. 2.14).

→ Is there any such ecotourism site in your State/UT? Collect some information and discuss with your friends.

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→ What should be a tourist's behaviour while visiting any tourist site? What should one be mindful of?

Fig. 2.14. Exploring forest canopy in Thenmala

Forests and Livelihoods

People, particularly rural and tribal communities, rely heavily on forests for their livelihoods. Forests provide essential resources such as food, fuel, fodder, medicine, shelter, timber, bamboo, and firewood. Non-timber forest products, such as honey, fruits, nuts, resins, and medicinal plants, help families meet daily needs and earn income. Several communities including the Gonds and Bhils perform rituals to honour the forest deities before collecting resources, reflecting sustainable cultural practices and a deep respect for nature (Fig. 2.15). Forests also support livestock grazing, small forest-based industries, ecotourism, and employment, making them vital for the livelihood, culture, and survival of many.

Fig. 2.15. Rituals to honour the forest deities

THINK ABOUT IT

Sacred groves are small forest patches protected by local communities for religious and cultural purposes. Across India, there are over 1,00,000 sacred groves, though many are not officially recorded. They are known by different names, such as *Devrai*, *Devarakadu*, *Kavu*, and *Sarna*.

Why are these groves so special? Despite being small, they are rich biodiversity hotspots, often preserving rare plants, medicinal species, and wildlife that may not survive elsewhere. Some groves are even linked to water sources and help to maintain streams and groundwater.

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Timber and bamboo (Assam, Tripura, Keralam)	Wild fruits, honey, mushrooms, and roots (Jharkhand, Odisha)
Tendu leaves, mahua flowers, sal seeds, gums (Madhya Pradesh, Chhattisgarh)	Ecotourism (Uttarakhand, Assam, Keralam)
Herbs like ashwagandha, neem, and tulsi (Himalayan region, Western Ghats)	Handicrafts and cottage industries (Rajasthan, Keralam)
	You may add some examples of forest produces from your own state/UT in the map above.

Fig. 2.16. Forest produces from different Indian states (map not to scale)

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Forest Dwellers

Forest dwellers have a deep and multifaceted connection to forests as they rely on them for household needs, cultural practices, and livelihood activities. Through their roles and responsibilities, they are often deeply involved in caring for and using forest resources. Women, in particular, play a vital role in collecting and managing forest resources for household needs, sustaining traditional knowledge and practices, and contributing to the livelihoods. Some of these connections are:

Knowledge Holders: They possess rich traditional knowledge of using herbs, plants, and forest resources for food, medicine, and cultural practices. This knowledge is passed down through generations and helps in biodiversity conservation.

Livelihood and Income: They are actively engaged in gathering and processing non-timber forest products (NTFPs) such as tendu leaves, mahua flowers, honey, lac, and bamboo, which provide income for households.

Forest Protection: They have been at the forefront of forest protection movements, such as the Bishnoi Movement in Rajasthan led by Amrita Devi. This also influenced the Chipko Movement in Uttarakhand, where village women hugged trees to prevent deforestation. Hence, they often play a key role in community forest management and eco-restoration.

Household Needs: In many rural and tribal areas, forest dwellers collect firewood, fodder for cattle, wild fruits, roots, tubers, and medicinal plants from forests to sustain their families.

Cultural and Social Role: In many communities, forests are linked to several rituals, festivals, and beliefs making them guardians of ecological and cultural traditions.

Fig. 2.17. Connections between forest dwellers and forests (Central image shows women carrying bundles of leaves and wood from a mountain forest in Sohra, Meghalaya)

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LET'S EXPLORE

Fig. 2.18. Traditional tribal livelihoods

Try to identify and classify the items for sale in the images given in Fig. 2.18.

Also, make a list of the food and medicinal varieties available in your local market.

Geographical Indication (GI)

Did you know that some forest products are unique to specific regions and are recognised as Geographical Indications (GI)? A **Geographical Indication (GI)** tag is a type of intellectual property right that identifies a product as originating from a specific place, where a given quality, reputation, or other characteristics of the product are essentially linked to its geographic origin.

In simple terms, it is a label that indicates that a product comes from a particular region and derives its uniqueness from that place. Thus, GI tagging helps protect region-specific products, promote cultural heritage, and generate economic benefits for local communities through exclusive market recognition.

GI-tagged Forest Products in India

India has several forest-based products that have been granted Geographical Indication (GI) tags because they are unique to their region, linked to traditional knowledge, and depend on the forest ecosystem (Fig. 2.19).

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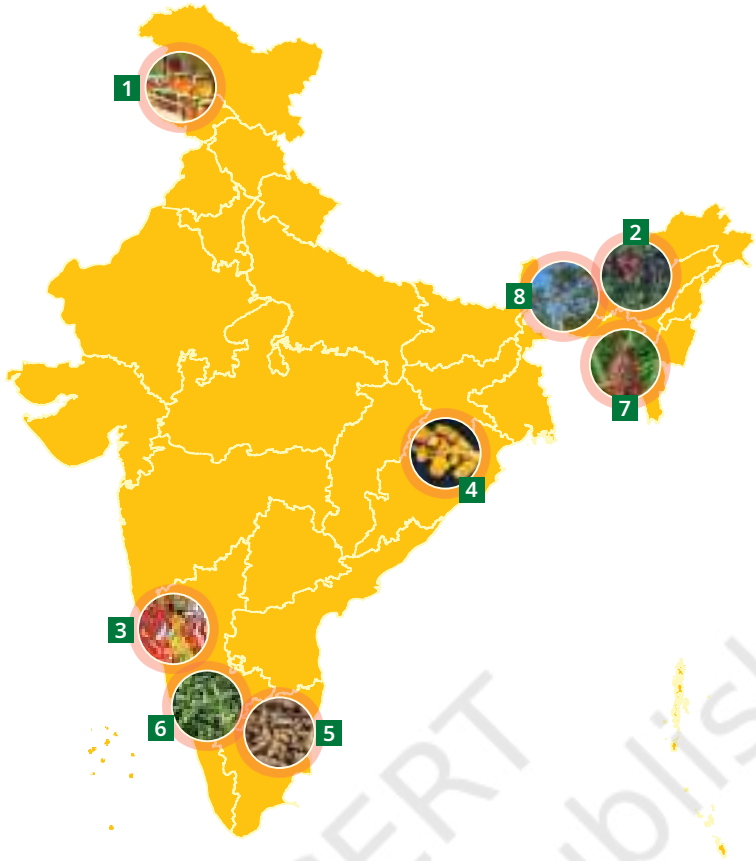

Fig. 2.19. Some GI-tagged forest products of India (map not to scale)

1. **Ramban Sulai Honey (Jammu and Kashmir):**
   It is highly nutritional with strong antioxidant and immuno-modulatory potential and has a distinct flavour, aroma, and taste.
2. **Tezpur Litchi (Assam):**
   - Grown in the forest-influenced soil and climate of Assam's Tezpur region.
   - It has a distinctive bright colour and high pulp content.
3. **Channapatna Toys (Karnataka):**
   The notable aspect of this forest-based craft is that they are made from *Wrightia tinctoria*, a forest tree locally known as 'Aale mara'.
4. **Kandhamal Haldi (Odisha):**
   - A variety of turmeric grown by tribal communities in the forest regions of Kandhamal.
   - It is known for its high curcumin content and medicinal value.
5. **Erode Manjal/Erode Turmeric (Tamil Nadu):**
   Cultivated in **forest-influenced soils**, it is used for cooking and ayurvedic medicine.
6. **Nilgiri Tea (Tamil Nadu, Keralam, Karnataka):**
   Though plantation-based, its quality depends on the forest ecosystem conditions of the Western Ghats.
7. **Tripura Queen Pineapple (Tripura):**
   With a forest-dependent cultivation, it is famous for its sweetness and aroma.
8. **Khasi Mandarin (Meghalaya):**
   A citrus fruit from the forest-rich hills of Meghalaya.

You may explore more GI-tagged forest products from other states through the School Bhuvan-NCERT Geoportal.

Forest-influenced soils: Soil profiles shaped over long periods by forest vegetation, root systems, and organic litter accumulation.

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LET'S EXPLORE

- → Can you explain how GI tagging of the forest products mentioned above connects geography, cultural traditions, local governance, and rural economy?
- → Can you name some GI-tagged forest products from your state or union territory?

Challenges Faced by People Living in Forests

People living in forests often struggle to access essential facilities such as healthcare, education, and clean water. Women are exposed to fuelwood fumes and face, many health issues, making their daily lives difficult. Forest dwellers depend heavily on forest produce such as firewood, honey, bamboo, and medicinal plants for their livelihood. Restrictions on forest use, deforestation, and the commercial exploitation of resources often limit their access to these resources. This creates a conflict between conservation laws and their survival needs, leading to economic insecurity.

Forested hilly regions are prone to natural disasters such as floods and landslides. Floods can wash away homes, crops, and roads, while landslides can bury villages and cut off access to essential services. Forest fires can quickly spread through dense vegetation, destroying homes and resources. Human care and preventive measures can help prevent many forest fires.

Crops are often damaged by elephants, monkeys, or wild boars, and sometimes people face direct attacks. Such conflicts not only cause economic losses but also create fear and insecurity among forest communities.

Many forest dwellers also face discrimination and neglect from mainstream society. They are often engaged in selling forest products at relatively low prices. Therefore, creating opportunities for processing and marketing these products is essential to enhance their income. Forest communities also require assistance in accessing government schemes and programmes, as well as greater awareness of their legal rights.

THINK ABOUT IT

You may already be aware that incidents of forest fires have been reported from different parts of India and are a matter of serious concern. According to the Forest Survey of India (FSI), about 36

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per cent of the country's forest cover is estimated to be prone to forest fires, of which nearly 4 per cent is extremely prone. Climate change as well as human-induced factors are the main reasons for such occurrences, some of which are quite severe and cause extensive damage to human life and the environment (Fig. 2.20). *Note: You may explore various fire locations on the School Bhuvan–NCERT Geoportal.*

Fig. 2.20. A forest fire

LET'S EXPLORE

You may have heard about the Pradhan Mantri Ujjwala Yojana (PMUY). How are the benefits of this scheme impacting the health of women?

THINK ABOUT IT

- Which wild animals (like monkeys or deer) have you seen in your residential area?
- Identify the causes behind these animals' presence in human settlements.

Fig. 2.21. Forest bridge to avoid roadkill

Government Efforts to Support Forest Dwellers

To reduce the hardships of people living in forests, the Government of India has taken several steps. One of the most important measures is the Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006, commonly called the Forest Rights Act (FRA). The Act recognises the rights of eligible communities over the forest land and forest resources that they have traditionally depended upon. It allows them to collect minor forest produce such as bamboo, honey, and medicinal plants, and grants ownership of small patches of land cultivated by them.

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Along with the FRA, the Government runs schemes for tribal welfare, including education scholarships, healthcare programmes, and skill development initiatives. The Joint Forest Management (JFM) programme encourages local communities to participate in protecting forests while benefiting from forest produce. Other initiatives, such as the Panchayats (Extension to Scheduled Areas) Act (PESA), the Dharti Aaba Janjatiya Gram Utkarsh Abhiyan (DAJGUA), and the Pradhan Mantri Janjati Adivasi Nyaya Maha Abhiyan (PM JANMAN), aim to promote the welfare of such communities.

Mendha Lekha: A Case Study in Community Forest Management

Mendha Lekha is a small tribal village located in the Gadchiroli district of Maharashtra, known for its effective community-based forest management and self-governance. The village, inhabited mainly by the Gond tribe, became the first in India to gain rights over its forest resources under the Forest Rights Act (2006). The people of Mendha Lekha follow the principle of *Mawa Nate Mawa Raj* (our village, our rule), ensuring that all decisions are made collectively through the Gram Sabha. They protect and sustainably utilise their 1,800 hectares of forest by regulating tree cutting, preventing forest fires, and promoting bamboo harvesting as a renewable resource. The revenue from the forest produce is used for community welfare activities such as education and infrastructure. This also upholds the fundamental rights of the people to participate in governance and control resources that sustain their livelihoods. Simultaneously, the villagers fulfil their fundamental duties by protecting biodiversity and preventing illegal exploitation. This model has not only strengthened local livelihoods but also demonstrates how traditional knowledge, rights, and duties, as well as democratic participation, can coexist to achieve sustainable forest conservation (Fig. 2.22).

Fig. 2.22. Community-led protection of forests through self-governance, traditional livelihood, and controlled grazing in Mendha Lekha

Ecological Commons—Shared Resources

The term **ecological commons** refer to natural resources like forests, rivers, lakes, grazing lands, air, and biodiversity that are shared by a community and used collectively for its survival and well-being. Unlike private property, these resources are not owned by any single individual but belong to the community as a whole, and are often managed through traditional rules and practices. For centuries, people have depended on the ecological commons for food, water, fuel, fodder, and medicinal plants. However, with population growth, industrial use, and privatisation, many of these commons are shrinking or degrading. Protecting ecological commons is important not only for the livelihoods of rural and tribal communities but also for maintaining ecological balance, regulating climate, and conserving biodiversity.

A Case Study of Commons-based Fisheries Management in Bihar

After independence, The Indian Fisheries Act, 1897, remained the basis for the governance of inland fisheries. Historically, control over perennial water bodies and *diara* waterfronts in Bihar was held by large landowners who marginalised the fishing community, rendering them the most vulnerable group in an environment abundant with aquatic resources.

Fig. 2.23. Fishermen at work in a maun (lake) at dawn

The historical insights from Bihar's 19th century case study suggest that a commons-based management model is best suited to riverine regions. To promote community-led management, legal safeguards for public access, and cooperative resource stewardship as

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sustainable alternatives—along with the preservation of cultural heritage associated with riverine waterscapes—the Bihar Fish Jalkar Management Act, 2006, and its subsequent amendments, were introduced. Under this model, local communities develop context-specific solutions to balance the productivity of local resources with sustainable rates of exploitation. The success of this model is reflected in the persistent growth of fish production in the region.

Case Study of Water Tanks Built by the Gond Kings

In India, for centuries, regions with seasonal rainfall have tackled the issue of water demand and management as a collective concern and used common public resources for the same. In fact, between 800 and 1200 CE, the Chandella and Bundela kings built around 8,000 tanks in the Bundelkhand region. Similarly, *keres* were built in Karnataka, *eri* tanks in Tamil Nadu, and *chevuru* tanks in Andhra Pradesh and Odisha. Bihar had the famous *ahar pynes* to store water and the Gond kings funded huge *katas*—three-sided reservoirs—in the Gondwana region. According to official estimates, there are more than 15 lakh functional tanks in India at present.

LET'S EXPLORE

Below is Fig. 2.24 from the School Bhuvan–NCERT Geoportal. Observe the image carefully and see if you can identify some of the tanks and mark the agricultural fields irrigated by them.

Fig. 2.24. Water tanks in Bhandara district, Maharashtra

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Before we move on...

- Biomes are large areas on Earth with specific vegetation and animal life. Biomes are classified based on temperature, precipitation, flora, fauna, etc.
- Humans interact closely with different biomes and adapt their lifestyles to the conditions and resources available.
- Traditionally, different human cultures have developed various practices for conserving nature. Today, there is a legal framework to support conservation efforts.
- Occupations have been closely associated with forests, right from primitive hunting to modern ecotourism. The variety of human interactions with forests highlights their significance for various communities.
- With growing population, there is increasing pressure on forests, which leads to problems like human–wildlife conflict, depletion of resources, etc.

Questions and Activities

1. Why does India have diverse biomes? How does the local geography connect to the diversity?
2. Make a list of economically valuable native plants in your region and comment on their abundance.
3. Prepare a map of your state or Union Territory showing national parks and sanctuaries.
4. Study the impact of forest fires on people dependent on forests. Try to include the causes, losses, and preventive measures.
5. Why are forests often called the 'lungs of the Earth'?
6. If large forest areas continue to be destroyed, what do you think would be its long-term effect on the global climate?
7. Discuss some traditional forest conservation methods with your grandparents or an elderly person.
8. How do forest dwellers contribute to the protection and conservation of forests?
9. What is a forest fire? Prepare a plan for prevention and mitigation of forest fires.

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Notes

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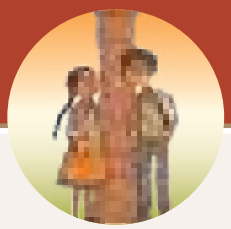
Chapter 3

Resistance and Resilience (1000–1700 CE)



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The Big Questions

1. *How did Indian rulers use resistance and alliances to safeguard their lands?*
2. *What factors contributed to the resilience of different kingdoms and communities during this period?*
3. *How did the Bhakti tradition foster a sense of solidarity?*
4. *How did the expansion of agriculture, trade, and commerce keep the economy sound?*

The period between 1000 CE and 1700 CE witnessed profound political, social, and cultural changes. The Indian subcontinent experienced repeated invasions and wars accompanied by religious discrimination and social distress. Regional powers frequently resisted invaders who were making inroads into their territories. This resistance took both military and cultural forms. Nonetheless, this period manifested the resilience of the Indian civilisation, which safeguarded its core ideas and values. Indian society not only defended itself from external threats but also preserved its traditions. In this chapter, we will see how resistance and resilience shaped this period of Indian history.

Let us begin by understanding the meaning of resistance and resilience in the context of the period under study and how they are interconnected.

Resilience refers to the capacity to, survive and adapt in challenging situations, including times of political unrest and instability. It is also about how communities and groups across diverse regions respond to challenges and opportunities. This capacity to absorb and adapt to hardship while preserving inherited socio-cultural and spiritual traditions can also be understood as resilience. The idea of India's resilient spirit extends beyond political and military spheres.

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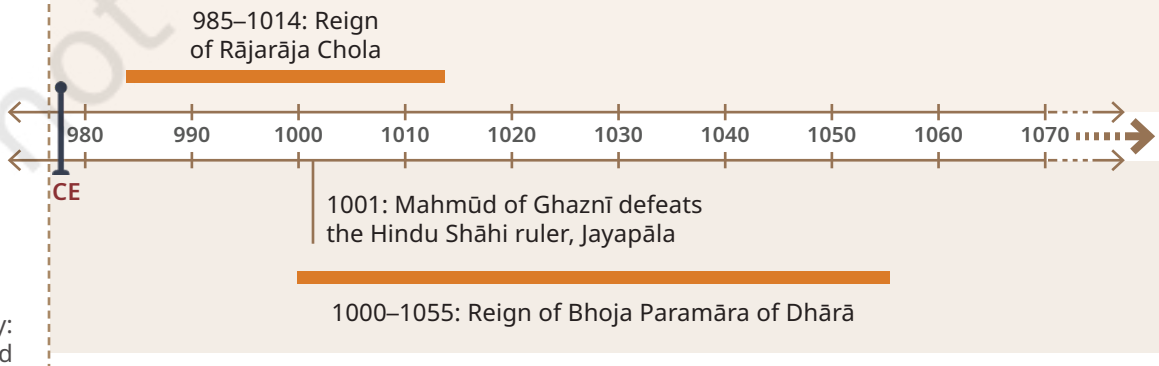
For instance, rulers and communities restored places of worship, modified customs to suit changing circumstances and, in many regions, established lasting patterns of coexistence across religious and cultural lines.

Resistance means actively opposing or challenging a force, authority, or threat in order to protect political power, territory, interests, or cultural and spiritual identity. Resistance may take the form of warfare, alliances, diplomacy, negotiation, or other political actions.

Resistance and resilience are, in fact, closely linked. Resilience is the backbone of resistance. While a political cause or leadership provides the condition for resistance, the ability to endure comes from resilience. Together, resistance and resilience help explain how Indian society sustained itself through different ages, particularly the period under study. This period is characterised by both conflict and cooperation, rather than being merely a story of invasion and resistance.

Safeguarding Sovereignty: Resistance, Alliances, and Confederacies

From the 7th century onwards, the Indian subcontinent witnessed successive phases of invasions driven by territorial expansion and religious motives. It may also be noted that the surplus agrarian produce of the fertile plains fed by perennial river systems contributed to the people's economic well-being and also became a major reason for the invasions. Against this backdrop, let us examine instances of resistance, as well as the strategic alliances and confederacies that emerged in these centuries at both the regional and pan-Indian level.



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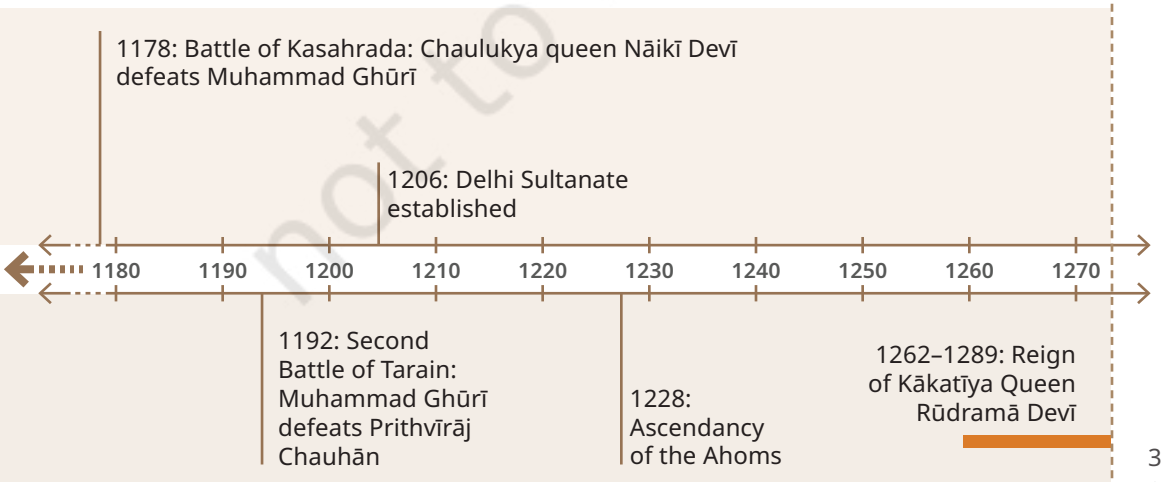
From the mid-7th century, Arab forces launched invasions by sea and land. These early invaders were confronted and repulsed by Indian rulers across the Deccan, Malwa, Kashmir, and Rajasthan. Mahmūd, the Sultan of Ghaznī (in present-day Afghanistan and northeastern Iran), launched his first invasion in 1001 against Jayapāla of the Hindu Shāhi dynasty. During the 11th and 12th centuries, rulers of the Hindu Shāhi kingdom, and the Chandellas, Solankīs, and Chauhāns opposed and fought against the Turkish invaders. In 1178, Queen Nāikī Devi of the Chaulukya kingdom defeated Muhammad Ghurī at Kasahrada, near Mount Abu. In northern India, some powers, such as the Chauhāns, Gahaḍavālas, and Senas, however came under Turkic control towards the end of the 12th century, even as Gujarat, Malwa, and Bundelkhand under the Chandellas put up stiff resistance for about a hundred years thereafter. The Turks failed to conquer present-day Odisha, and parts of present-day Assam. Several states of Rajputana, in particular, remained centres of sustained resistance throughout the period under study. Thus, during the 11th and 12th centuries, northern and western India bore the brunt of Turkish invasions, while peninsular India remained free from Turkic control.

LET'S RECALL

You may recall learning about the invasions of Mahmūd of Ghaznī and Muhammad Ghūrī in the 11th and 12th centuries, covered in Grade 7 Social Science textbook. These invasions had seriously disrupted the political, social, and religious fabric of the Indian subcontinent.



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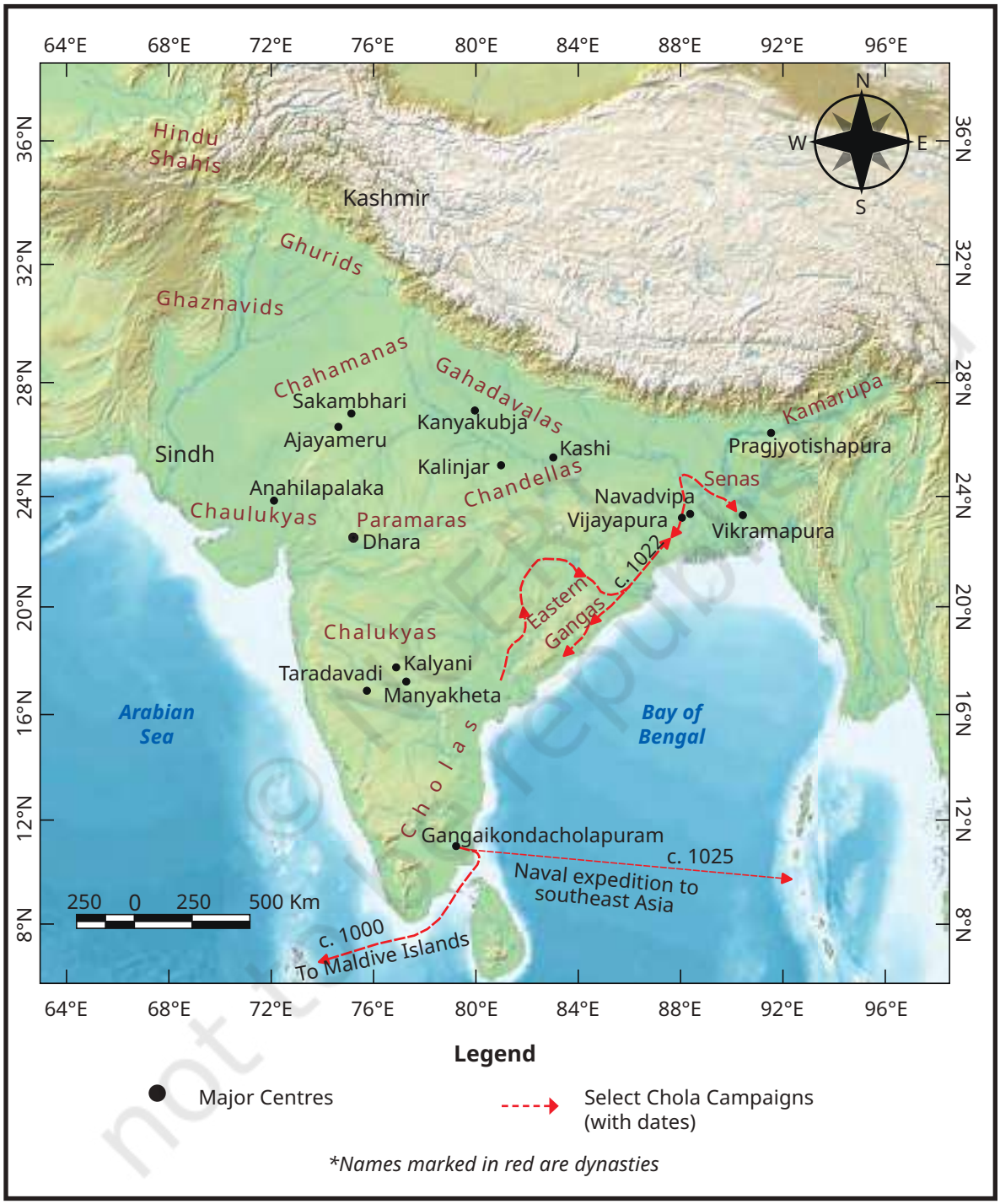
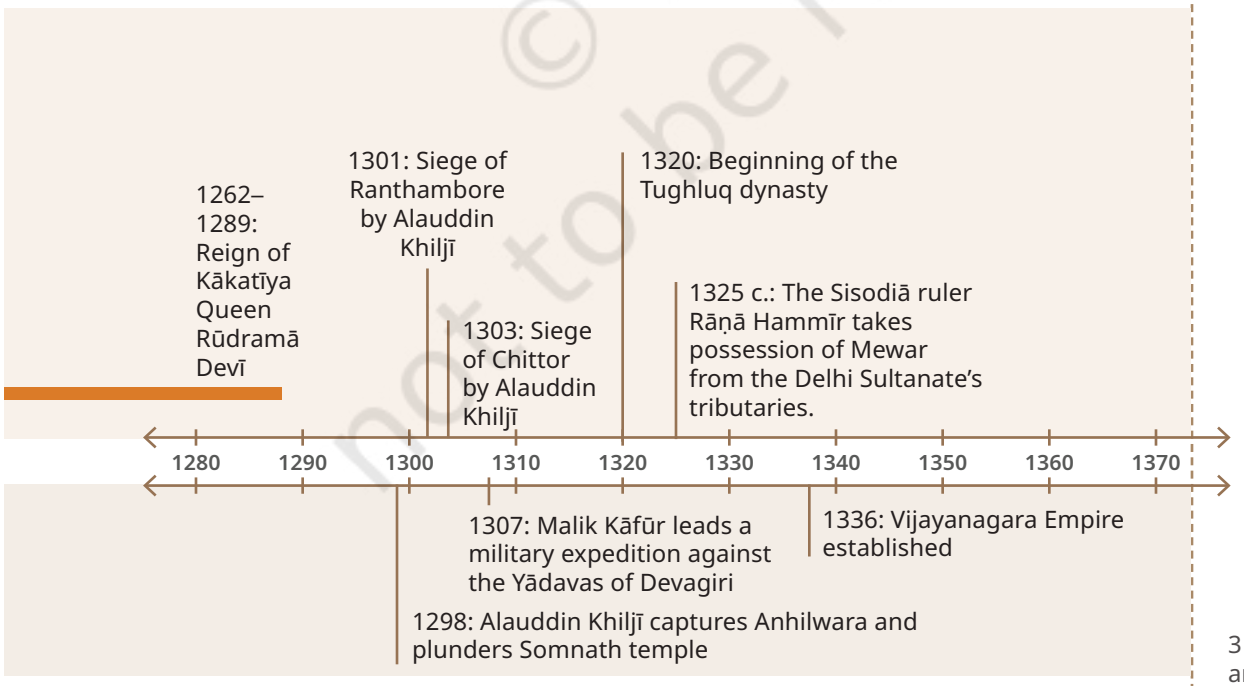


Fig. 3.1. Some political kingdoms of Indian subcontinent in the 11th and 12th centuries

Rajput Dynasties: Confederacies and Resistance (1200–1500 CE)

The establishment of the Delhi Sultanate in 1206 was a turning point in Indian history. The new state introduced administrative practices that led to greater centralisation of authority. For example, the *iqta* system, under which territories were assigned to military commanders in lieu of salaries, became a mechanism for land revenue collection and a system that increased the burden on the peasantry. Furthermore, during their military campaigns for territorial expansion, the forces of the Sultanate demolished sites sacred to the local communities. This led to stiff resistance from several kingdoms and chieftains. Therefore, resistance throughout the Delhi Sultanate was fuelled by both political factors and the determination to preserve socio-cultural continuity. For instance, the Yādava dynasty of western Deccan (Devagiri), several Rajput dynasties of present-day Rajasthan, the Kākatīya rulers of eastern Deccan (Warangal), and the Hoysālas of present-day Karnataka repeatedly resisted the Sultans of Delhi. These states played an important role in checking the expansion of the Sultanate.

Over the years and under different Sultans, the Delhi Sultanate faced opposition and defiance from several states and regions including Mewar, Gwalior, Malwa, Awadh, Etawah, Ajmer, Jalore, Kalinjar, and Kannauj.



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Rajput–Mughal Dynamics: Politics and Complexities of Rule (1500–1700 CE)

In the early 1500s, the weakening and subsequent decline of the Delhi Sultanate under its last ruler, Ibrāhim Lodī, opened the door to two parallel developments. First, Zahiruddin Muhammad, also known as Bābur, from Central Asia launched an invasion on India. Second, Indian rulers began to assert their own authority. In 1526, Bābur defeated the Lodi forces in the First Battle of Panipat, bringing the Sultanate to an end. In 1527, Rāṇā Sangrām Singh I, popularly known as Rāṇā Sāngā, of Mewar led a Rajput confederacy against Bābur in the Battle of Khanua. The battle ended in favour of Bābur and is seen as an important turning point in Indian history.

As Mughal rule consolidated its power, relations between the Rajputs and the Mughals evolved over time, shaped by strategic considerations. Some Rajput kingdoms such as Mewar, maintained stiff resistance to Mughal authority, while others like the Kachhwāha Rajputs of Amber, negotiated cooperative relationships. The Mughals, for their part, had a strong incentive to collaborate with the Rajputs—they needed the Rajputs' military might, their loyalty as well as their administrative strength to consolidate and stabilise the Mughal Empire. Towards this end, the Mughals integrated some of the Rajputs into the empire through the grant of **mansabs** and other understandings.

Mansab: It means 'rank' or 'office', and its holders were known as mansabdars. Akbar introduced the mansabdari system, which combined military organisation with civil administration. It established a clear hierarchy among the nobility and helped strengthen imperial control.

Mewar's Resistance against Mughals

Akbar attacked the kingdom of Mewar and laid siege to the fort at Chittor in 1567–1568. The Mughal army paid a heavy price in the campaign. Under Rāṇā Udai Singh, the Rajputs put up a stiff resistance for more than five months before the fort was captured

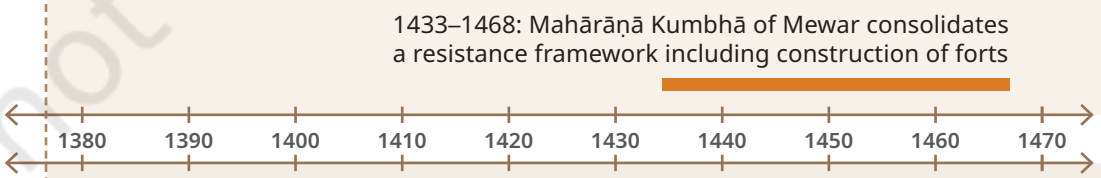

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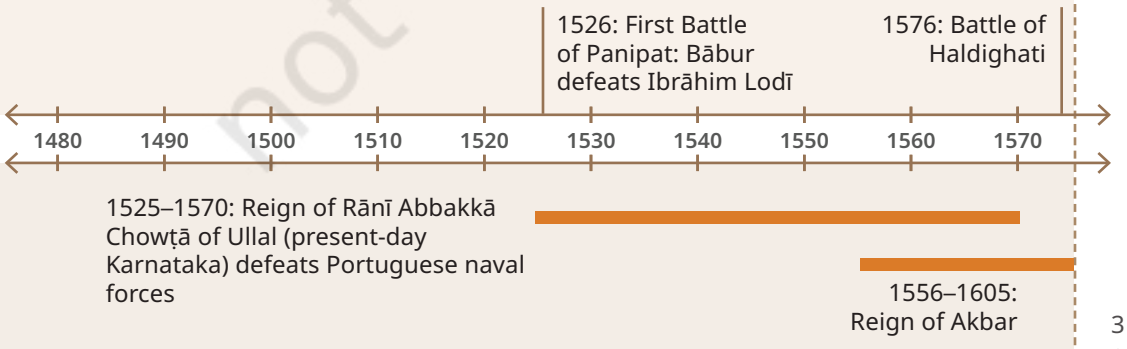
Fig. 3.2. A painting depicting the Battle of Haldighati

by the Mughals. By this time, Akbar had already concluded several successful campaigns and the empire was at the peak of its power. In a strategic move, Rāṇā Udai Singh retreated to the Aravalli Hills along with a large number of his subjects.

In 1576, Akbar launched another attack on Mewar. This time, Mahārāṇā Pratāp, the son and successor of Rāṇā Udai Singh, fought against Mughal forces in the Battle of Haldighati. Continuing his fierce resistance, Mahārāṇā Pratāp modified his military strategy and adopted guerrilla warfare. He used the rugged Aravalli terrain to his advantage and conducted surprise attacks, raids, and ambushes on the Mughal forces and their networks. He continued to resist and eventually reclaimed most of his ancestral territories in the 1580s.



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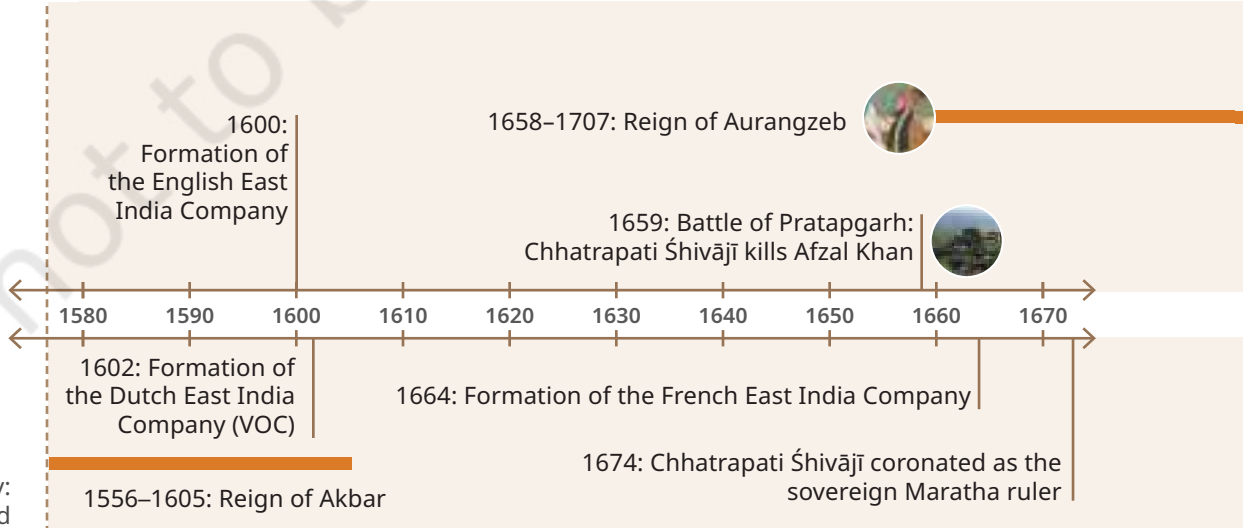
The Battle of Dewair fought near Udaipur in 1582 was significant as it resulted in the collapse of key Mughal military outposts in Mewar and helped Mahārāṇā Pratāp reclaim control over Mewar.

Mahārāṇā Pratāp's son and successor, Mahārāṇā Amar Singh, continued Mewar's tradition of military resistance against the Mughals. Jahāngīr made military advances towards Mewar in 1605, 1608, and 1609. But each attempt was countered by a strong and organised resistance from the Rajputs. Eventually, the Mughals were compelled to acknowledge Mewar's distinct political position.

Fig. 3.3. Victory memorial in Dewair, Rajasthan

Fig. 3.4. A 19th century miniature painting of Mahārāṇā Pratāp



On one hand Mewar continued its defiance in the form of a strong, organised, and determined resistance, and on the other hand, the Mughal state began to recognise and appreciate Mewar's unique geo-political position with trade routes leading to the major port of Surat. Needing peace with the rulers of Mewar more than ever, the Mughals were eventually compelled to negotiate a treaty. Under the terms of the treaty, Mewar leadership was exempted from personal attendance at the Mughal court, thus clearly acknowledging Mewar's continued autonomy.

LET'S RECALL

Do you remember reading about the Battle of Haldighati in your Grade 8 textbook? You may recall that the Bhil community provided crucial, tactical, and logistical support to Mahārāṇā Pratāp in his resistance against the Mughals. Their extensive knowledge of the Aravalli terrain, expertise in guerrilla warfare, skills in archery, role as guides, and above all, their bravery strengthened Mewar's resistance. This enduring alliance was later commemorated in Mewar's state emblem (Coat of Arms), which depicts a Rajput warrior and a Bhil warrior with the local deity Eklingji (Shiva) shown at the centre.

Fig. 3.5. Mewar's Coat of Arms

1658–1707: Reign of Aurangzeb

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LET'S EXPLORE

After the death of Mahārāṇā Pratāp, renowned poet Dursā Arhā gave a vivid account of Mahārāṇā Pratāp in the following lines:

> *O Pratap, you kept your horses unbranded, your head unbowed, your fame untarnished. You were strong enough to carry on your work against heavy odds. You never participated in the Nauroz festival, nor did you mount guard for the imperial presence during Jharokha Darshan. You attained a very high place in this world.*

Fig. 3.6. Source: Rawat Saraswat, Dursā Arhā (New Delhi: Sahitya Akademi, 1983)

Identify any two specific actions mentioned in the text which show that Mahārāṇā Pratāp maintained his independence from the Mughal court.

DON'T MISS OUT

Other Rajput rulers like Kānhaḍa Deva Chauhān (Jalore), Rāṇā Hammīra Deva (Ranthambore), Rāṇā Kumbhā (Mewar), and Rāo Surjan Hāḍā (Bundi) put up a strong resistance at different points of time.

The Ahom Kingdom: Centuries of Endurance

The Ahom kingdom of Assam maintained its political independence for almost six centuries (1228–1826) and successfully resisted several invasions and attempts at annexation. The military strategy of the Ahoms relied on exploiting the local terrain rather than engaging in direct confrontation. Ahom commanders possessed detailed knowledge of Assam's forests, river systems, and monsoon cycles, which invading forces generally lacked. Despite repeated attempts by the Mughals to annex the Brahmaputra valley, the Ahoms successfully defended their territories against Mughal invasion.

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Fig. 3.7. A monument at Guwahati, Assam commemorating the Battle of Saraighat

The first Ahom–Mughal clash took place in 1615 near Samdhara. Mughal forces under Abu Bakr made early gains but were soon compelled to retreat. By 1616, the Ahoms had recovered the lost territory, setting a pattern that would recur over the next sixty years. In 1662, Mīr Jumlā captured the Ahom capital, Garhgaon. Subsequently, a treaty was imposed on King Jayadhvaj Singha. However, the king defied the Mughals and renounced the treaty within a year of the Mughal withdrawal, and the Ahoms retained their independence.

A decisive check on Mughal ambitions in the region came in 1671, when the Ahom commander Lachit Borphukan defeated a Mughal fleet at the Battle of Saraighat, securing the Brahmaputra as an effective territorial boundary. The Ahoms' successful resistance against the might of the Mughal empire is a remarkable story of bravery, pride, and endurance.

Ahom endurance rested on a combination of strategic alliance-building, guerrilla tactics, Assam's challenging geography, and the *Paik* system (about which you have read in Grade 8) of military-administrative organisation. The Ahom navy also played a key role, using light, fast boats, called *bāchārī*, built specifically for river warfare, which gave them an advantage over Mughal fleets.

Fig. 3.8. Model of a bāchārī boat

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Fig. 3.9. An aerial view of Hampi, the capital of the Vijayanagara Empire

Vijayanagara Empire: Resilience against Expansion

Resistance to foreign invasions was not confined to northern India. In the Deccan and beyond, the expanding campaigns of the Delhi Sultanate steadily weakened the older kingdoms of the south. Malik Kāfūr, Alauddin Khiljī's general, attacked major cities and temples, including those at Warangal, Belur, and Madurai. Later, Muhammad bin Tughlaq's rise to power extended the Sultanate's influence even farther south, ultimately contributing to the decline of the Pandya Kingdom and the eventual emergence of a new sultanate territory known as the Madurai Sultanate.

It was against this backdrop that the Vijayanagara Empire rose in 1336 to check the expansion of the Sultanate into peninsular India. Harihara and Bukkā established the independent kingdom of Vijayanagara—meaning 'City of Victory'—located near present-day Hampi, Karnataka. The region's geography, including a formidable river and rugged, rocky terrain, provided Vijayanagara with a strong natural defence and a secure base for military operations. They made determined efforts to defend and expand their state's frontiers. The empire gradually expanded by annexing territory south of the Krishna and Tungabhadra rivers and became one of the richest and most influential states, encouraging the flourishing of literature in Kannada, Telugu, Tamil, and Sanskrit. Vijayanagara's capital city became a centre of trade, culture, and political power in southern India.

During Bukkā's reign, Vijayanagara made major territorial gains, including the subjugation of the Madurai Sultanate and the annexation of Arcot. In 1342, the ageing Hoysāla king Vīra Ballāla III

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fought the Madurai Sultanate, resulting in the Battle of Kannanur, which was a pivotal episode in this struggle. However, the Sultan launched a surprise counter-attack and executed the Hoysāla ruler. For almost four decades, Vijayanagara wrestled with the Sultans of Madurai. Later, the campaign against Madurai Sultanate was led by Kumāra Kampana, the son of Bukkā. He killed the Sultan and brought the Sultanate of Madurai to an end. Kumāra Kampana also reinstated the divine images in the Rajasimheshwara Temple at Kanchipuram and the Sri Ranganathaswamy Temple at Srirangam.

LET'S RECALL

In Grade 8, Part I, you have read about how the Vijayanagara Empire reached its peak in the 16th century, under its celebrated ruler, Kṛiṣhṇadevarāya, who expanded and secured the empire's dominance over the Deccan. Under his rule, the empire achieved both military power and cultural glory. Kṛiṣhṇadevarāya provided grants to many temples, including Tirupati in Andhra Pradesh and the Vitthala temple in his own capital Vijayanagara, which had many grand temples, palaces and other buildings. After winning many wars against his neighbours, Kṛiṣhṇadevarāya died of illness in 1529.

Decades later, the Vijayanagara army was defeated at the Battle of Talikota in 1565 by an alliance of the Deccan sultanates. In the aftermath, the city of Vijayanagara was sacked and largely abandoned, while the empire gradually fragmented. Several of its territories came under the rule of smaller Nāyaka kingdoms.

DON'T MISS OUT

By the end of the 16th century, the Vijayanagara Empire had split into smaller kingdoms ruled by the Nāyakas. They also continued the tradition of resistance—not always through military action, but sometimes through bold acts of defiance aimed at Mughal authority. An interesting account is of Muthu Vīrappa Nāyak III of Madurai. Local tradition holds that the Mughal emperor would dispatch a royal slipper to rulers as a symbol of imperial supremacy and domination. Recipient rulers were expected to receive the slipper ceremonially at the borders of their territories and pay tribute. When the slipper arrived at the borders of Madurai, Muthu Vīrappa Nāyak refused to observe this protocol. In an act of defiance against

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this ritual of subordination, he refused to meet the procession at the border, compelling the Mughal emissary to carry the token right up to the palace hall. Even then, Muthu Vīrappa Nāyak refused to bow before the slipper. Instead, he forced the Mughal representatives to place it on the floor, put his own foot inside it, and mockingly asked why the Badshah had not sent a matching pair. He then had the messengers expelled from his kingdom.

Fig. 3.10. A painting of Chhatrapati Śhivājī with his personal guards

Marathas: The Quest for Swarājya

The Marathas were a formidable force in the Deccan particularly during the 17th and 18th centuries. The establishment of an independent Maratha state in 1674 is closely associated with the rise of Chhatrapati Śhivājī. The Marathas had been involved in the region's political affairs for centuries. Their knowledge of the Deccan's complex geography and landscape, along with their bravery and martial abilities, made them a formidable force in both administrative and military affairs.

LET'S RECALL

You have already learnt about the foundation of Maratha power and the rise of Chhatrapati Śhivājī in your Grade 8 textbook, Part I. Think back to his struggle and the ways in which he built and strengthened his kingdom. What do you remember about the strategies and institutions that helped the Marathas establish their power?

Over time, the Marathas emerged as a prominent political and military force in India. In the Maratha state, territorial control and sovereignty were closely linked to the idea of *swarājya*, or self-rule. Chhatrapati Śhivājī is regarded as one of the earliest proponents of

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the spirit of *Swarājya*—the idea of an independent, self-governing territory free from external control.

As you have read in your Grade 8 textbook, the Marathas fought many battles against the Mughals and other powers. The prolonged warfare exhausted the Mughal Empire economically and politically, to the point that Aurangzeb spent more than two decades in the Deccan in an attempt to control the region and subdue Maratha resistance. As political boundaries shifted constantly, the Marathas relied on strategic alliances to extend their control over diverse territories. Through these alliances, they built a vast network of tributary relationships that extended their influence far beyond the areas under their direct rule. The key to the success of the Marathas was their bravery and military skills.

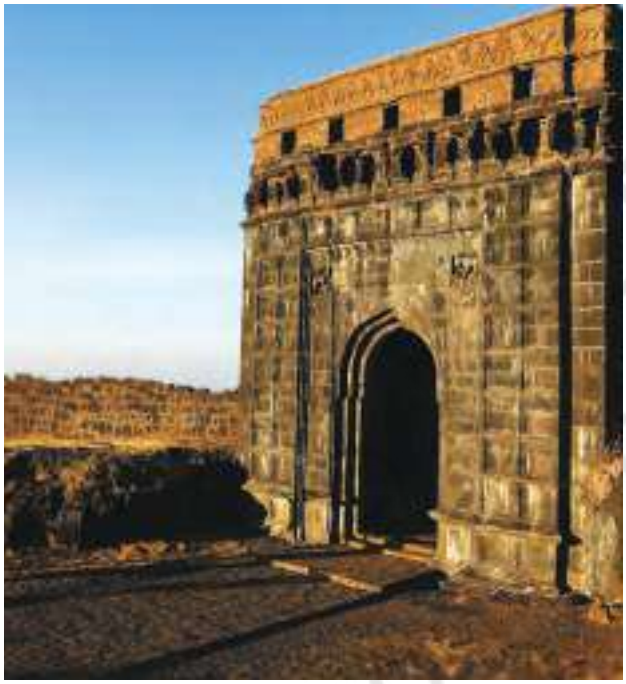

Fig. 3.11. Raigad fort

Fig. 3.12. Sinhagad fort

The Marathas strategically built forts such as Raigad and Sinhagad which played a crucial role in resistance against the Mughals. The Maratha territories were characterised by hills and deep valleys. Maratha forts, therefore, functioned as strategic nuclei for guerrilla warfare. This network of forts enabled the Marathas to launch swift attacks from different directions and provided safe retreats. Different types of forts were constructed to serve distinct purposes, and while some functioned as advanced outposts, those on the frontiers served

as fallback capitals to avoid total defeat. Many formed part of the wider defensive network, and most were designed to withstand prolonged sieges. Forts also served as centres of administration and judicial authority. Strong defences, water sources, and armouries provided resilience and self-sufficiency, enabling the Marathas to sustain prolonged resistance during sieges.

DON'T MISS OUT

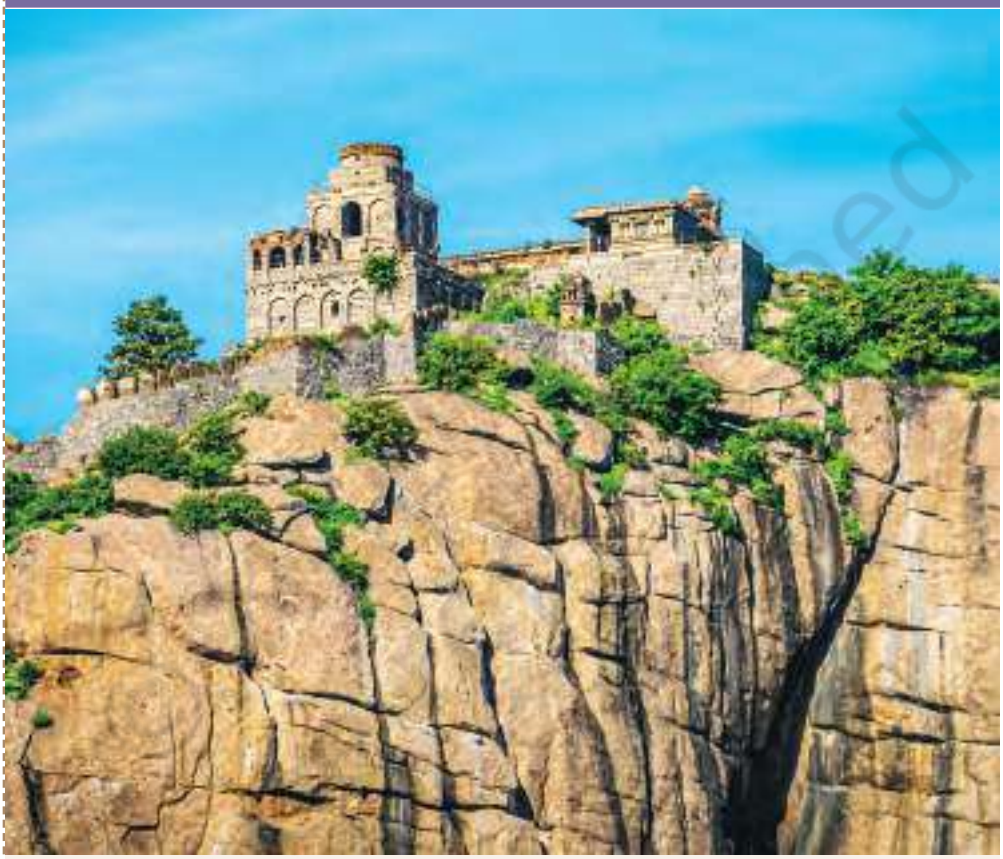

Fig. 3.13. Gingee Fort or Senji Fort in Tamil Nadu

The Fort of Senji, or Gingee, is situated in the Villupuram district of Tamil Nadu. It is considered a marvel of military and civil engineering and military architecture. One of its gateways is said to have been designed so that a small contingent of soldiers could defend it against an army of ten thousand. The fort came under the control of several powers, including the Vijayanagara Nayakas, Bijapuri Sultans, Mughals, Marathas, French, and English. Its changing control also highlights the fort's strategic importance and centrality to the political affairs of the region.

In 2025, the Fort was declared a UNESCO World Heritage Site during the 47th session of the World Heritage Committee in Paris.

Fig. 3.14. Some political dynasties in the Indian subcontinent in the 17th century

The Sikhs: A Story of Spiritual and Political Resilience

A major spiritual movement was founded by Guru Nānak in the late 15th century which later came to be known as Sikhism. It emphasised devotion to one God, social equality, and the rejection of rituals. Under Guru Nānak and his successors, the Sikh community expanded and flourished across the subcontinent, particularly in the Punjab, Sindh, and Gangetic plains. During this period, Punjab was largely under the Lodī Sultans and later under the Mughals. It was a fertile, densely populated region, with a large peasant population.

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Fig. 3.15. A 19th century image of Sri Harmandir Sahib (Golden Temple), Amritsar

Due to the growing socio-cultural influence of the Sikh Gurus, the Mughal state increasingly intervened in their affairs from the beginning of the 17th century. As the Guru's influence was increasing, Jahāngīr ordered the execution of Guru Arjan Dev (the fifth Sikh guru) on the pretext of the Guru's alleged support to Jahāngīr's rebellious son, Khusrau Mirza. The martyrdom of Guru Arjan Dev marked a significant turning point in the history of the Sikhs and contributed to a gradual shift in their organisational and political orientation. Later, Guru Hargobind (the sixth Sikh Guru) was detained by Jahāngīr in Gwalior Fort. During this period, the Sikhs visited Gwalior in large numbers and paid their respects to the Guru from outside the fortress. In 1675, Aurangzeb ordered the execution of Guru Tegh Bahādur, the ninth Sikh Guru. The martyrdom proved a turning point in Indian history. The brutality of the Mughal state against the Sikh Gurus particularly Guru Arjan Dev, Guru Tegh Bahādur, and the tenth Guru, Guru Gobind Singh has few comparisons in history. The valour and sacrifice of the Gurus for the cause of the faith became the defining features of Sikhism. Beyond political motivations, the act was also deeply rooted in the rigid religious and discriminatory practices of the Mughals. The martyrdom of the Sikh Gurus to uphold their faith and principles firmed the conviction of the Sikhs that persecution had to be met with firm resistance and resilience. Such persecution and political pressure could only be countered through collective organisation, and armed resistance as and when required. Many armed conflicts took place between the Sikhs and the Mughals. Guru Hargobind established the *Akāl Takht*, a seat of temporal authority, to administer justice and address matters of governance. Scholars suggest that the establishment of the *Akāl Takht* was intended, in part, to counter Mughal political dominance.

The concept of *mīrī-pīrī* became vital in this context, representing the Guru's authority over both spiritual (*pīrī*) and worldly (*mīrī*) affairs. In Amritsar, the Harmandir Sahib (Golden Temple) represented spiritual authority (*pīrī*), while temporal power (*mīrī*) was represented by the *Akāl Takht*. In 1699, Guru Gobind Singh established and developed a

Sikh order known as the Khalsa. The Khalsa adopted a distinct martial code in response to the changing socio-political environment. Through this process, spiritual identity became closely intertwined with the pursuit of political identity.

Due to sustained Mughal hostility, including the execution of Guru Gobind Singh's sons, the Sikhs organised themselves into armed groups known as *misals* to defend themselves and the territories through military and administrative structures (Eventually, these *misals* became a network of confederacies that united the Sikhs against the Mughals.). In this process, many people from peasant communities became soldiers and fought against Mughals in Punjab.

Jat Resistance

During the 17th century, the Jats, a farming community from Ganga–Yamuna Doab were burdened with heavy taxes and exploitation by Mughals. These economic hardships combined with Aurangzeb's religious policies led to growing resentment amongst the Jats. In 1669, under the leadership of Gokul Jat, they rebelled against the Mughal Empire. Mughal forces struggled for many months to suppress the uprising.

However, the Jats continued their resistance. In 1686, they took advantage of Aurangzeb's prolonged military campaigns in the Deccan and rebelled once again under the leadership of the *zamindars* of Sinsani and Sogar. The Mughals sent several armies to suppress the rebellion, leading to a long and fierce struggle in which both sides suffered heavy losses. By 1691, the Mughals had gained the upper hand. Yet, the Jat resistance continued.

DON'T MISS OUT

Niccolao Manucci, an Italian traveller, wrote an extensive account about India during his journey (17th century) including descriptions of Jat resistance against the Mughals.

Bhakti Tradition: Affirmation of the Spirit of Resilience

India underwent a profound transformation in its religious and spiritual landscape between the 11th and 17th centuries. Bhakti—a tradition of personal devotion to the divine—was expressed through devotional

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Fig. 3.16. Saint Kabīr with Nāmdeva, Rāidās, and Pipājī, Jaipur, early 19th century

hymns composed in local languages. From the 7th century onwards, the rise and expansion of Bhakti in peninsular India shaped social, religious, and political life. Bhakti tradition emphasised inclusivity, socio-religious equality, simplicity, and connecting with people's everyday life. For instance, Eknāth composed in Marathi and propagated a devotional path that was accessible to all without dependence on institutions.

LET'S EXPLORE

Explore the contributions of the Nayanar saints (devotees of Śhiva) like Karaikal Ammaiyar and the Alvar saints (devotees of Viṣhṇu) like Andal to the spread of the Bhakti tradition in peninsular India. Discuss their teachings and devotional practices and the ways in which they connected with people through their hymns.

DON'T MISS OUT

Bhakti saints sang divine lores in different languages, but the unifying factor was devotion to the ultimate reality.

From the 14th century onwards, a strong wave of Bhakti spread across northern India. Rāmānanda, a prominent saint of this period, helped bridge the Bhakti traditions of southern and northern India. The tradition developed into two broad streams—*Nirguṇa bhakti*, devoted to a formless divine, and *Saguṇa bhakti*, centred on a divine with attributes. One of the remarkable features of the Bhakti tradition was the diverse social and cultural background of its saints.

The Bhakti tradition not only provided a spiritual refuge, but also played a role in preserving and transmitting core cultural traditions. When the construction of temples faced restrictions or existing temples were being destroyed, devotees could turn to non-material forms of worship, such as *nām kīrtan* (the recitation of the divine Name). This made devotional practice less dependent on

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physical structures that could be destroyed. Bhakti flourished through local songs and community participation (*satsang*), making it accessible to wider sections of society and more difficult to suppress. For instance, Sankardev composed the *Kīrtana Ghosha,* a highly respected work that remains central to present-day Vaishnava traditions.

Fig. 3.17. King Pratap Rudra of Odisha bowing to Chaitanya Mahāprabhu

Bhakti saints such as Chaitanya Mahāprabhu and Guru Nānak used community singing to bridge social divides. Community singing, or *sankīrtan*, drew large gatherings and fostered a shared sense of community and collective emotion among devotees. In some instances, it also became an act of defiance. For example, Chaitanya Mahāprabhu led public *sankīrtan* in Bengal to protest against restrictions imposed by a Qazi on devotional practices. In Rajasthan, religious orders such as the Dādu Panth and Niranjani Sampradāya provided a form of stability. Some of them even maintained standing armies of warrior ascetics to protect trade routes, facilitate the movement of saints, and provide educational and medical services.

DON'T MISS OUT

The widespread popularity of Indian literature also strengthened social resilience over time. The two great epics, *Rāmāyaṇa* and *Mahābhārata*, were translated into many regional languages. Tulsidas, a well-known Bhakti saint from northern India, composed the *Rāmcharitmānas* in Awadhi, a widely spoken variety of Hindi in the Awadh region of present-day Uttar Pradesh. By doing so, he brought the story of Rāma into households across the region. The work played a transformative role in the Bhakti tradition by making the spiritual knowledge and teachings of the *Rāmāyaṇa* accessible to wider sections of society.

The Bhakti tradition's emphasis on equality and inclusion gradually encouraged greater participation of women. Many rose to prominence as saints, coming from diverse socio-economic backgrounds. They expressed their spirituality through devotional hymns composed in their own languages. The use of local languages strengthened shared identities and fostered a sense of solidarity.

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DON'T MISS OUT

Fig. 3.18. A stamp commemorating Atukuri Molla

Atukuri Molla, a Telugu poet, composed the *Rāmāyaṇa* in Telugu, now known as the *Molla Rāmāyaṇam.* Her father was a potter in Gopavaram, a village in Badvel Mandal, Andhra Pradesh. She also recited the *Rāmāyaṇa* at the court of Kṛiṣhṇadevarāya.

Some Bhakti saints left their homes and devoted themselves entirely to the divine. Akkā Mahādevi (from present-day Karnataka) and Lāl Ded (from present-day Kashmir), both Shaivites, renounced worldly comforts and attachments, living as ascetics. In contrast, Bahināi, a saint from Maharashtra, fulfilled her familial duties while following the path of devotion to the deity Vithoba.

Northern and Central Region

- Lāl Ded
- Guru Nānak
- Rāmānanda
- Kabīr
- Mīrābāī
- Rāidāsa
- Vallabhāchārya
- Sūrdās
- Tulsīdās

Eastern Region

- Jayadeva
- Chaitanya Mahāprabhu
- Krittibāsa
- Mādhava Kandalī
- Sankardev

Southern Region

- Basavannā
- Mādhavāchārya
- Annamāchārya
- Pothanā
- Poonthanam Namboothiri

Western Region

- Ekanātha
- Jñānadeva
- Muktābāī
- Nāmadeva
- Janābāī
- Chokhamelā
- Narasimha Mehtā
- Bhalana
- Pīpājī
- Tukārām

Fig. 3.19. Some prominent Bhakti Saints from 12th–17th centuries

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LET'S MAP

On a map of India, locate and label the present-day states associated with the saints mentioned in the above infographic. Also, find and mark a saint from your own state who is not on the list. Are you familiar with any songs, poems, or verses composed by these saints? Share with the class and explain what it means or what you like about it.

THINK ABOUT IT

Vakh (_vachan_ or saying) of Lāl Ded

To learn the scriptures is easy, to live them, hard.
The search for the Real is no simple matter.
Deep in my looking, the last words vanished.
Joyous and silent, the waking that met me there.

Fig. 3.20. Translation of a vakh composed by Lāl Ded

Lāl Ded says that it is 'easy' to learn scriptures but 'hard' to live them. What is the difference between knowing a rule and actually following it? Can you give an example from your daily life?

During this period, many devotees from diverse backgrounds contributed to the Bhakti traditions. Syed Ibrāhim Khān, popularly known as Raskhān, from present-day Uttar Pradesh, wrote in Braj and is known for his compositions like *Prem Vāṭikā* and *Sujān Raskhān*. Abdul Rahīm Khān-i-Khāna, popularly known as Rahīm, a Mughal military commander, wrote poetry in Braj to express his devotion to Kṛiṣhṇa. Bibi Lachimar (of Turkish origin) devoted herself to the Bhakti of Viṣhṇu.

DON'T MISS OUT

Worship of goddesses was a long-standing tradition in India, and during the period under study, kings continued to invoke goddesses such as Kālī, Durgā, Bhavānī, and other forms of Shakti as protectors in times of conflict. Annamarājā, ruler of Bastar kingdom, is believed to have received his sword from Mānikyadevī (Danteśhvarī), while Kumāra Kampana was believed to have used a goddess-given sword against the Sultanate forces at Madurai. Similarly, Śhivājī is also believed to have received a sword from the goddess Tuljā Bhavānī to resist the Mughals.

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Fig. 3.21. A Mughal painting depicting Akbar worshipping the Sun

Sulh-i-Kul

During the later part of Akbar's reign, the philosophy of *Sulh-i-kul* was adopted as a state policy. It represented 'peace with all', 'universal toleration', or 'absolute peace'. This policy had political implications, as it appears to have been intended to secure the support of political powers other than Muslims. Over time, Akbar also began to distance himself from certain orthodox elements and adopted practices that included reverence for the Sun. The Mughal state adopted a relatively inclusive approach during the later part of his reign.

Within the Indian religious, spiritual, and philosophical landscape, the Bhakti tradition emerged as a transformative force. It spread across the Indian subcontinent, taking distinct forms in different regions and social contexts. It played a crucial role in shaping and preserving India's spiritual landscape and social fabric.

DON'T MISS OUT

Sufism began to expand in the subcontinent with the arrival of the Turks. It was a mystical dimension of Islam that generally adopted a comparatively liberal outlook towards the world. It emerged in West Asia as a form of spiritual response to the worldly orientation of some *Ulema* (Islamic theologians and jurists) and the orthodoxy of *Shariat* (Muslim jurisprudence and law). Sufism spread through various orders, or *silsilahs*. During the Delhi Sultanate, the Chishti and Suhrawardi orders became prominent, while under the Mughals, the Qaddiriyya and Naqshbandiya orders gained importance. Sufi *khanqahs* (hospices) and shrines developed as centres of popular religious life and also played a role in social and political affairs. Sufis also contributed to the spread of Islam. In India, Sufis adopted many local practices to connect with the wider population, including using Indian languages such as Hindavi and Bengali. Some Sufis also incorporated yogic practices, such as *prāṇāyāma*, known in Persian as *habs-i-dam*.

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Temples: The Resilience of Sacred Sites

Temples functioned as central pillars of the region's socio-economic and political life. These religious institutions evolved into major economic hubs that managed extensive land grants, provided employment to hundreds of specialised workers, and served as financial institutions for local traders. Rulers used temple patronage to strengthen their political legitimacy. Beyond their spiritual and religious functions, temples facilitated social integration and served as centres of education and culture, where artistic traditions such as Bharatanatyam and many other art forms were preserved. They also exercised quasi-judicial authority, collecting local taxes and administering legal penalties.

Temples also served as sources of legitimacy for political authority. The deities of these temples were regarded as protectors of the state and its people. In some cases, the deity was regarded as the ruler, while the actual ruler governed as the deity's deputy. For example, the Gajapati rulers of Odisha called themselves the *Rauta* (deputy) of Jagannath. Similarly, the Kachhwāhā rulers of Jaipur governed on behalf of Govindadevajī, while the Rajput rulers of Kota regarded themselves as representatives of Rāma.

Temples in India also functioned as significant economic institutions. They provided employment to a wide range of groups, including architects, artisans, sculptors, priests, musicians, and administrative staff. As substantial landholders, temples received donations, endowments, and land grants, often given as acts of

Fig. 3.22. Sri Ranganathaswamy Temple, Srirangam

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Fig. 3.23. Brihadeeswara Temple, Thanjavur

religious merit. The Brihadeeswara (Rajarajeshvara) temple at Thanjavur exemplifies this, having received extensive donations of gold, ornaments, and vessels. This wealth enabled temples to support irrigation works and agrarian development, reinforcing their role in economy and the society. Temples also extended credit to cultivators, village assemblies, and merchant groups. This convergence of economic and religious activity contributed to the emergence of temple towns such as Thanjavur, Kanchipuram, and Puri, as prominent urban centres.

Between the 7th and 13th centuries, temple construction proliferated across the subcontinent under the patronage of various dynasties. However, from the 11th century onwards, we begin to see many instances of temple destruction, particularly during the Ghaznavid and Ghurid invasions. From the 12th century onwards, many temples and *maṭhs* were not only plundered or destroyed, but mosques were also sometimes constructed on the same sites.

While there were instances of the destruction of sacred sites, local rulers, merchant guilds, and religious communities actively maintained and restored temples during this period, allowing sacred centres to continue functioning. Local rulers who negotiated alliances with the central authority were generally allowed to build major structures. A royal temple built by Vīra Someśhvara at Kannanur, an earlier Hoysāla regional capital, was converted into a

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Fig. 3.24. Kashi Vishwanath Temple, Varanasi

mosque under the Sultanate rule before being restored as a temple by Kumāra Kampana. Further examples of temple construction and restoration include the reconstruction of the Kashi Vishwanath Temple by Toḍar Mal, revenue officer of Akbar, and Rājā Mān Singh, chief of the Rajput Kachhwāha family (the temple was later demolished during the reign of Aurangzeb. The present temple was built by Ahilyābāī Holkar adjacent to the original site in 1780). The latter also patronised the construction of the Govind Devji Temple at Vrindavan, one of the largest temples built in northern India during this period. The Bundela king, Vīr Singh, patronised the construction of the new Keshav Dev Temple at Mathura.

From the 16th century onwards, a shift can also be seen in the design and location of Indian temples. Temples built during this period were often situated in less visible locations, along narrow lanes, with entrances that did not open onto main streets. Many lacked the conventional appearance of temples, and instead, resembled ordinary houses. Detailed exteriors became less common, and tall *shikharas* were frequently avoided. This pattern is captured in the term 'havelī temple', used by some historians to describe temples constructed in the style of *havelī* residences.

It reflects how communities adapted temple architecture to avoid destruction. Such examples can be found in Delhi, Varanasi, and Rajasthan, including the Jagat Shiromani Temple.

The Jagat Shiromani Temple, also known as Mīrābāī Temple, was built by Queen Kanakwati, wife of Mān Singh I, in Amer. The temple is dedicated to Kṛiṣhṇa and Mīrābāī.

Fig. 3.25. Jagat Shiromani Temple, Amer

DON'T MISS OUT

Jagat Singh challenged the Mughal authorities through his architectural projects. In addition to fortifying Chittor, he built a large Jagdish temple near his palace in Udaipur. Scholars suggest that the temple was a religious and political response to the Jama Masjid constructed by Shah Jahan in Delhi.

Fig. 3.26. Jama Masjid, Delhi

Fig. 3.27. Jagdish temple, Udaipur

Indian Economy: Agriculture, Trading Networks, and Merchant Communities

The expansion of the agrarian economy and the gradual proliferation of centres of artisanal and craft production were significant developments of the early second millennium CE.

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The main source of state income was land revenue, and the expanding frontiers of empires resulted in an increase in the area under cultivation. Over time, the collection of land revenue became increasingly strict and exploitative at various levels of administration. For example, during the reigns of Alauddin Khiljī and Mohammad bin Tughlaq, the land tax was raised to as much as 50 per cent of total agricultural production, in addition to other taxes. Moreover, the Sultans of Delhi, and later the Mughals had limited role in maritime trade. Their concerns regarding coastal areas and port cities were largely confined to political and military matters and, to some extent, the collection of customs duties. As a result, coastal areas and their commerce remained largely in the hands of regional power holders and local communities.

Irrigation technologies such as rope pulleys, spinning wheel, and Persian wheel were widely used by local populations, resulting in enhanced agricultural and economic output. In the Vijayanagara Empire, agricultural production varied greatly in both form and scale. Consequently, irrigation systems ranged from extensive canal networks to small rain-fed patches of land. A defining feature of the Vijayanagara economy was its investment in small-scale irrigation.

DON'T MISS OUT

During this era *hunḍī* was widely used as a financial instrument and can be compared in some respects to modern bank cheques. It functioned as a bill of exchange, facilitating large-scale commercial transactions without the need to carry large amounts of currency. It was also a trusted means of secured long-distance remittance. *Hunḍīs* were operated through networks of money-changers known as *sarrāfs* and their agents, called *gumasthas*.

Fig. 3.28. A hunḍī stamped by the Bank of Bengal, Agra, with a denomination of one rupee and eight annas

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THINK ABOUT IT

What do you think made *hunḍī* a safer and more convenient choice than physically transporting money across long distances?

Fig. 3.29. A painting of a sarrāf or money changer

In addition to agricultural expansion, the period witnessed significant growth in trade and commerce. India's inland regions were connected to trade routes in Central Asia, facilitating a substantial flow of goods and artisanal knowledge, much of it carried by large groups of traders known as caravans. Inland trade networks expanded alongside the growth of urban centres. Local trading communities such as the *banjārās*, along with revenue-collecting officials, played a crucial role in connecting rural regions with urban centres.

From the 13th century onwards, seaborne trade grew significantly, with trading networks linking western India to the Persian Gulf and the Arabian Sea. Gradually, India became a major centre of the Indian Ocean trading network, with the eastern Mediterranean and China forming its vital axes until the arrival of Europeans in the subcontinent.

DON'T MISS OUT

Kṛiṣhṇadevarāya, the Vijayanagara ruler, greatly emphasised in his *Amukthamālyadā* the importance of oceanic trade, maintenance of ports, and fair treatment of foreign traders. He writes,

> *Manage your ports well, and let commerce increase in horses, elephants, and gems, pearls, and sandal-paste. When drought, sickness, and calamity make foreigners seek refuge in your lands, shelter them in keeping with their station. Give out gardens, yards, and mines to those whom you wish to favour...*

Fig. 3.30. A translated excerpt from the Amukthamālyadā

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The World of the Indian Ocean: Continuity and Change

With the turn of the 16th century, two major developments occurred—the arrival of Europeans for trading ventures and the Mughal conquest of northern India. The Europeans, particularly the Portuguese, and the Mughals entered an already established economic system in which local institutions and networks played a crucial role. The Portuguese attempted to establish greater control over oceanic trade and commerce, while the Mughals sought to centralise control over the land revenue system. European involvement, which began with the Portuguese and was later followed by the Dutch, English, Danes, and French, developed into a complex system of interactions between European trading companies and Indian merchant networks. Subsequently, the Indian subcontinent became a major global intersection of commerce and cross-cultural exchange.

Simultaneously, new merchant groups emerged in coastal and hinterland areas. They controlled wealth, production system, supply chains, and at the same time, funded states and supported their political allies. The peasantry was involved in the production of raw materials, while weavers, dyers, bleachers, and other groups were working regularly to ready the deliveries of cotton textiles. Before the 15th century, the Indian Ocean trading network had a huge presence of Arab, Chinese, and Indian merchants. In subsequent centuries, Gujarati merchants in particular emerged as significant

Fig. 3.31. A painting of a woman weaver

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beneficiaries of the changing commercial order. Alongside them, numerous merchant groups flourished in Malabar, Coromandel, and Bengal, adapting to the growing influence and competition of the British, Dutch, and French.

These merchant groups were not merely involved in the buying and selling of goods, but were closely connected to the political economy, facilitating and financing trade. Indian merchants functioned as traders, shippers, bankers, money-changers, brokers, *dubhaśhīs* (interpreters and translators), diplomats, and bureaucrats.

Mercantile activities were not limited to a few, select communities. For instance, *vellalār*, who were traditionally agriculturists and pastoralists, participated in the Indian Ocean trading network in various capacities. From the 16th century onwards, many peasant groups began to participate actively in commerce. Some of them even learnt new languages, including Portuguese, French, English, and Persian, to facilitate their commercial activities.

THINK ABOUT IT

Identify some cultural and economic pillars that exist in any community (places of faith, traditions, business, and trade, etc.). How do you think these contribute to resilience within a society facing invasions/external pressures?

Between 1000 and 1700 CE, despite frequent invasions, shifts in political authority, and unrest, the Indian economy remained dynamic. The economy was driven by adaptations of agricultural innovations and irrigation systems, markets, powerful merchant groups, advanced financial instruments like the *hunḍī*, and India's pivotal position in transcontinental Indian Ocean trade. This economic strength contributed to resilience and stability, giving people the resources they needed to resist politically and culturally. Thus, this period is best understood as a complex interplay of conflict and cooperation, defined by the deep interconnection between resistance and resilience. The Rajputs, Ahoms, Vijayanagara rulers, Marathas, Sikhs, and Jats, including many others, consistently defended their lands and identities. To safeguard their sovereignty, they used a strategic mix of direct military resistance, highly adaptive guerrilla warfare, knowledge of local terrains, diplomacy, and alliances.

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The resilience of Indian civilisation extended far beyond the battlefield to preserve its socio-cultural continuity.

Before we move on...

- Resilience is the ability of individuals, communities, and societies to endure difficulties, adapt to changing circumstances, and sustain their way of life despite challenges.
- Cultural traditions, shared beliefs, community support, knowledge, and economic resources can strengthen resilience.
- Resistance means actively opposing or challenging a force, authority, or threat, in order to protect political power, territory, interests, or cultural identity.
- Resistance may take the form of warfare, alliances, diplomacy, negotiation, or other political actions.
- The Bhakti tradition demonstrated resilience by adapting devotional practices, using local languages, encouraging community participation, and preserving traditions even when circumstances changed.
- Economic strength, through agriculture, irrigation, markets, merchant networks, financial practices, and trade, also contributed to resilience by providing resources that helped communities withstand periods of instability.
- The period from 1000 to 1700 CE was not simply a story of invasions and battles. It was also a story of how people resisted threats, adapted to changing circumstances, preserved their cultural identities, and built new forms of cooperation and exchange.

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Questions and activities

1. Choose the most appropriate answer.

   Resistance and resilience are closely connected because:

   a. Resistance always involves warfare.

   b. Resilience enables communities to endure difficulties and sustain resistance.

c. Resistance is possible only when a kingdom has a large army.

d. Resilient societies avoid all forms of conflict.

2. Why and how did groups such as the Sikhs, Marathas, and Rajputs assert their political power and identity during periods of changing political authority?

3. Why did some Rajput rulers decide to negotiate alliances with the Mughal power?

4. Imagine a market in a port town of India such as Surat, Masulipatnam, or Puducherry. You are a foreign trader from Italy or France and want to make a deal of cotton textiles. How would you secure the product in the local market? Whose help would you seek and what kind of challenges would you face in terms of language, quality of product, prices of commodities, etc.?

5. Watch any video at the given link (*https:// www.indianculture.gov.in/forts-of-india/tales-of-forts*) and analyse the various functions of forts and fortresses. How did rulers perceive the importance of forts in terms of strategic concerns.

6. What architectural changes occurred in temple construction during the period under study? How do these changes reflect upon the resilient nature of Indian civilisation?

7. A kingdom has limited military resources but possesses fertile agricultural land, strong local support, and detailed knowledge of its terrain.

   Explain two ways in which these resources could contribute to the kingdom's resilience when faced with a threat.

8. Consider the following statement:

   "A tradition becomes more resilient when people can carry it with them."

   Do you agree? Explain using examples from the Bhakti tradition.

9. Voices from the Past

   Choose any two of the following and write a short first-person account explaining what resilience means from that person's perspective:

   - A Bhakti saint
   - A Rajput soldier

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- An Ahom commander
- A Maratha administrator
- A merchant
- A Jat leader

10. Design a Fort

You are a military architect advising a ruler.

Draw a simple, labelled plan of a fort that includes the following:

- Residential complex
- Defensive walls
- Water source
- Food storage
- Escape route
- Watchtower
- Administrative area

Now, write a few sentences explaining how your design would help the kingdom withstand a long siege.

Notes

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Chapter 4

Governance and Public Policy

The Big Questions

1. *What is governance, and how have its meaning and practice evolved over time?*
2. *What is public policy? How does it affect our day-to-day lives, and what makes it 'public'?*
3. *What are the various stages of policy-making and policy implementation?*
4. *Who are the major contributors to policy-making and implementation?*
5. *What role do citizens play in shaping public policy and strengthening good governance?*

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In the chapters 'Democracy' and 'Elections', we learnt that democracy is a system of self-governance that involves institutions, rules, and practices that enable people to participate in decision-making. However, in a well-functioning democracy, it is not enough to participate only in elections to form the government. A robust democracy requires transparent institutions, responsible representatives, and the continuous participation of people in public life.

Governance

Governance in a democratic system is based on transparency, accountability, participation, and the rule of law. Democratic governance places citizens at the centre of the political process. It is more people-centric and welfare-oriented.

LET'S EXPLORE

Let us see how a government and citizens come together to address various challenges.

Imagine that your town experiences several days of unusually heavy rainfall. Roads become flooded, schools remain closed,

electricity is disrupted in some areas, public health concerns rise, and many people find it difficult to reach schools, workplaces, or even hospitals. During such situations, organisations and individuals come together to help all in need.

The municipal corporation clears drains and removes garbage. The Public Works Department repairs damaged roads. The police manage traffic. School authorities inform parents that schools will remain closed. Health workers prevent the spread of diseases. Local representatives raise people's concerns, while residents volunteer to help vulnerable neighbours and report on flooded areas. The situation improves with collective efforts of all.

Form groups and discuss the following.

- → Who are the different people and organisations involved in dealing with this situation?
- → Why cannot a single department solve the problem on its own?
- → What would happen if these departments did not communicate or coordinate with one another?
- → What role can citizens play before, during, and after the heavy rainfall?
- → What makes the collective response effective?

Fig. 4.1. Collective response to resolve the crisis

Share your ideas with the class. Make a list of the qualities that help the town respond effectively. These qualities are essential whenever governments and society work together to solve public problems.

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Solving public problems requires much more than a government issuing orders. Different institutions and citizens need to work together, make decisions, implement them effectively, and remain accountable. This process is known as **governance.**

Understanding Government and Governance

Government	Governance
Refers to the institutions that exercise political authority	Refers to the process of decision-making and implementation
Includes the Legislature, the Executive, and the Judiciary	Includes government, citizens, media, civil society, and private organisations
Focuses on who governs	Focuses on how governing is carried out

The terms 'government' and 'governance' are closely related. However, they are not synonymous. Government refers to the institutions and authorities that exercise political power, while governance refers to the process by which decisions are made, implemented, and monitored in the public interest. The government formulates and implements people-centric policies, and good governance ensures that these policies are transparent, accountable, effective, and responsive to people's needs. Governance is the system through which decisions are made, implemented, and monitored for the welfare of the people. It refers to the way a country, state, city, village, school, or organisation is managed. Therefore, understanding governance is essential to know how government impacts our lives.

LET'S EXPLORE

Government refers to the institution that has the authority to provide public services. For example, the municipal corporation or panchayat is responsible for installing street lights. Governance is the process by which this street lighting is planned, implemented,

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and maintained, involving people in identifying dark spots, reporting non-functional lights, and providing feedback.

It ensures the continuous delivery of public services—in this case, well-lit streets and greater public safety.

Can you identify more such examples from your day-to-day life to understand the difference between government and governance?

Evolution of Governance in India

In the previous section, we studied that while the government is the institutional aspect, governance refers to its functional aspect. The government's institutional setup influences the nature of governance, and governance initiatives, in turn, require institutional adaptations to implement policies and schemes.

We are familiar with the structure of government at different levels in India. But the same structure has not always existed as it does now; it has evolved from certain historical experiences. Let us understand this historical context by looking at the 'big picture' of the world history.

Do you remember the phenomenon of colonialism that you learnt about in Grade 8? We know that European powers colonised many societies, particularly those in Asia, Africa, and the Americas. Before colonial rulers forced their own models of governance and policy-making on colonised people, these societies governed themselves in accordance with local and organic systems. Although most colonies had become independent by the end of the 20th century, colonial institutions largely persisted, and their structures continued to dominate even in contemporary times. India is not an exception to the historical legacy of colonialism. However, India stands out in two important ways. Firstly, unlike many societies that experienced colonial rule, India has preserved most of its knowledge of its pre-colonial institutions and traditions of governance through texts and practices. Secondly, after Independence in 1947, it was not possible to replace all colonial institutions at once. The leaders of independent India therefore framed the Constitution in the spirit of India's own values and ideals.

Since Independence, Indian institutions have continued to draw upon traditional ideas and values of governance and have adapted them to the needs of a modern democracy. For example, the British 'Order of the Star of India' used the phrase, 'Heaven's Light, Our Guide' as its motto but after Independence, we have adopted '*Satyamev Jayate*'

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(Truth alone triumphs), from the *Muṇḍaka Upaniṣhad* as our national motto. This reflects the continuing importance of Indian ethical values in public life. The motto of the Supreme Court of India was chosen to be '*Yato Dharmastato Jayaḥ*'. This phrase is taken from the *Mahābhārata,* and it means 'Where there is *Dharma* (righteous duty), there is victory'. These changes are not just symbolic or superficial. They signify the core values as well as the ideals of our institutions, which have been bestowed along with the responsibility of ensuring good governance.

LET'S EXPLORE

The *Sengol*—a cultural marker of rulership and good governance—was placed in India's newly inaugurated Parliament building in the year 2023 by Prime Minister Narendra Modi. The tradition of using the *Sengol* as a mark of righteous governance began more than a thousand years ago during the reign of Chola dynasty. To mark the occasion of India's Independence, at the stroke of midnight on 15th August 1947, the first Prime Minister, Jawaharlal Nehru, received the *Sengol* from the Adheenam Saints (an old Shaivite tradition) of Tamil Nadu. Try to find out more about the journey of the *Sengol* through its historic pictures.

Fig. 4.2. Sengol

Mahatma Gandhi visualised an ideal model of governance as *Rām Rājya. Rām Rājya* is rooted in the principles of probity, transparency, compassion, welfare, and people-centric orientation in governance and public policy. Gandhi's views on *Rām Rājya* reflect the collective imagination of Indians, from their distant past to the present day, that *Rām Rājya* is one of the best examples of good governance and a good collective life, as enshrined in the *Rāmāyaṇa* and the sixteenth-century text of *Rāmcharitamānas.*

From the Vedic period to the *Rāmāyaṇa* and the *Mahābhārata,* Indian tradition has consistently viewed governance as a sacred responsibility rather than merely a means of exercising power. *Rājadharma,* the foremost duty of an ideal king, was to protect the people by upholding *Dharma* (righteous duty) and promote the welfare of society. Good governance was measured not only by the ruler's authority but by the peace, prosperity, justice, and well-being in society. Kauṭilya's *Arthaśhāstra* says the following:

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प्रजासुखे सुखं राज्ञः प्रजानां च हिते हितम्

Prajāsukhe sukhaṁ rājñaḥ prajānāṁ cha hite hitam

(In the happiness of the people lies the happiness of the ruler, and in the interest of the people lies the interest of the ruler.)

LET'S RECALL

You have read in the first part of this textbook that the *Ṛigveda* refers to the *Sabhā* and *Samiti* as important institutions where matters of governance and public welfare were discussed.

Good Governance

Governance affects many aspects of our daily life. It helps maintain law and order so that people can live safely and peacefully. It protects the rights and freedom of citizens, which are essential in a democratic society. It also helps people access important public services, such as education, healthcare, transport, and so on.

Good governance promotes justice and fairness and ensures that people are treated equally before the law. It also helps build trust between citizens and the government.

Governance is the government's responsibility, but good governance depends on how different sections of society and government perform their responsibilities and work with one another. For example, the government makes laws and policies and provides public services. Citizens elect their representatives, follow the law, and participate in public life. The media helps people understand public issues and brings important matters to public attention. Civil society organisations work with communities and raise awareness about social concerns. The private sector contributes through economic activity and innovation.

Therefore, features of good governance include responsiveness, transparency, participation, the rule of law, accountability, effectiveness, and inclusiveness.

When people and institutions perform their responsibilities honestly and work together, governance becomes more effective and public welfare is achieved.

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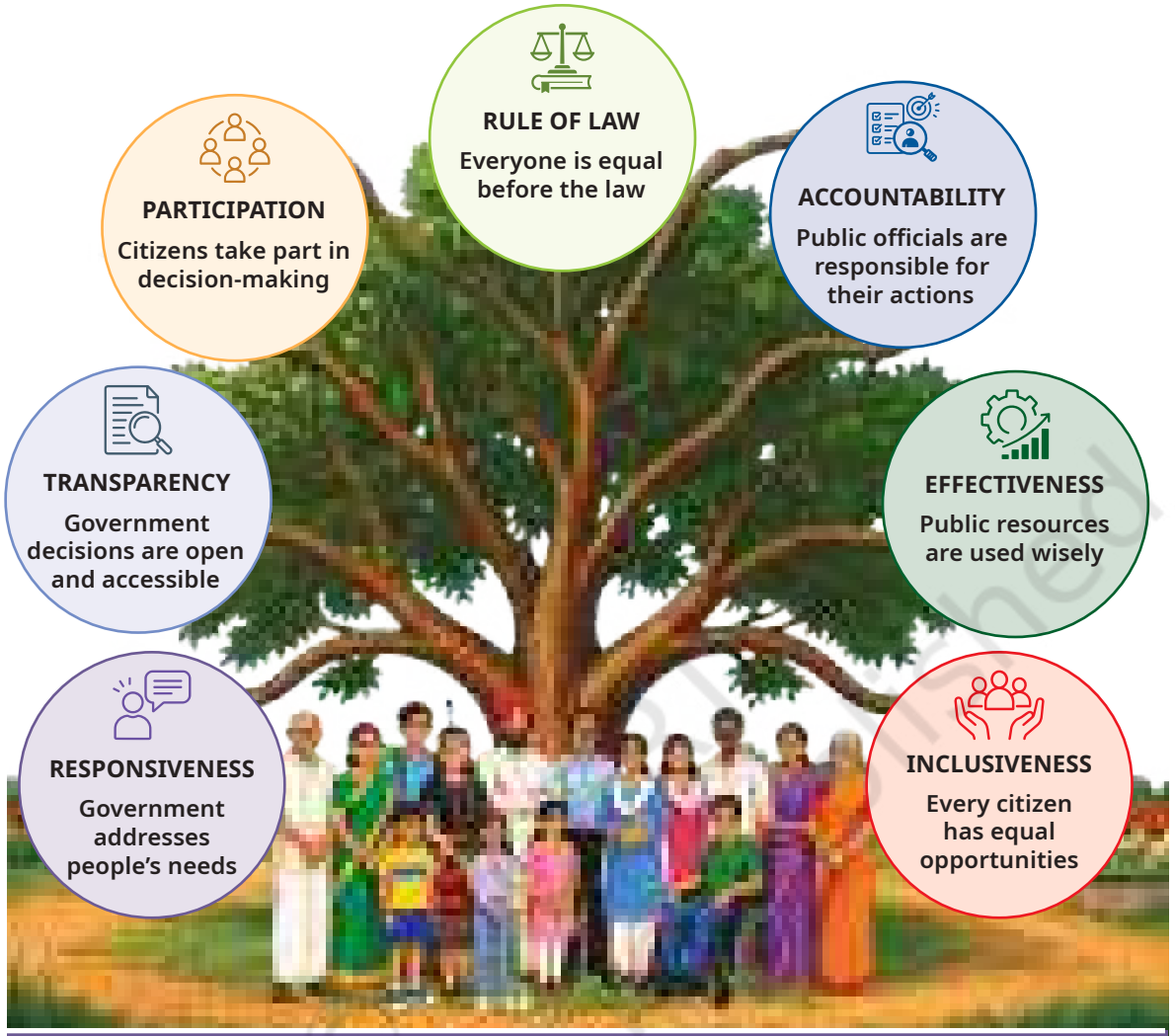

Fig. 4.3. Features of good governance

LET'S EXPLORE

Select two cities (one should be from a region other than yours) in India that have received awards for being the cleanest city. Make a visual presentation (chart paper or slideshow) to explain the 'good governance model' used in those cities by looking into the role played by government agencies (local, state, and national), residents, market associations, civil society organisations, etc. Also look into the steps they are taking to continue being clean cities.

LET'S EXPLORE

Form pairs or groups and assign one State or Union Territory to each group. Prepare a short classroom discussion paper on:

1. Good governance initiatives being implemented.
2. Suggestions for further improving governance.

Challenges to Good Governance

The task of governance is always ongoing. With time, the needs, desires, and aspirations of people also evolve, creating new challenges to which the government and the institutions have to respond. For example, weak coordination between different institutions and departments can affect the implementation of government programmes. At the same time, unequal access to public services can leave some groups or communities behind, particularly those who face social, economic, geographical, or other barriers. Malpractices and misuse of public offices can prevent resources and benefits from reaching the people. Misinformation and fake news also pose a serious challenge to good governance. Addressing these challenges requires effective institutions, responsible officials, active citizens, and a continued commitment to the principles of good governance. The initiative of e-governance has contributed significantly towards good governance. Let us understand details about e-governance in the following section.

LET'S EXPLORE

Did you know that 25th December is celebrated as National Good Governance Day? Can you find out why this date was chosen? How is National Good Governance Day celebrated in your district? Discuss in your class.

E-Governance: Technology for Better Governance

Today, digital technology has changed the way the government interacts with people. **E-governance** refers to the use of information and communication technology by governments to provide public services, share information, communicate with citizens, receive feedback, and improve administrative processes. It can make

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governance more transparent, accessible, and convenient and quick for the citizens. E-governance substantially reduces paperwork and allows people to access services without visiting government offices.

LET'S EXPLORE

Below is a chart of a few e-governance initiatives. Complete the chart and also add two such initiatives undertaken by your state government.

Initiative	Objective
UMANG	
Aadhaar	
GeM (Government e-Marketplace)	
DigiLocker	
CPGRAMS	
APAAR	

E-governance is particularly valuable in a large country like India, as it enables governments to deliver public services more efficiently, transparently, conveniently, and quickly to citizens across vast geographical areas.

However, e-governance is only effective when people have access to digital devices, reliable internet, and the skills needed to use them. Therefore, while technology can strengthen good governance, the government must also ensure that digital services remain inclusive and accessible to everyone.

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LET'S EXPLORE

Let us try to understand the quality of life of citizens before and after e-governance. Form five groups and interview your seniors, teachers, security guards, administrative staff, and parents regarding the issues listed below. Did they have access to the internet and a smartphone or computer to make use of these e-governance facilities? Find out what problems they had faced before e-governance and whether they felt the same issues were easily handled after e-governance.

1. Obtaining a birth certificate
2. Obtaining a admit card
3. Collecting and storing marksheets
4. Receiving salary
5. Booking rail/flight tickets
6. Registering a vehicle

So far, we have understood that governance is the process through which decisions affecting people are made, implemented, and monitored with governments, institutions, both public and private, and citizens playing an active role in the process. But governance ultimately needs concrete instruments to address public problems and meet people's needs. Public policy provides that link; it translates governance into planned action by turning public concerns and government priorities into programmes, decisions, and solutions.

Public Policy

The Meaning of Public Policy

Every day, we experience the impact of public policies, often without even noticing it. The roads we travel on, the traffic signals that help us cross safely, the clean drinking water supplied to our homes, the electricity that lights our classrooms, the public parks where we play, and the waste collected from our neighbourhood are all examples of public services made possible through public policies. These policies help make our daily lives safer, healthier, and more convenient.

Suppose your school notices that many students bring single-use plastic bottles and wrappers in their lunchboxes and schoolbags

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every day. As a result, a large amount of waste is generated, and the campus becomes dirty, leading to environmental degradation in the long run. To address this problem, the principal discusses the issue with teachers, students, and parents. Together, they decide to introduce a 'Plastic-Free Campus Policy'. Under this policy, students are encouraged to carry reusable water bottles; plastic bags are discouraged; dustbins for waste segregation are placed around the campus; monthly awareness programmes are organised; and teachers are assigned with the responsibility of monitoring the campaign. These decisions form a policy because they provide a planned course of action to solve a problem and achieve a sustainable goal.

Similarly, governments make public policies to address problems that affect society as a whole. Public policy is a result of a course of action adopted by the government to solve public problems, meet people's needs, and promote the welfare of society. Through this course of action, governments decide what should be done, why it should be done, who will do it, how it will be implemented, and how its success will be measured. Further, public policy is monitored and evaluated through a feedback loop, and necessary changes are made to the policy if required.

What is 'Public' about Public Policy?

The word 'public' refers to all the people living in a society or community. A policy becomes public policy when it is made for the public benefit, implemented by public institutions, and often financed with public money collected through taxes. Public policies address issues that concern society as a whole, such as education, healthcare, transportation, sanitation, environmental protection, and public safety.

Public policy is about responding to the needs and aspirations of the people. Therefore, people play an important role in shaping them. People identify issues, express their concerns, participate in consultations, provide suggestions, and share feedback after a policy is implemented. Governments use this feedback to improve policies over time.

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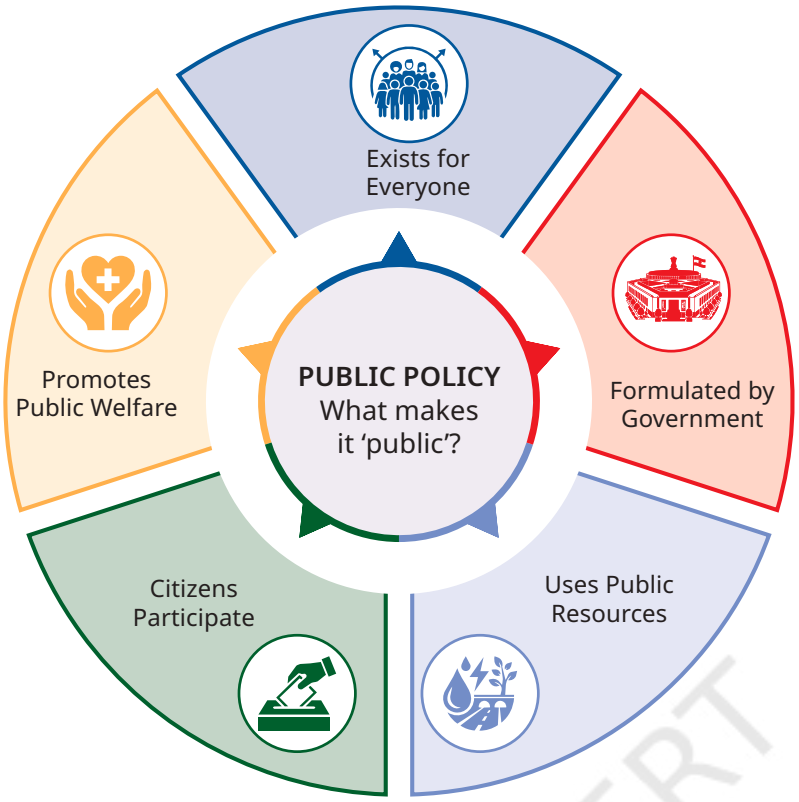

Fig. 4.4. Public policy: What makes it 'public'?

However, not every policy is a public policy. Schools, companies, hospitals, sports clubs, and even families often have their own policies to guide the way they function. For example, a school may have a policy on the use of mobile phones, a company may have a work-from-home policy, or a housing society may have rules for parking vehicles. These policies apply only to a particular organisation or a limited group of people. Public policies, on the other hand, are formulated by governments to address issues that affect society on a larger scale.

LET'S EXPLORE

Read each of the following examples carefully. Identify whether they are a **policy** or a **public policy**. Give a reason for your answer.

Example	Policy/ Public Policy	Why?
The government introduces a sports policy.		
The government launches a campaign to provide safe drinking water to every household.		

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Example	Policy/ Public Policy	Why?
A company allows employees to work from home twice a week.		
A municipality introduces waste segregation in the city.		
A housing society fixes visiting hours for guests.		
The government introduces free vaccination for children.		

Participants in the Public Policy Process

Have you ever wondered who decides whether new schools are to be built, roads are to be repaired, and taxes are to be decreased or increased? Is it only the government? No, it takes the combined efforts of many individuals, organisations, officials, and even ordinary citizens in identifying issues, providing solutions, and putting them into action.

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Table 4.1. Key Stakeholders in Policy-making and Implementation

Contributors	What do they do?	Example
Citizens	Identify problems, vote, participate, give suggestions, and provide feedback.	Residents request a zebra crossing near a school.
Elected representatives (MPs, MLAs, and local representatives)	Represent people, raise public issues, participate in law-making, and approve budgets.	An MLA raises the issue of drinking water shortage in the Assembly.
Government officials, departments, and ministries (Central, State and Local Governments)	Prepare and implement policies and programmes, coordinate departments, deliver public services, and monitor their progress.	A district collector coordinates the implementation of a government scheme in the district.
Civil society organisations, NGOs, experts, researchers, and the media	Study problems, suggest solutions, create awareness, monitor programmes, and provide feedback.	An NGO conducts a water-conservation campaign, and the media highlights pollution concerns.
Private sector	Facilitate technical know how, innovation, resources, and expertise, often working with the government.	A company installs solar panels in government schools through a public-private partnership.
International organisations	Bring in international experience, facilitates technical know how and financial support.	The World Health Organization (WHO) assists countries in controlling infectious diseases.

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LET'S EXPLORE

Your school's main gate faces a road that connects residential areas and a major market area. There is heavy traffic congestion on the road, especially during morning and afternoon hours. Pedal rickshaws, electric rickshaws, school buses, private vehicles, and pedestrians—all struggle to get their way.

Form groups in your class and choose a course of action from the options given below.

1. Form a group of volunteers to regulate traffic in rush hours.
2. Write an application to the principal about the issue.
3. Meet the principal, along with all the class representatives of the school, to discuss the gravity of the situation and then impress upon the principal the need to take up the issue with the local people's representative so that a traffic signal can be installed.
4. Request the school management to construct and open another gate on the other side of the school.
5. Suggest that the school shift its pick-up and drop-off point to a location slightly away from the main gate.
6. Change school timings so that the school's opening and closing hours do not coincide with the busiest traffic hours in the market.

Answer the following based on the information given above.

→ Which option did you choose and why?

→ Can you suggest a better solution to address the given problem? Compare your suggestion with other options.

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→ Which stakeholders would be involved in solving this problem in your suggested solution? Explain their role in solving the problem.

→ Suggest a feedback loop to ensure that the problem does not recur.

Case Study—A Ramp to Better Healthcare

Smt. Shanti Devi, a 72-year-old woman, visits the hospital for a health check-up. On arriving the hospital, she finds that she has to climb stairs to reach the Outpatient Department (OPD). There is no ramp or handrail to support her. At the same time, a man in a wheelchair arrives but is unable to enter the building because the entrance has only steps, with no wheelchair-accessible ramp. A visually impaired person may be able to climb the steps to enter the hospital but finds it difficult to locate different hospital departments because there are no tactile paths or Braille signs.

Many patients and their family members realise that the hospital infrastructure is not accessible to everyone. They decide to raise this issue with the hospital authorities and local representatives. A local disability rights organisation also highlights the problem through newspapers and social media. The district administration takes note of the issue and asks the Public Works Department (PWD) and the Health Department to inspect the hospital. After consulting doctors, hospital staff, architectural experts, organisations working with persons with disabilities (PwDs), senior citizens, and residents, a plan is prepared to make the hospital more accessible. The government approves the funds needed to construct ramps, lifts, handrails, tactile pathways, and accessible toilets, and to install clear signboards. Once the work is complete, the hospital becomes easier to use for older persons, persons with disabilities, pregnant women, patients, and children.

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This is an example of how a public policy moves from identifying a problem to implementation.

LET'S EXPLORE

Read the case study and follow the journey of the issue from identifying the problem to implementing a solution. As you read, identify the problem, the people and institutions involved, the actions taken, and the steps through which the solution was developed and implemented. Then, answer the questions below:

1. Arrange the various stages of the policy process in sequence for upgrading the hospital to be more patient-friendly.
2. Which stakeholders were involved in solving this problem? Identify at least three.
3. Can you think of any public building in your locality that could be made more accessible? What improvements would you suggest?

The Making of a Public Policy

A public policy is developed through a systematic process rather than a single decision. It usually begins when citizens, government officials, researchers, or civil society organisations realise that a new need has to be met, an existing problem has to be resolved, or a facility has to be made better in the interest of the people. Then the government studies the issue, collects information, and consults experts, stakeholders, and the public to understand different viewpoints and possible solutions. Based on these discussions, the government authorities prepare, review, and approve a draft policy. Once the policy is approved by the concerned authorities, it is translated into programmes and schemes, which provide a clear framework for different ministries, departments, and local governments to take specific action. The implementation of any policy is continuously monitored and evaluated to assess whether it is achieving its objectives. Feedback from citizens and other stakeholders is used to improve the policy over time. Thus, public policy is a continuous cycle of problem identification, consultation, policy formulation, implementation, monitoring, evaluation, and improvement. This process helps the government respond effectively to the changing needs of society.

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Fig. 4.5. Exemplar of a policy cycle

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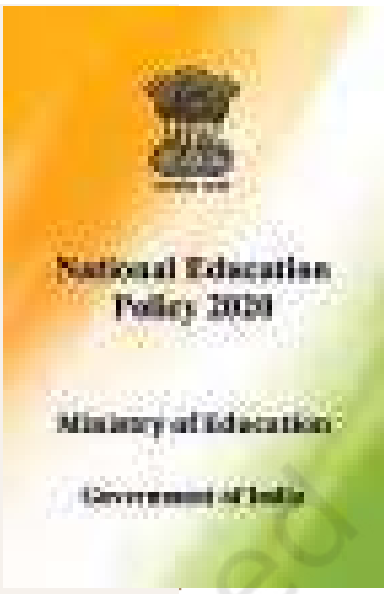
DON'T MISS OUT

National Education Policy (NEP) 2020: A Collaborative Journey

The National Education Policy (NEP) 2020 is an excellent example of how public policy is developed through consultation, expert advice, and public participation. India's previous National Policy on Education was introduced in 1986.

Over time, society changed rapidly. These changes created a need to rethink India's education system.



Recognising this need, the Government of India initiated the process of developing a new education policy in 2015. For drafting this policy, the government set up an expert committee in 2017, which was chaired by renowned scientist Dr K. Kasturirangan. The committee included experts from education, science, technology, social sciences, and public administration. A unique feature of NEP 2020 was the large-scale public consultation. The draft policy was made available for feedback by citizens and all stakeholders. During the formulation of the policy, 2.5 lakh Gram Panchayats, 6600 blocks, 6000 Urban Local Bodies, and 676 districts were consulted, and about 2 lakhs suggestions were received. The expert committee carefully examined the suggestions. Several recommendations from citizens and institutions were incorporated before the final policy was prepared. After discussions within the Government of India, the National Education Policy 2020 was approved by the Union Cabinet in July 2020.

Stakeholder Role-Play

Divide the class into groups. Each group represents one of these stakeholders:

- The Union Government, a state government, a school principal, a teacher, a parent, a student, an NGO, and a journalist
- Discuss how the stakeholder group you belong to can contribute to the successful implementation of the NEP. Present your ideas to the class. You can access the policy document at *https://static.pib.gov.in/WriteReadData/userfiles/NEP_Final_English_0.pdf*

THINK ABOUT IT

If any step in policy-making and implementation is skipped, what would be the implications? Discuss with a suitable example.

For a policy to be implemented effectively, it is always further organised into programmes and schemes. A policy is a broad goal

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or direction adopted by the government to address an important public issue. It tells us what the government wants to achieve and why. A programme explains the plan or approach for achieving that goal. A scheme is a specific action or initiative through which the programme is implemented so that its benefits reach the people. In simple terms, policy provides direction, programmes provide the plan, and schemes put the plan into action. Let us understand this through the example of the National Health Policy 2017, given in the following infographic.

Fig. 4.6. Difference between a policy, programme, and scheme

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Policy Coordination: Key to Good Governance

Public policymaking and implementation require multi-level planning and decision-making. This exercise becomes more complex and challenging in a country as vast and diverse as India.

Let us try to understand this with an example. The Government of India, recognising the transformative power of sports in nation-building and holistic development, launched the Khelo Bharat Niti in 2025 with the aim of transforming India's sports landscape. The government may plan a policy, but its success depends on many stakeholders—in this case, states, districts, local bodies, schools, coaches, families, and athletes. They all have different yet important roles to play. Families need to support a sports career for their children, especially their daughters; national or state governments need to encourage students, including specially abled students to take up sports; local authorities have the responsibility to maintain sports grounds or parks; scientists and innovators are expected to conduct research on better nutrition or equipment; and the Central Government has to visualise the importance of sports for holistic health as well as for international diplomacy.

Thus, the success of a policy, like Khelo Bharat Niti 2025, depends on the coordination between various

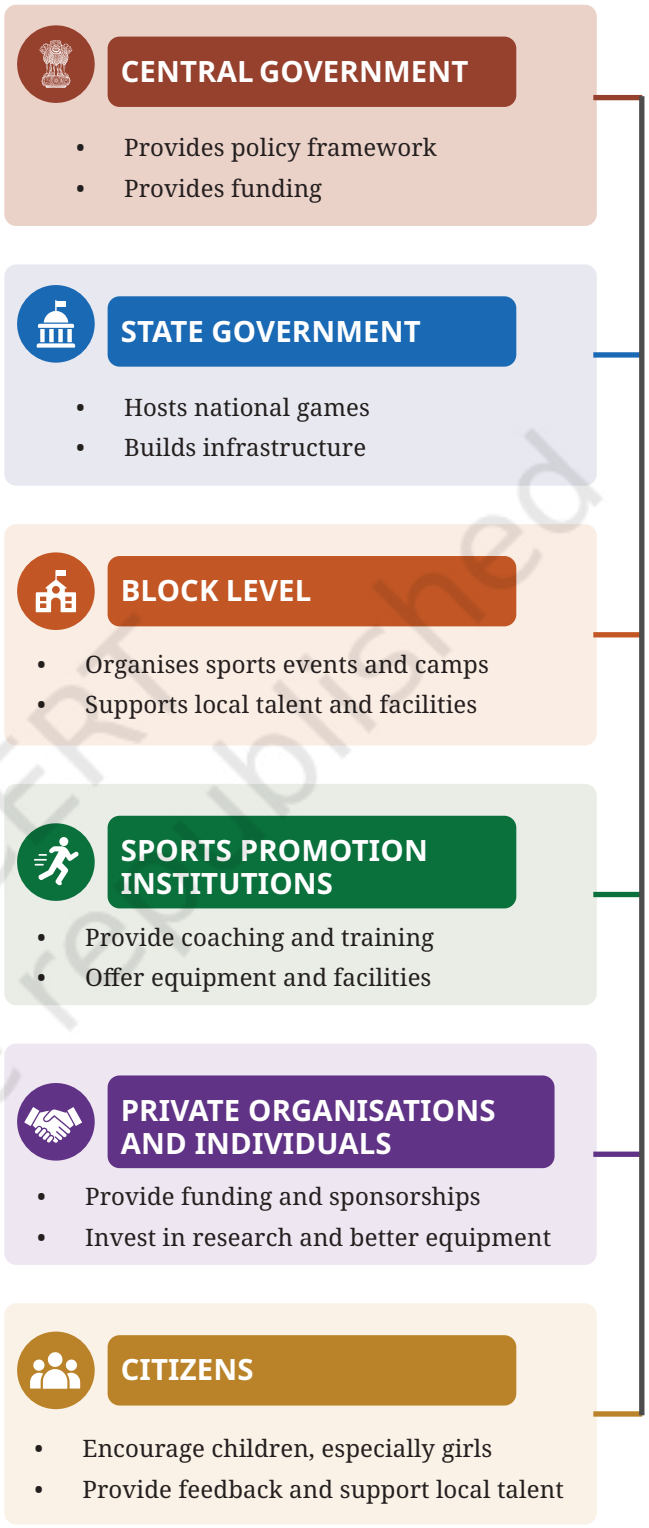

Fig. 4.7. Khelo Bharat Niti: A policy coordination exemplar

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stakeholders. The policy document of Khelo Bharat Niti says, "Strengthening the sporting culture and ecosystem from the block level to the national level is a key focus of the Khelo Bharat Niti 2025". At the national level, the role of the Union Government is primarily focused on providing the larger policy framework, including the policy objectives. Also, it is supposed to provide funding as well as identifying the sources of funding for its agencies located at lower vertical levels of states, districts and blocks. Khelo Bharat Niti mandates state governments to strengthen sporting ecosystems through initiatives such as 'Model Sports State'. State governments are expected to host major national games and tournaments and attract private investments through the public-private partnership (PPP) model to build sports infrastructure. Similarly, individual districts are encouraged to become 'Model Sports Districts' by enhancing their sports capabilities. Further at the level of blocks, sporting programmes are to be organised on a regular basis to attract mass participation and facilitate talent scouting.

Apart from the role of the government and its agencies, the Khelo Bharat Niti 2025 seeks an active role of private organisations and individual citizens as well. Private organisations can come forward to sponsor sportspersons and sports teams. They can invest in supporting scientific research programmes to develop better nutrition and equipment, and better medical facilities for sportspersons.

LET'S EXPLORE

1. As a student, what role can you play in the successful implementation of Khelo Bharat Niti 2025?
2. Your school is designated as the 'state's sports ambassador'. What actions should it take to fulfil its role as sports ambassador?
3. Which ministries of the government can contribute towards the success of Khelo Bharat Niti 2025? How?

Governance may sound like government, but it is more about how decisions are made, implemented, and connected to people's everyday lives. Public policy provides an important link between governance and public life. Through public policies, governments decide how resources should be used and how public services should be provided. Elected representatives, government officials, local institutions, civil society organisations, researchers, the media, the

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private sector, and people, all have different but connected roles in good governance. People, therefore, have an important contribution to make in the process of governance. By voting, following laws, raising concerns, participating in public affairs, and giving feedback on government services, they can help improve the way public institutions function. In a democratic country like India, governance is guided by the constitution, which lays down the framework for governing the nation.

Good governance is essential to strengthening democracy, promoting sustainable development, and ensuring national progress. It depends on transparent institutions, accountable leadership, active citizen participation, and a strong commitment to the rule of law. Public policy is integral to good governance. Good governance and public policy are crucial foundations for achieving national development goals. Good governance is, therefore, an ongoing effort that will help realise the vision of making India a developed, prosperous, and self-reliant nation.

Before we move on...

- Governance refers to the process of making, implementing, and monitoring decisions for the welfare of people, while government refers to the institutions that exercise political authority to govern.
- Good governance is based on principles such as participation, transparency, accountability, the rule of law, responsiveness, and efficiency, ensuring that public institutions serve the people effectively.
- India's understanding of governance has evolved from the ideas and values reflected in *Sabhā* and *Samiti*, *Rām Rājya*, and *Rājadharma* to the modern democratic system based on the Constitution and citizen participation.
- Public policy consists of decisions and actions taken by the government to solve public problems and promote the welfare of the people. Effective public policies are developed, implemented, monitored, and improved through a systematic policy process.
- Governance and public policy involve the combined efforts of many actors, including the government, institutions, civil society organisations, private sector, the media, and citizens, each performing complementary roles.

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- Policy coordination among different government departments, institutions, and stakeholders is essential for addressing public problems, responding to emergencies, and delivering public services efficiently.
- Active citizen participation through voting, providing feedback, obeying laws, community involvement, and holding public authorities accountable strengthens democracy and promotes good governance.

Questions and activities

1. "Good governance is the responsibility of the government alone." Do you agree with this statement? Give reasons to support your answer.
2. Do you think students can contribute to good governance? Write a short note describing any five ways in which students can become more responsible and active citizens.
3. Choose any one public policy [for example, Right to Education (RTE), Swachh Bharat Mission, PM POSHAN, Ayushman Bharat, Digital India, or Jal Jeevan Missio] and prepare a chart showing the following aspects:
   - The problem it addresses
   - The major stakeholders involved
   - How different actors coordinate
   - The role of citizens in its success

   Present your findings to the class.
4. Answer the following in one or two sentences.
   a. What is governance?
   b. How is governance different from government?
   c. What is meant by public policy?
   d. Name any four principles of good governance.
5. Compare the following and present your answer in a tabular form.
   a. Government and governance
   b. Policy and public policy

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6. Imagine you have been appointed to advise the district administration on improving governance. You have proposed six points (given below), but the administration decides to prioritise only three areas for improvement. Which three would you recommend as priority areas, and which one out of the three do you feel is the most important?

   a. Improve drinking water facilities in schools.

   b. Improve sanitation and public toilets in schools and market areas.

   c. Provide tricycles for orthopaedically challenged students.

   d. Install open-area swings and open-air gym in public parks.

   e. Ensure the vaccination of stray dogs to prevent rabies.

   f. Ensure distribution of iron and folic acid tablets for female students in schools.

7. Match the policy action point to its corresponding stage in the policy cycle.

S. no.	Policy action point	Stage of policy cycle
1.	Assessing the impact of the policy	Policy implementation
2.	Putting the policy into action	Policy evaluation
3.	Framing the legislation	Policy adoption
4.	Multi-dimensional consultation with stakeholders	Policy formulation
5.	Addressing a new need of the people	Policy desirability (problem identification)

8. Read the following case and answer the given questions.

   The government launches a programme to plant one lakh trees in a city. Funds are allocated, but only a few saplings survive because they are not watered or protected.

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a. Which stage of the policy cycle appears to be weak?

b. Which organisations should coordinate to improve the programme?

c. Suggest two measures that could improve the outcome.

9. Read each statement carefully. Do you agree or disagree? Give reasons for your answer.

a. A good public policy can fail if it is not implemented properly.

b. Citizen participation sometimes makes decision-making slower, but it can improve governance.

c. Technology alone cannot ensure good governance.

10. Design a public policy.

You are a member of your school's student representative body. Identify one issue, for example, plastic waste, water wastage, road safety, energy conservation, or littering in public places. Prepare a simple policy proposal that includes the following aspects of policymaking:

- The reason for selecting the issue
- Your proposed solution
- Who should implement it?
- How will students participate?
- How will you know whether the policy has succeeded?

Present your proposal to the class as a chart or slideshow presentation.

Notes

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Chapter 5

Managing Your Personal Finances

The Big Questions

1. *What is personal finance and why is it important for individuals?*
2. *What are the pillars of personal finance?*
3. *What are some common savings and investment options?*
4. *How can you protect your savings and assets?*
5. *What is income tax?*

Have you ever received pocket money or money from relatives during festivals or special occasions? Think about how you usually use that money.

- Do you spend it immediately or save some of it?
- Do you keep your savings in a piggy bank *(gullak)* or somewhere else?
- How do you manage expenses, such as travelling to school, buying stationery, or occasionally purchasing food from the school canteen?
- Have you ever planned your spending to save for something you wanted later, like visiting your grandparents during summer break?

Fig. 5.1. Savings in a piggy bank

Our journey into managing money begins in childhood. As we grow into adulthood, we earn income, and spend it in many ways to fulfil our needs and wants. We also save money and make investments to achieve goals, such as buying a house or funding higher education.

Individuals and households in today's world have to make many decisions involving money. Money is, therefore, an important resource that should be utilised efficiently. This is where an understanding of personal finance becomes important. Personal finance refers to the planning and management of an individual's money to meet their needs and wants. It involves making smart financial decisions, such as how much to spend, how much to save,

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where to invest, and how to prepare for future financial needs. For instance, if someone wants to buy a house in the next five years, in order to achieve this goal, they will have to plan and manage their money well.

Pillars of Personal Finance

Income
The money you receive (salary, business income, etc.).

Budgeting
Deciding how much to save, how much to spend, and how one spends that money on needs and wants.

Saving
Setting aside money for future needs or emergencies.

Investing
Deploying your savings in assets like stocks, mutual funds, etc.

Protection and risk management
Safeguarding your money and assets through insurance and debt management.

Fig. 5.2. Pillars of personal finance

Income (Earnings)

Income is the money a person receives regularly or occasionally in return for products sold or services rendered. It is the starting point of personal finance because without income, it is not possible to spend, save, or invest. For example:

- A teacher earns a monthly salary.
- A shopkeeper earns profits from a business.
- A person earns rent from letting out property.
- An investor earns interest from bank deposits.
- A shareholder receives dividends and interest from securities.

THINK ABOUT IT

- Have you ever thought about earning your own money? What kind of skill would you require and which work would you choose?
- There are many 'Earn While You Learn' programmes that allow college students to continue their education while earning an income through part-time work, internships, or apprenticeships. This not only provides financial support but also helps them gain practical skills, work experience, and a better understanding of the real-world workplace.

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Budgeting (Planning Expenses)

Budgeting means planning in advance how income will be spent. A **budget** is a written plan that aims to balance income and expenses. People often plan to save and make financial decisions with a specific purpose in mind to be achieved in future. For example, a family may decide how much money to save each month or spend monthly on food, rent, school fees, transport, etc. Other examples include:

- Saving for a school trip next month.
- Buying sports equipment after six months.
- Saving for higher education in the future.
- Setting aside money for emergencies.

Setting such financial goals helps people prioritise their needs and make careful decisions regarding spending, saving, and investing.

One simple way to set goals is to ask:

- What do I want to achieve?
- How much money will I need?
- By when do I want to achieve it?
- What steps can help me reach this goal?

Budgeting helps control unnecessary spending, such as excessive eating out or impulse shopping, and enables people to prioritise important expenses. It also helps people live within their means by balancing income and expenditure.

THINK ABOUT IT

Suppose you and your friends need to buy new badminton rackets for an upcoming tournament because the old ones have worn out. How would you manage your budget to cover the cost? Would you cut back on other expenses, use your savings, or borrow the amount from your parents or friends?

LET'S EXPLORE

Think of all your expenses and prepare your monthly/weekly budget using the template on Page 122.

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Steps in Budgeting

1) Identify your expenses

Begin by listing all your monthly expenses, both large and small. This helps you understand where your money is being spent.

2) Set a savings goal

Decide how much you want to save, what you are saving for, and by when you need to have it saved up. A clear goal makes saving easier. For example, to save ₹3,000 in six months for a new bicycle, you would need to save ₹500 each month.

3) Track your spending

Regularly compare your actual spending with your budget. This helps you identify areas where you may be overspending and make corrections if needed.

4) Review and adjust your budget

A budget should be reviewed periodically to reflect changes in income, expenses, or financial goals. For example, rising transport costs or unexpected medical expenses may require adjustments to the budget. Regular reviews keep the budget realistic and effective.

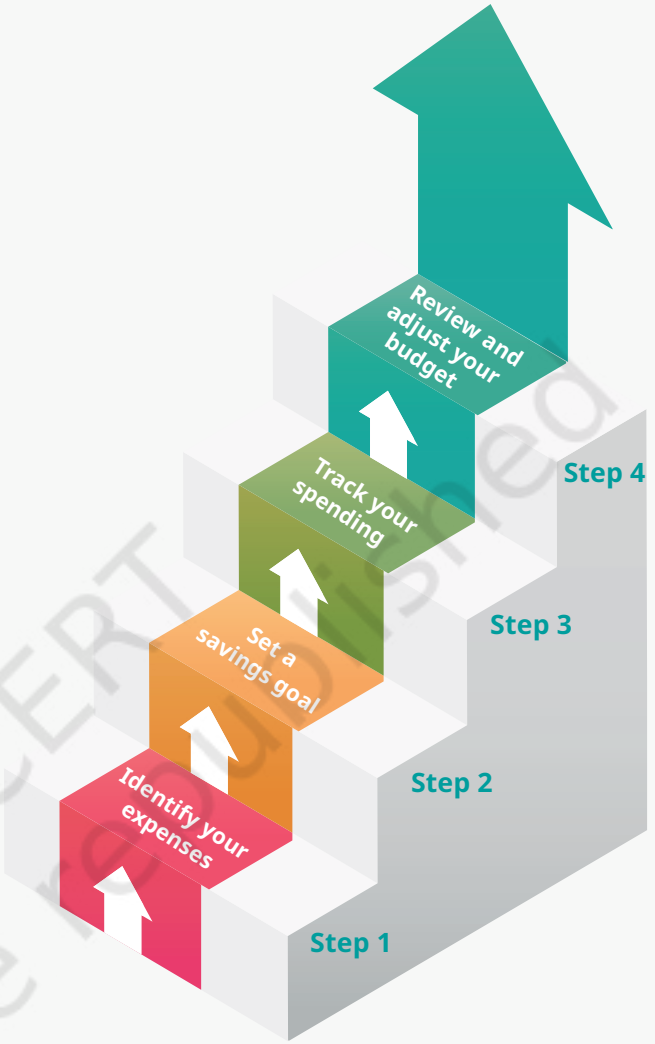

Fig. 5.3. Steps in budgeting

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Saving

Saving means keeping aside a part of one's income for future use and represents a financial surplus accumulated over time. Savings are important for comfortable living, and because life is uncertain and emergencies can arise at any time. People also save to make future investments in assets, such as fixed deposits, stocks, mutual funds, and real estate.

THINK ABOUT IT

- What would you do if you received ₹100 today? Would you save some or all of it, or spend it right away?
- In the Grade 9 Part 1 Chapter 'Building Blocks in Economics: The Problem of Choice', you learnt about wants and needs. Would you prefer spending the money on a want or a need?
- Financial experts often suggest saving first and spending what is left. Do you agree?
- Do you think individuals with low income can also save? Why or why not?

Investing (Growing Wealth)

Priya and Sameer both receive ₹500 as a birthday gift. Priya keeps her ₹500 in a piggy bank at home. One year later, she still has ₹500. Sameer puts his ₹500 into a savings scheme that earns 8% interest a year. One year later, he has ₹540. Both are being responsible with money, but are they doing the same thing?

Investing means putting your savings into assets with the aim of protecting their value and generating returns. It is like planting a mango tree—you do not get fruit immediately, but once you do, it continues to provide fruit for many years. While keeping savings in cash provides greater **liquidity**, investing helps increase wealth and protects the value of money. It also allows money to grow faster than the rate of **inflation**. For example, if a school bag costs ₹1,000 today, and inflation is 6% per year, the same bag may cost about ₹1,060 next year. If your money is simply kept at home and does not grow, you may not be able to buy the bag as your **purchasing power** will fall over time.

Do you remember the story of the king and the sage from the Grade 7 Chapter 'Banks and the Magic of Finance'? It explained the principle of compounding, where returns are earned not only on the original

Liquidity: Refers to how quickly and easily an asset can be converted into cash or cash equivalent without affecting its value.

Inflation: The increase in the prices of goods and services over time, which reduces the purchasing power of money. As prices rise, the same amount of money can buy fewer goods and services than before.

Purchasing power: The real-world value of money, measured by the quantity of goods and services a single unit of currency can buy. It determines your standard of living, as higher purchasing power means you get more goods and services with the same amount of money, while a decrease means you can afford less with the same amount of cash.

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investment (the principal) but also on the returns already earned, thereby accumulating high returns over time. Let us understand this again with an example.

Compounding: The process where an investment's earnings, such as interest or dividends, are reinvested to generate their own additional earnings over time.

Power of Early Investments

Suppose two people, Bhavisha and Dhruv, each save ₹1,000 at the end of every year and earn 7% annual interest. The only difference is that Bhavisha starts saving at the age of 25, while Dhruv begins at the age of 35. Although both save the same amount each year, but since Dhruv invested much later, by the age of 65, he accumulates only about 40% as much money as Bhavisha. This happens because **compounding** allows earlier investments to earn returns for a longer period, and those returns continue to generate additional returns over time.

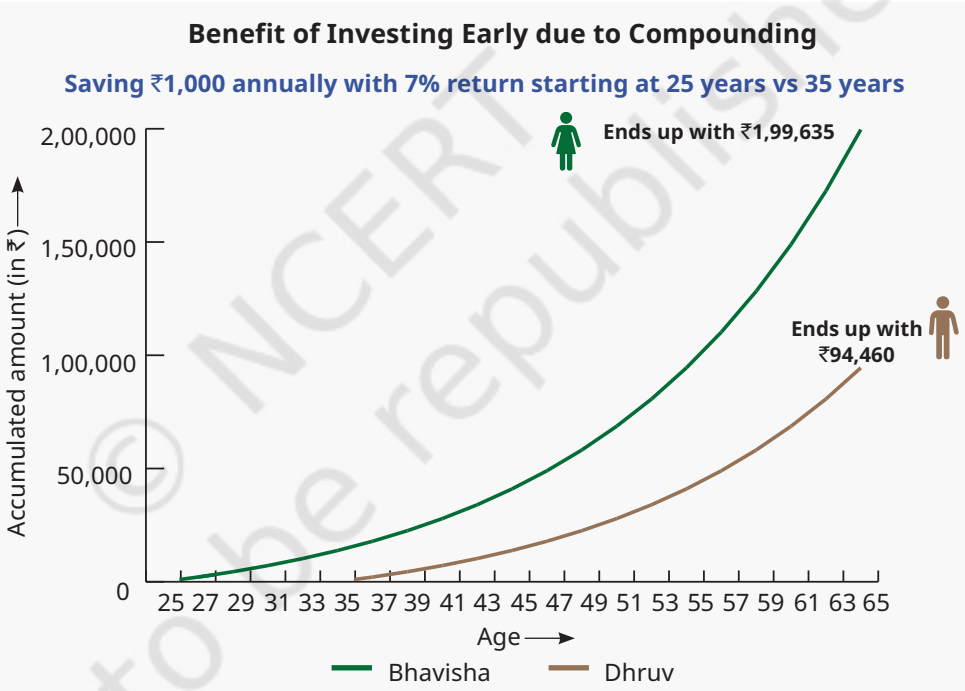

Fig. 5.4. Power of compounding

LET'S EXPLORE

→ In Fig. 5.4 above, the curves are flatter in the beginning and tend to become steeper in the later years. Why do you think that happens?

→ Talk to your parents and find out the fixed deposit rates of their bank. You can also refer to the bank's website.

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But where can one invest? Let us explore some common investment options.

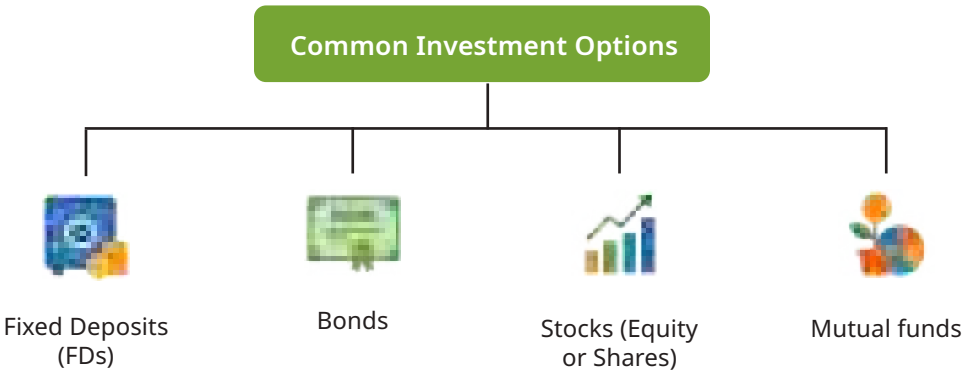

Fig. 5.5. Common investment options

a) Fixed Deposits (FDs)

Fixed deposits are among the most common and safest investment options offered by banks and post offices. In a fixed deposit, a person deposits a lump sum amount for a fixed period, such as one, three, or five years. During this period, the bank or post office pays a fixed rate of interest on this amount, which remains unchanged throughout the investment period. Fixed deposits offer assured returns and are suitable for people who prefer safety over high returns.

Other government-backed investment options include the Public Provident Fund (PPF) and small savings schemes, such as the National Savings Certificate (NSC) and the Sukanya Samriddhi Yojana. These schemes encourage long-term savings while providing safe and reasonable returns.

b) Bonds

Bonds are investments through which people lend money to the government or companies (issuers of bonds) for a fixed period. Bondholders earn regular interest. For example, if you buy a bond worth ₹10,000 that pays 5% interest per year, you will receive ₹500 as interest after one year. If the bond matures after one year, you will receive ₹10,500 in total. In reality, the **maturity period** ranges from 1 day to 30 years or more. You may also be able to sell the bond before maturity if you need the money earlier, making them more liquid. Government bonds are generally considered safer than **corporate bonds** and are suitable for those seeking a stable, predictable income.

Maturity period: The duration between its issue date and the point at which the issuer repays the principal amount.

Corporate bonds: Debt securities issued by companies to raise capital from investors. When you buy a corporate bond, you lend money to the company. In return, the company promises regular interest payments and to repay the initial principal amount when the bond matures.

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BSE SENSEX: The BSE SENSEX (Stock Exchange Sensitive Index) is the benchmark index of the Bombay Stock Exchange (BSE). It tracks the performance of 30 large, financially sound, and actively traded companies from different sectors of the economy and serves as an important indicator of the overall performance of the Indian stock market.

c) Stocks (Equity or Shares)

Shares represent ownership in a company. When a person buys shares, they become a part-owner of that company and may earn returns through dividends as well as an increase in the share price. Investing in shares involves higher risk because share prices fluctuate due to market conditions, company performance, and broader economic changes. However, shares also have the potential to generate higher returns over the long term. They are, therefore, suitable for investors who are willing to take greater risks for higher returns and who invest keeping the long term in mind.

LET'S ANALYSE

Look at the graph below that displays the performance of **BSE SENSEX** over the years. It fluctuates over short time periods; however, it can go up significantly in the long run. Why do you think these fluctuations happen? Discuss possible reasons with your classmates and find out more from newspapers, the internet, or by speaking to your family members.

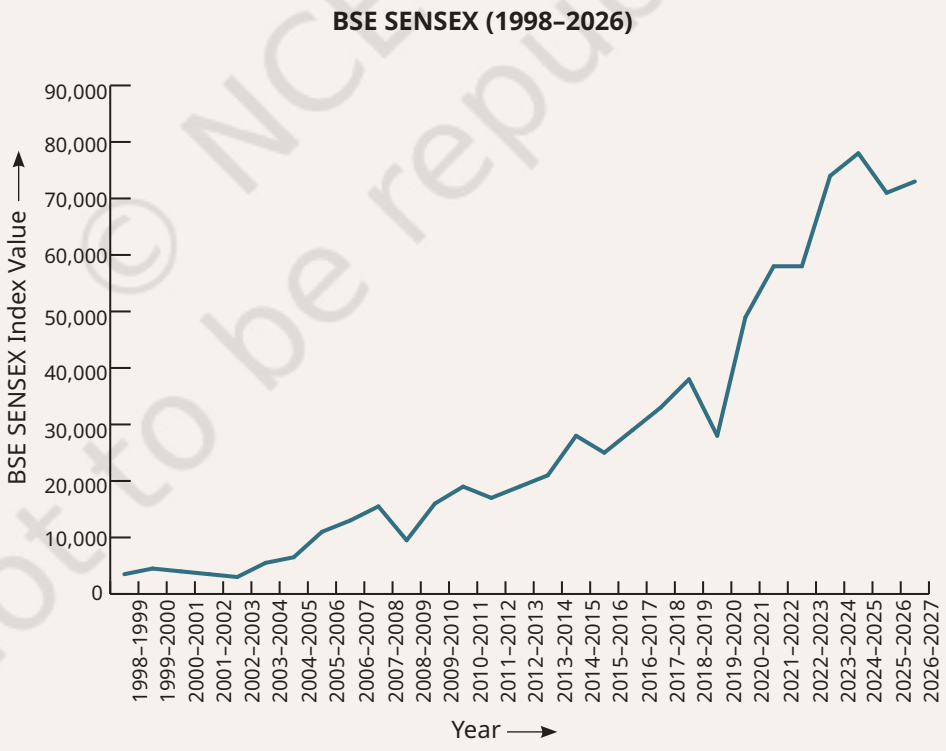

Fig. 5.6. BSE SENSEX over the years

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d) Mutual funds

Mutual funds pool money from many investors and invest it in a basket of shares, bonds, or other financial assets. These investments are managed by professional fund managers who decide where to invest the money. If the investments perform well, investors earn returns in proportion to the amount they have invested. However, if the investments perform poorly, investors may incur losses.

Mutual funds are popular because they reduce risk by spreading investments across different types of assets, a strategy known as **portfolio diversification.** They are suitable for people who may not have the time, knowledge, or experience to manage investments on their own.

THINK ABOUT IT

Have you heard this disclaimer in mutual fund advertisements on the television?

"Mutual fund investments are subject to market risks. Please read all scheme-related documents carefully before investing."

Why do you think this is included?

Protection and Risk Management

Life is full of uncertainties, such as illness, accidents, loss of income, or damage to property. These unexpected events can lead to significant financial losses. Therefore, an important aspect of personal finance is protection—safeguarding your income, savings, and assets against such risks.

Insurance

One of the most effective ways to achieve financial protection is through insurance. Insurance is a legal contract in which an individual pays a regular amount, called a premium, to an insurance company. In return, the company agrees to provide financial compensation for specific losses or expenses covered under the policy. For example, a family may pay ₹500 every month for health insurance. If a family member requires medical treatment, the insurance company covers a large part of the hospital expenses. Without insurance, the family might have to use a substantial portion of its savings for the expenses incurred towards the treatment.

Insurance works on the principle of risk sharing. Many people pay premiums into a common pool. In case of any unexpected loss,

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those insured receive financial support from the insurance company. Therefore, insurance provides financial security during emergencies, reduces the impact of accidents, illness, or property damage, helps families recover from financial setbacks, and encourages people to save and plan their finances through regular premium payments.

Types of Insurance

a) Health Insurance

Health insurance helps cover medical expenses, such as hospitalisation, medicines, diagnostic tests, and treatments. It reduces the financial burden of illness or accidents and allows families to focus on recovery rather than worrying about medical costs. In India, government schemes such as Ayushman Bharat Pradhan Mantri Jan Arogya Yojana provide health insurance coverage to eligible low-income households.

b) Life Insurance

Life insurance provides financial support to a family if the policyholder, who may be the breadwinner of the family, passes away. It helps dependents meet daily expenses, continue children's education, and repay loans. Life insurance is an important tool for ensuring long-term financial security.

c) General Insurance

General insurance protects against losses related to vehicles, homes, businesses, and other property. For example:

- Motor insurance covers repair costs and damages caused by vehicle accidents.
- Property insurance protects homes and businesses against losses caused by fire, theft, flood, earthquake, and other disasters.
- Crop insurance protects farmers from financial losses as it covers damages caused by natural disasters, pests, and market price changes.
- Cattle insurance protects farmers and dairy owners against financial loss in case of death or disability of their animals.

Fig. 5.7. A person applying for a loan in a bank

Borrowing and Credit

Sometimes, people may need to borrow money from banks, relatives, or other sources as loans. Borrowing or using credit can be helpful when it is used for productive purposes, but it can become harmful if misused. For instance, an education loan can help a student pursue higher education, whereas borrowing money to purchase

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unnecessary consumer goods may lead to a **debt** burden and financial difficulties. Thus, responsible borrowing involves taking loans only when necessary and ensuring that they can be repaid on time.

Debt: Amount of money owed to somebody.

THINK ABOUT IT

Have you ever heard your parents say, "This month's EMI has been paid"? An EMI (equated monthly instalment) is the fixed amount of money a person pays every month to repay a loan.

Suppose a family buys a refrigerator costing ₹30,000 but does not want to pay the entire amount at once. In such a case, they may take a loan and repay it through an EMI of ₹3,000 every month for 12 months. Each EMI includes a portion of the original loan amount (the principal) as well as the interest charged by the lender. EMIs are commonly used to purchase mobile phones, household appliances, two-wheelers, homes, and cars.

- Why might a family choose to buy something on EMI even if it costs more in the long run?
- Can having too many EMIs affect savings and emergency funds?

Risk and Return

When deciding where to invest money, it is important to consider four factors—**risk, return, time horizon, and liquidity**.

Risk

Risk refers to the chance of losing money.

Return

Return is the profit earned from an investment.

Time horizon

Time horizon is the length of time for which money can remain invested.

Liquidity

Liquidity refers to how easily an asset can be converted into cash when needed.

Fig. 5.8. Factors to consider before investment decision

Different investment options have different levels of these factors. For example, fixed deposits (FDs) are generally low-risk options that provide steady but relatively lower returns. Mutual funds may offer higher returns, while shares (equity) have the potential to generate the highest returns over the long term but also involve the highest level of risk.

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In general, lower-risk investments tend to offer lower returns, while higher-risk investments may offer higher returns. Therefore, people should choose investments based on their financial goals, the amount of risk they are willing to take, and when they are likely to need the money.

Fig. 5.9. Relationship between risk and return

Taxes

Taxes are compulsory payments made by individuals and businesses to the government to fund public services, such as schools, hospitals, transport, defence, and other essential facilities. There are two main types of taxes:

- **Direct taxes,** which are paid directly on a person's or organisation's income or profits, such as the income tax.
- **Indirect taxes,** which are added to the price of goods and services, such as the Goods and Services Tax (GST).

Personal Income Tax

Personal income tax is a tax paid on the income a person earns in a year (called a financial year, say 1 April 2001 to 31 March 2002). All citizens of India with a personal income are required to pay income tax, with the sole exception of individuals who earn below a certain limit and are exempt from paying the tax. As income increases, the tax payable to the government also increases. This system is known as **progressive taxation,** which aims to reduce income inequality by imposing higher tax rates on individuals with higher incomes.

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The underlying principle is that people with greater financial capacity can afford to contribute a larger share of their earnings towards funding public services, such as schools, hospitals, roads, and security. Paying taxes honestly and on time is an important responsibility of every citizen as it contributes to the well-being of society and the development of the country.

Fig. 5.10. Filing income tax

India follows a slab system for calculating income tax, as shown in Table 5.1. Under this system, taxable income is divided into different ranges, called tax slabs, and each slab is taxed at a specified rate, as seen in Table 5.1. The tax slabs are proposed by the Ministry of Finance as part of the Union Budget presented to the Parliament every year. The Central Board of Direct Taxes (CBDT), under the Ministry of Finance, is responsible for administering the laws relating to direct taxes in India.

Table 5.1. New Tax Regime (Applicable for FY 2025–26)

Annual Income (₹)	Tax Rate (%)
Up to ₹4 lakh	Nil
₹4 lakh–₹8 lakh	5%
₹8 lakh–₹12 lakh	10%
₹12,00,001–₹16,00,000	15%
₹16,00,001–₹20,00,000	20%
₹20,00,001–₹24,00,000	25%
Above ₹24,00,000	30%

Let us understand how a person's income tax liability is calculated. Suppose Priya, a business analyst, earns ₹14 lakh per year. Firstly, her income to be taxed will be calculated as follows:

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Step 1: Calculate Taxable Income

Particulars	Amount (₹)
Annual Salary	14,00,000
Standard Deduction	75,000
Taxable Income	13,25,000

Standard deduction: A fixed amount that salaried employees and pensioners can deduct from their income before calculating income tax, as allowed under the Income-tax Act, 2025

Step 2: Applying the Tax Rate as per the Tax Slabs

Different portions of income are taxed at different rates. For Priya, the first ₹4 lakh of her income are not taxed; the next ₹4 lakh are taxed at 5%; the next ₹4 lakh are taxed at 10%, and so on, as seen in Table 5.2.

Table 5.2. Calculation of Income Tax as per the Tax Slabs

Portion of Income in the Tax Slab	Tax Rate	Tax on that Portion of Income
Up to ₹4,00,000	0%	0
₹4,00,001–₹8,00,000 (next ₹ 4,00,000)	5%	5% of ₹4,00,000 = ₹20,000
₹8,00,001–₹12,00,000 (next ₹4,00,000)	10%	10% of ₹4,00,000 = ₹40,000
₹12,00,001–₹13,25,000 (remaining ₹1,25,000)	15%	15% of ₹1,25,000 = ₹18,750
Total Tax		**₹78,750**

LET'S ANALYSE

→ Sai's taxable income is ₹3,80,000 in a financial year. How much income tax does he have to pay according to the above tax slabs?

→ Ask your parents, relative or a person in your neighbourhood if they file personal income tax returns. Make a small report on the procedure to file an income tax return. You may refer to the following website: *https://incometaxindia.gov.in/Pages/tax-services/file-income-tax-return.aspx*

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In conclusion, managing money wisely is not just about earning more—it is about making careful financial decisions at every stage, whether it involves spending, saving, investing, protecting against risks, borrowing responsibly, or paying taxes. The choices individuals make today shape their financial future and determine their ability to achieve financial stability while managing financial opportunities and challenges.

Before we move on...

- Making informed financial decisions from a young age builds financial security, independence, and confidence for the future.
- Personal finance rests on the five pillars—income, budgeting, saving, investing, and protection—to ensure financial security for individuals.
- Budgeting helps individuals plan their expenses, control unnecessary spending, and save for future needs. Saving and investing are important for building wealth, meeting goals, and protecting money from the effects of inflation.
- Different investment options involve different levels of risk and return, so investment choices should match an individual's goals and risk tolerance.
- Insurance protects individuals and families from financial losses during contingencies.
- Income tax helps governments provide essential public services and infrastructure. Paying taxes honestly is a citizen's responsibility, and helps the government provide public services and promote national development.

Questions and activities

1. Why is managing money wisely important for all individuals irrespective of their income?
2. How can budgeting help people balance their present needs with their future financial goals? Give examples.
3. **School Reopening Budget Activity**

   Your parents have given you ₹3,000 to prepare for the new school year. Make a list of the items you need, find out their prices (by visiting local stores, or asking a parent or teacher),

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and prepare a budget. Calculate the total cost and check whether it stays within ₹3,000.

- If your total is less than ₹3,000, you may add one additional item of your choice.
- If your total exceeds ₹3,000, decide which item(s) you can postpone or avoid buying to stay within your budget.

Use the budget template provided and ensure that all expenses are accurately calculated.

School Reopening Budget

Total Amount	₹3000
Expenses	**Amount**
School bag	₹
Books	₹
Uniform and shoes	₹
Notebooks	₹
Pens	₹
Pencils	₹
Art supplies	₹
Total Expenses	₹
Savings	₹

4. Why is saving alone often not enough to build wealth in the long run? How do investments help increase the value of money over time?

5. **Investment Planning Activity**

   Suppose you received cash gifts worth ₹3000 on your last birthday and decided to save ₹1500. With the help of your teacher, calculate the returns you might have earned if you had invested this amount in different combinations of investments. For example, what returns would you have received if you had invested ₹500 in a fixed deposit, ₹800 in the stock market, and ₹200 in gold? Try other combinations and find information on the returns from different investment options using reliable sources, such as newspapers, magazines, or official websites. Based on your findings, prepare an investment plan explaining why you think each investment option is suitable or unsuitable.

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6. How does inflation affect the value of money over time? Why should people consider inflation while making financial decisions?
7. Different investment options involve different levels of risk and return. Do you think taking higher risks is always the best way to earn higher returns? How should a person decide which investment option is most suitable for them?
8. Why do people buy insurance policies even though they may never face a major loss? How does insurance provide financial security?
9. Imagine your family receives ₹1,00,000 from farming, a small business, livestock sales, and savings. Help your family decide how to use the money wisely.

Step 1: Prepare a Budget

Allocate the ₹1,00,000 among the following categories:

Category	Amount (₹)	Percentage (%)
Savings		
Education		
Household expenses		
Health and emergency fund		
Farm or business investment		
Home improvement		
Festivals and recreation		
Other		
Total	**1,00,000**	**100%**

Step 2: Needs vs Wants

Classify each expense as a **need** or a **want** and give a reason.

Examples:

- School fees
- New mobile phone
- Better seeds or livestock feed
- Festival celebrations
- Water storage tank

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Step 3: Save for the Future

Suppose your family saves **₹30,000** at **8% compound interest per year.**

Using the formula **A = P(1 + r)^n**, where **A:** Amount, **P:** Principal, **r:** Rate of compound interest, **n:** number of years, calculate:

1. The amount after 5 years.
2. The interest earned.

Step 4: Save First, Spend Later

Many financial experts recommend:

Expenses = Income – Savings

If your family receives ₹1,00,000 and saves ₹20,000 first, only ₹80,000 is available for spending. This helps ensure that future needs and emergencies can be met.

Step 5: Reflect

1. What percentage of the money did you save?
2. Which expenses were needs and which were wants?
3. How can savings improve a family's financial security?
4. How does compound interest help money grow?

Challenge: Create a pie chart showing how the ₹1,00,000 was allocated across different categories.

Notes

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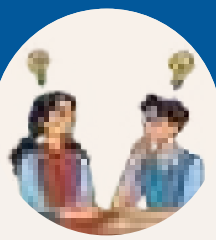
Chapter 6
From Ideas to Startups



The Big Questions

1. *How does entrepreneurship impact society and the economy?*
2. *What are some functional areas of a business?*
3. *How to develop a business plan?*

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Imagine your school is hosting an annual festival, and you participate in the 'Young Entrepreneurs Challenge', where you have to develop a product or service for parents and children visiting your stall. You come up with an idea for a lemonade *(nimbu pani)* and set up a stall with your classmates. At the end of the day, you prepare and submit a detailed report on the expenses incurred and the **profit** made to your class teacher.

What do you think is required to run the stall? You will have to decide how much to spend on lemons and sugar, where to procure glasses, how much to charge for each glass, and so on. Also, you will have to assess how many people liked the drink and if you made a profit or loss. This simple activity captures the essence of running a business—planning, organising, and learning by doing. This is **entrepreneurship**, a process through which individuals, or a group of individuals, convert a potential idea into a business venture.

Every great business starts with an idea, a need, or a desire to innovate

Fig. 6.1. Elements of entrepreneurship

Profit: Money that is earned in trade or business after paying the costs of producing and selling goods and services.

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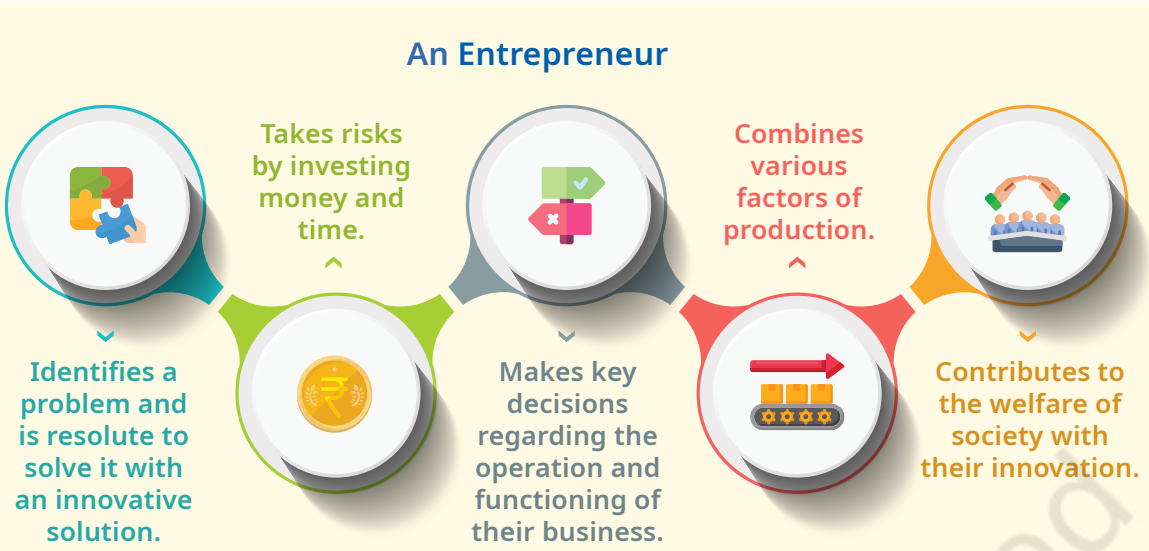

Fig. 6.2. Tasks performed by an entrepreneur

and meet the needs of society. Entrepreneurs are curious individuals who see opportunities where others see challenges and turn that spark into action. Whether it is leading a business that manufactures drones for traffic surveillance or creating eco-friendly jute bags that can replace plastic, an entrepreneur works hard, and is driven by passion and a commitment to creating value for society. Let us understand this through a story.

From a Dead Mouse to Riches

One day, a rich merchant told his king, "Even a dead mouse can be the start of a fortune, if someone works hard and uses their brain." A poor young man overheard this and decided to test it. He picked up a dead mouse and sold it to a shopkeeper for two coins since his cat was hungry. With those coins, the young man bought fruits and water, which he sold to a tired flower picker in exchange for flowers. He then sold the flowers in the city.

Fig. 6.3.

Step by step, he grew his small earnings by clearing fallen branches after a storm and selling them as firewood, trading grass with horse dealers, opening a refreshment shop to later even striking business with foreign merchants.

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With intelligence, dedication, and hard work the man became very wealthy. But he did not forget how it all started—with the merchant's words. Out of gratitude, he gave the merchant half his fortune. Also, the man used his wealth not just for himself, but to help communities in the city as well.

Fig. 6.4.

The story above is from the famous *Jataka Tales*. What can you take away from the tale? Is it about spotting opportunities or working smart? What kind of traits do you think entrepreneurs need to have? Discuss with your classmates.

Let us learn about stories of a few entrepreneurial minds who turned their ideas into reality.

From Rooftop to Revolution

In 1959, seven ordinary homemakers from a Mumbai **chawl** gathered on a rooftop with borrowed equipment, a recipe, and ₹80. Their goal was modest—earn some extra income by making and selling *papads.* They had no business degrees, no investors, and no master plan; instead, what they had was grit and unity. There were no factories at first; each woman took raw dough home every morning, rolled *papads* through the day between household tasks, and returned the finished pieces by evening with the condition that 'the quality had to be consistent'. They decided to start a cooperative society where every member who joined would be an equal owner, and profits would be shared equally regardless of seniority.

Chawl: A traditional, multi-storey residential tenement building, typically consisting of dense, single-room units.

In 1962, they registered the venture as Shri Mahila Griha Udyog Lijjat Papad, a cooperative society, where 'lijjat' means tasty in Gujarati. Afterwards, many branches opened across India and garnered thousands of members. In the 1980s and 90s, Lijjat expanded into other products like flour, masalas, and soaps, and exports began too, crossing a turnover of ₹100 crore. By 2023, the cooperative had over 45,000 women members, 83 branches, and an annual turnover of ₹1,600 crore.

The women begin work early in the morning, facilitated by pick-up and drop-off bus services. Lijjat also provides scholarships to

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Fig. 6.5.

the school-going children of the members to encourage them to pursue further education. The organisation has helped thousands of women become financially independent, empowered communities, and built a large-scale business while being completely self-reliant.

THINK ABOUT IT

- What problem was Lijjat solving beyond just selling *papads*?
- What challenges do you think the founders could have faced at the start? How might they have overcome them?
- Can you think of a similar cooperative idea you could start in your community?

The Man who Transformed the Hospitality Industry

Born in Punjab in 1898, Mohan Singh Oberoi faced hardships early in life when his father passed away while he was still an infant. After somehow completing his education, he moved to Shimla and took a job as a hotel desk clerk, earning just ₹50 a month.

Rather than limiting himself to his assigned duties, Mohan learned every aspect of hotel operations, from guest service to billing and administration. His dedication, curiosity, and willingness to take initiative set him apart.

In 1934, he took a bold step by **mortgaging** his wife's jewellery and investing all his savings to buy a small hotel. Despite the risks, he believed in his vision. Through hard work and determination, he

Mortgage: A mortgage is a special type of loan used to buy a house or land with some collateral.

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transformed struggling hotels into successful businesses, and went on to build one of India's largest hotel chains. He later established luxury hotels, founded a hotel management school, and expanded his business across the world. In 2001, the Government of India accorded him the Padma Bhushan. He passed away in 2002; by then, the Oberoi Group, named after him, had become one of the world's most respected hotel brands.

His journey shows how entrepreneurship is not only about having a good idea. It involves planning, teamwork, decision making, and the effective management of different areas of the business. For example:

- He **hired and trained skilled** staff to provide excellent hospitality services, and constantly worked to improve hotel facilities and guest experiences.
- He carefully **managed the business's finances** to support growth and expansion.
- His hotels were **renovated and operated efficiently** to maintain high standards.
- The company **built a trusted brand** known for luxury and quality service.
- **Customer satisfaction** remained the top priority in all hotel operations.

These are some of the functions that entrepreneurs and their teams perform. There are departments or groups within a company that focus on such distinct tasks and shared skills, also known as functional areas. Let us understand them in detail and learn how a business plan is prepared.

Business Plan—A Road Map for the Business

Let us assume that you and your friends decide to open a bakery together that offers snacks and desserts. How would you go about it? You have to consider many crucial questions, identify challenges, and plan at every stage before starting the bakery to turn this simple idea into a successful business venture. This is when a business plan is developed. It is a written document, a blueprint explaining the goals of the business and the steps to achieve them. It guides the entrepreneur from the initial idea stage to successfully starting and running the business. It also serves as a guide and roadmap for

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decision making, helping them stay organised, set priorities, and work systematically to grow the business.

Exploring aspects such as those listed below could be especially helpful while creating a business plan.

Fig. 6.6. Key considerations in making a Business Plan

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1. **Needs and idea:** To understand what people want, you would have to understand customer needs.
    - Do customers prefer fresh bread, cakes, pastries, or healthier options like whole wheat products?
    - Will the bakery provide fresh, affordable, or unique baked goods that were not available in the area before?
2. **Potential customers:** Knowing your customers helps you decide what kind of products to offer. To identify the target consumers, one may think about:
    - Decide whether to specialise in offering cakes, cupcakes for children, or savoury snacks for young adults, or everyday breads and confectionery items for families.
3. **Money and costs:** Your business needs a robust financial start, which begins with planning finances and estimating the initial investment required. This includes the cost of equipment, such as an oven, rent for the premise, staff salaries, ingredients, such as flour and nuts, and packaging.
    - How should funds be arranged? (through savings, loans from banks, friends, relatives, or investors)?
    - How much will the expenses be?
4. **Location:** You may want to choose the right place for your bakery.
    - Can choosing a busy area, such as near schools, colleges, or markets, help attract more customers?
    - Can selling online or taking orders through mobile apps help you reach more people?

Fig. 6.7.

5. **Human resources (people and skills):** You will have to decide how many people have to be hired and trained, and the required skills for the job.
    - Identify people for roles, such as bakers, helpers, and staff for billing or delivery.

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6. **Pricing and profits:** You may consider fixing prices carefully so that they not only cover your costs and generate profit but also remain affordable for customers.
    - How to balance profitability with affordability when setting prices of goods and services?
    - Can offering a range of pricing options help attract a wider set of customers, for example, cheaper mini cake jars?
7. **Promotion or marketing:** To spread awareness among people about the bakery, you may want to promote it through social media, posters, and provide special offers.
    - What kind of packaging can enhance the appeal and shelf-life of the bakery products?
    - How can customer service be improved to retain customers?

Fig. 6.8.

8. **Challenges and risks:** These are threats that disrupt operations or cause losses, for example, high competition, rising cost of raw materials, changing customer preferences, volatile cash flows, etc. These are an inevitable part of every business venture.
    - How can perishable **inventory** be managed effectively?
    - How can consistent quality in the products be ensured?
    - How should fluctuating customer demand be handled?
    - Are there other bakeries in the locality and how can we compete with them?

Inventory: The goods, materials, and assets a business holds for the purpose of manufacturing, production, or resale.

Accounting: Recording and tracking all the money related to the business so the owner knows how it is performing.

Functional Areas of a Business

The questions in Fig. 6.6 about a business can be grouped into a few main functional areas, like management and operations, finance and **accounting**, human resources, and marketing and sales. Each area has a specific function and helps the business run smoothly. They improve decision making and ensure resources are used properly. Together, they provide structure, better coordination, and help the business grow successfully. Let us learn about them in detail.

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FUNCTIONAL AREAS OF A BUSINESS

Every successful business runs on four key pillars. Together, they turn goals into growth.

1 Management and Operations

Looking after day-to-day matters

Key Focus: Efficient processes and smooth operations

Examples: Production, quality, supply chain, etc.

2 Finance and Accounting

Managing money wisely

Key Focus: Financial health and smart budgeting

Examples: Budgeting, reporting, cash flow, etc.

3 Human Resources

Managing people

Key Focus: Right people and skills

Examples: Hiring new staff, managing employee benefits, and handling workplace conflict resolution, etc.

4 Marketing and Sales

Spreading awareness and driving sales

Key Focus: Customer awareness and business growth

Examples: Advertising, promotions, sales, etc.

Fig. 6.9. Functional areas of a business

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Fixed cost: Business expenses that remain constant regardless of production volume or sales levels. They do not change with output in the short term and act as indirect or overhead costs, such as rent, salaries, or insurance.

Variable cost: Expenses that change in direct proportion to the volume of goods or services a business produces or sells. They increase as production rises and decrease as production falls, covering raw materials, packaging, etc.

Investment: The allocation of money or resources into an asset, venture, or project with the expectation of generating future income, profit, or value appreciation.

Management and Operations

Fig. 6.10.

Management and operations are essential for running a business smoothly. Management involves planning, organising, and guiding the use of resources to achieve business goals, such as increasing market share through customer satisfaction. Operations focuses on producing and delivering goods or services efficiently and on time. For example, in a bakery, you must choose the right equipment, arrange ingredients, monitor quality, and plan daily production to ensure fresh and high-quality products are made regularly without delay or wastage.

Finance and Accounting

Finance is the management of money in a business. It involves planning, obtaining, and using funds to help the business operate and grow.

For the bakery, you would have to plan for the following:

- Paying **fixed costs:** rent, equipment costs (ovens, mixers, refrigerators), etc.
- Paying **variable costs:** cost of ingredients, salaries, packaging material, etc.
- Investing in growth opportunities
- Managing profits and cash flow

Accounting is the process of recording, organising, and reporting a business's financial transactions. Every entrepreneur needs to understand basic accounting concepts to know how their business is doing. It means keeping track of the money coming in (revenue) and the money going out (expenses), and making sure all records or documents are correct. One important record is the profit and loss account (or income statement). It shows whether the business has made a profit or a loss over a period of time, such as a month or a year. This helps to make informed decisions about spending and **investment**. We can track what the business owns and how much has been borrowed. It also enables compliance with tax laws and regulations of the country.

Profit = Total Revenue – Total Expenses

Revenue is the total money earned by a business from its main operations or sale of goods and services. For example, ABC bakery

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may earn ₹2,95,000 in a month, as seen in its profit and loss account shown in Fig. 6.11.

Expenses include the money spent on running and maintaining a business. These include salaries to employees, electricity bills, cost of raw materials, etc. Further, there may be some interest to be paid on a loan and lastly, taxes to be paid to the government.

PROFIT AND LOSS ACCOUNT
*ABC Bakery (Teen enterprise) | For the month ended March 2026

Income		
	Particulars	**Amount (₹)**
	Revenue from operations (sales)	
	Bread and buns	₹1,20,000
	Pastries and cookies	₹75,000
	Cake orders	₹85,000
	Delivery charges	₹15,000
	Total Income	**₹2,95,000**
Expenses		
	Particulars	**Amount (₹)**
	Cost of goods sold (COGS)	
	Raw materials: flour, sugar	₹1,00,000
	Packaging material	₹12,000
	Rent	₹40,000
	Salaries and wages for staff	₹30,000
	Utilities (electricity and gas)	₹15,000
	Advertising and marketing (social media advertisements, posters)	₹8,000
	Miscellaneous expenses	₹7,000
	Interest on loan	₹10,000
	Total Expenses	**₹2,22,000**
Profit Calculation		
	Particulars	**Amount (₹)**
	Total sales	₹2,95,000
	Total expenses	– ₹2,22,000
	Profit before tax (PBT)	₹73,000
	Tax to be paid	– ₹13,000
	Profit after tax (PAT)	₹60,000

Fig. 6.11. A sample profit and loss account for ABC bakery
**This is a fictional name and any similarity in real world is purely coincidental*

Note: Students need not memorise the above categories.

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Human Resources

People are the heart of every business. Human resources are the individuals employed by the entrepreneur who contribute to the business. For a bakery, functions, such as hiring bakers, helpers, cleaners, and a cashier; training staff in hygiene and baking standards, and ensuring a comfortable and motivating workplace fall under the human resources department. They helps build a strong team with required skills that fosters business growth.

Marketing and Sales

The marketing and sales team helps attract customers and convinces them to buy the products. In a bakery, this includes choosing a catchy name, promoting the business on social media, and offering discounts.

Some simple strategies include:

- Creating unique or seasonal items and customised desserts like mango-flavoured cake.
- Offering discounts, deals, or special offers.
- Providing easy online ordering and delivery.
- Using social media to share photos and videos for awareness among people.
- Promoting eco-friendly practices of recyclable packaging material.

LET'S EXPLORE

→ Interview two people in your vicinity who run a business (tailor, shopkeeper, factory owner, etc.) and find out about their journey of starting the business by asking them the following questions:

- Why did they start the business?
- What were the challenges they faced?
- What were the risks taken?
- What did they learn from the process?

→ Prepare a short report on the tasks an entrepreneur performs at each stage, with reference to Fig. 6.2.

→ Based on the case study of Mohan Singh Oberoi on Page 128, which functional area helped improve the quality of the company's products? How did different functional areas work together to make the business successful? Now that you know about the functional areas of a business, create a business plan for your business idea using the template given on Page 142.

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India's Traditional Entrepreneurial Community

Entrepreneurship in India has deep roots among traditional communities, where skills, occupations, and occupational knowledge have been passed down through generations. Long before modern startups, these communities practised self-employment, innovation, and trade, forming the backbone of India's local economy.

The Chettiar (Nattukottai Chettiar) community from Tamil Nadu is a well-known traditional entrepreneurial community. Historically, Chettiars were traders, bankers, and moneylenders, operating not only across India but also in Sri Lanka, Burma (now Myanmar), Malaysia, and other parts of Southeast Asia. They were known for:

- Strong business ethics and trust-based lending
- Excellent accounting and record-keeping skills
- Reinvesting profits into trade, temples, education, and community welfare

The Chettiar community built wealth through trade in salt, rice, textiles, and finance, often without formal banks. Even today, Chettiar entrepreneurship inspires modern India by showing how risk-taking, discipline, and long-term vision can build sustainable businesses rooted in tradition.

Fig. 6.12. A traditional Chettiar home

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LET'S EXPLORE

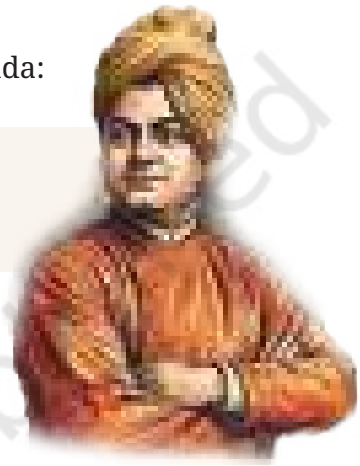
Find out more about other regional communities who have excelled as entrepreneurs.

Startups and Their Contribution to the Nation

Startup: A young, early-stage company founded by entrepreneurs offering an innovative product or service.

India's vibrant **startup** ecosystem is built not only on innovation but also on perseverance. All entrepreneurs should be aware that there is a chance of their business ending in failure. So while some startups grow rapidly, others struggle or even close down. As with other things in life, there is nothing wrong with failure if an honest effort has been made. Hence, these experiences contribute to entrepreneurial learning, and every unsuccessful attempt offers lessons that help entrepreneurs develop stronger ideas and build better businesses in the future.

As was famously said by Swami Vivekananda:



> *"Arise, awake and stop not till the goal is reached.*

Atmanirbhar Bharat: A national policy framework launched by the government in 2020 to make the country self-sustaining, boost domestic manufacturing, and reduce import dependency.

With consistent hard work and perseverance, today, India is the third-largest startup ecosystem in the world, powered by its young and talented population. The startup world is growing fast, driven by creativity and technological support, with around 120 unicorns (startups valued at over 1 billion dollars) in 2025. These are working in exciting fields, like e-commerce (online shopping), fin-tech (digital payments), health-tech (online healthcare), and agri-tech (farming technology). Cities like Bengaluru, Hyderabad, Mumbai, and Delhi NCR have become major startup hubs. These ventures are creating jobs, supporting sustainability, and driving economic growth.

Viksit Bharat 2047: The Government of India's vision to transform India into a fully developed nation by 2047, marking 100 years of independence.

In recognition of the importance of promoting entrepreneurship, the government has introduced many schemes, such as Startup India, which supports the vision of **Atmanirbhar Bharat**, focusing on making India self-reliant through innovation and local solutions. As India moves towards the vision of **Viksit Bharat 2047**, entrepreneurship will continue to play a vital role in creating employment, driving innovation, empowering communities, and building a more prosperous, self-reliant, and sustainable nation. The next generation of entrepreneurs may well be sitting in your classroom today, ready to transform ideas into solutions that make a meaningful difference to society.

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IMPROVING QUALITY OF LIFE

Medicines, transportation, technology, and innovations

CREATING JOBS AND INCOME

Employment for households

PAYING TAXES TO THE GOVERNMENT

Funding roads, schools, and hospitals

PRODUCING GOODS AND SERVICES

Boosting economic activity

MAKING INNOVATIONS

Driving economic growth and offering solutions

CORPORATE SOCIAL RESPONSIBILITY

Education, environment, and healthcare

Fig. 6.13. Contribution of businesses to society and economy

Before we move on...

- Entrepreneurship is the process of turning ideas into action by identifying opportunities, taking risks, and creating value for society.
- A business plan acts as a roadmap that helps entrepreneurs set goals, organise resources, and make informed decisions.
- Every business has key functional areas—management and operations, finance and accounting, human resources, and marketing and sales—that work together for success.
- Understanding revenue, expenses, profit, and cash flow helps businesses track performance and remain financially sustainable.
- Challenges, risks, and even failures are part of the entrepreneurial journey and provide valuable opportunities for learning and improvement.
- Startups and entrepreneurs drive innovation, create jobs, solve real-world problems, and contribute to the vision of a Viksit Bharat 2047.

Questions and activities

1. Explain why a business can fail even with a great idea if one functional area [finance, human resources (HR), operations, or marketing] is weak. If you were running a bakery, which functional area would you prioritise first and why?
2. Do you think innovation alone is enough to make a business successful? Why or why not?
3. "A business plan is a dynamic roadmap for a startup." Explain this statement with suitable examples. When should one change the business plan?
4. Do you think every entrepreneur should think about the impact of their goods or services on society and the environment? Explain using relevant examples.
5. In 1931, two brothers, Digambar and Govind Dandekar, started a small business in South Mumbai selling ink powder. They began with limited resources, using recycled glass bottles to sell liquid ink. At first, they faced financial problems and struggled to convince customers to trust their products. Instead of giving up,

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they focused on providing good-quality stationery at affordable prices. As the business grew, they noticed that artists in India did not have access to good art materials. Seeing this as an opportunity, they travelled to Glasgow to study colour chemistry and improve their products. After returning to India, they set up a laboratory to develop colours and art supplies suited to Indian needs. Over time, the company expanded into pencils, crayons, geometry boxes, and paints, eventually becoming a well-known brand.

- What entrepreneurial qualities did the Dandekar brothers display?
- What opportunity did they identify in the market?
- Was travelling to Glasgow a risky decision?
- Which functional area helped the company improve its products?
- How did they serve society apart from earning profits?

6. How do you think entrepreneurship is important for India's goal of becoming a developed nation by 2047?
7. Choose a product or service and develop a business plan for it using the following template:
   - Business name
   - Business idea
   - Target customers
   - Product/service and price
   - Marketing idea
   - Basic cost and expected profit
8. Find out about various government schemes to promote entrepreneurs in India. Learn about the eligibility requirements of these schemes. Prepare a write-up and discuss in the class.
9. Take any popular brand and identify:
   - The problem it tackles
   - Its initial target customers
   - How it differentiated itself from other competitors or brands
   - The resources it may have started with

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10. Over the next week, observe your daily routine at school and home. Identify 2–3 problems or inconveniences you or others face. For each problem, answer the following questions in a tabular format:
    - Who faces this problem?
    - How often does it occur?
    - Why hasn't it been solved yet?
    - What business idea could solve it?
    - Would people pay for your solution?
11. Think of your startup idea and fill in the business plan template given below:

    a. Business idea

    Name of the company or shop: _____

    Product/service: _____

    Tagline (optional): _____

    b. Entrepreneur profile

    Founder(s): _____

    Why did you choose this business idea?

    _____

    _____

    What problem does your business solve?

    _____

    _____

    c. Target customers

    Who will buy your product or service?

    ☐ Students ☐ Tourists

    ☐ Parents ☐ Farmers

    ☐ Local residents ☐ Businesses

    ☐ Others: _____

    Describe your target customers:

    _____

    _____

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d. Market research

Find answers to the following questions about potential customers.

Question	Findings
What do the customers need?	
What products or services already exist?	
How is your idea different?	

e. Product or service description

Describe your product or service. List its key features and benefits:

_____

_____

_____

_____

f. Resources required

Resource	What is needed and how much?
Human Resources	
Finance	
Materials	
Technology	
Infrastructure	

g. Operations plan

How will the business work? Write the steps. You may depict this using a flow chart.

h. Marketing plan

How will customers learn about your business?

- ☐ Posters
- ☐ Social media
- ☐ Word of mouth
- ☐ School events
- ☐ Local market
- ☐ Newspapers
- ☐ Other: _____

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i. Financial plan

Source(s) of finance:

Estimated monthly cost on various items:

Item	Cost (₹)
Total Cost	

Details of expected monthly revenue sources:

Source of Income	Amount (₹)
Total Revenue	

Expected Profit:

Revenue – Cost

₹ _____

j. Risks and challenges

What challenges might your business face?

_____

_____

k. Social and environmental impact

How will your business help people or the environment?

_____

_____

l. Conclusion

Why do you believe your business will succeed?

_____

_____

_____

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12. **Visit a factory or a production unit.**

Explore how a business operates by visiting a factory or production unit that manufactures a product. Prepare questions for the owner or manager, such as:

- What product does the factory produce?
- Who are its customers?
- How did the business start?
- What challenges did the business face in its early years?
- How does the business manage its finances?
- How are employees recruited and trained?
- How are products supplied to wholesalers, retailers, or consumers?

During the visit, observe the production process carefully. Note the resources used, such as raw materials, machinery and technology, workers, and managers. Interact with the owner or manager and record key information. Take notes and photographs (if permitted).

After the visit:

- Understand the manufacturing process and create a process map showing the journey of the product from raw materials to the final customer.
- Identify barriers to growth, such as:
  - → **Financial Barriers:** Lack of capital, difficulty obtaining loans or credit, etc.
  - → **Technical Barriers:** Outdated machinery, limited access to technology, etc.
  - → **Human Resource Barriers:** Skill shortages, difficulty hiring or retaining workers, etc.
  - → **Market Barriers:** Competition, limited access to suppliers or customers, etc.
  - → **Regulatory Barriers:** Complex procedures, licences, approvals, or paperwork, etc.

Recommend some practical solutions through which the business could overcome these challenges. Share your findings by preparing

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a presentation and a report highlighting your observations, analysis, and recommendations. The format of the report may be as follows:

- Introduction
- Production or service delivery process (with diagrams or flowcharts)
- Resources required
- Barriers identified (supported by observations or evidence)
- Suggested solutions
- Reflection: key lessons learned from the visit

Notes

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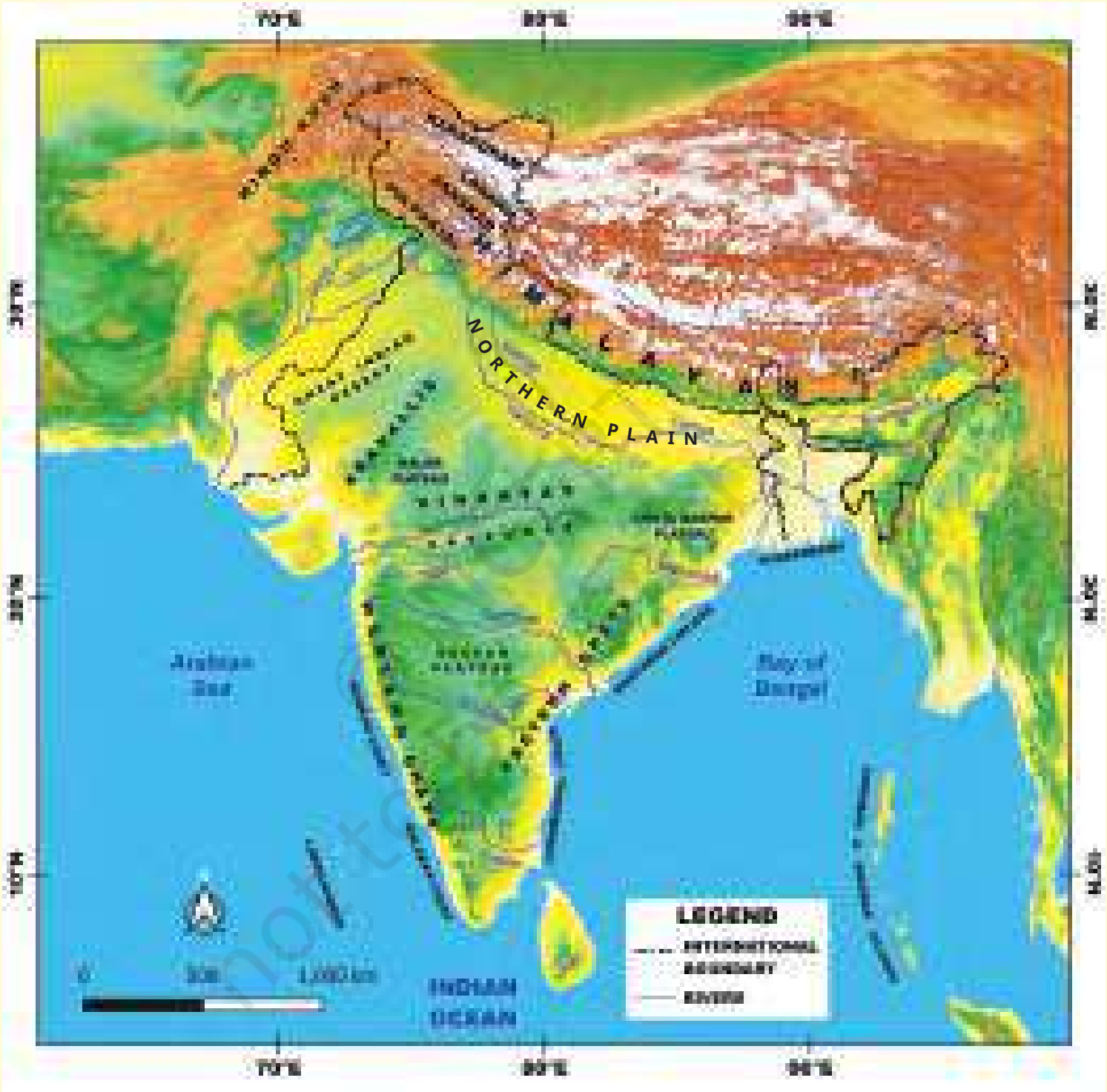
Political Map of India



Physical Map of India

Images and maps from external sources

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- Fig. 3.28. Ministry of Culture; https://museumsofindia.gov.in/repository/record/vmh_kol-R6787-23482
- Fig. 3.29. Courtesy: Bibliothèque Nationale de France - https://gallica.bnf.fr/ark:/12148/btv1b10538068s/f89.item#
- Fig. 3.31. Courtesy: BnF or Bibliothèque Nationale de France - https://gallica.bnf.fr/ark:/12148/btv1b10538068s/f71.item
- Fig. 4.2. Courtesy: Vummidi Bangaru Jewellers India Pvt Ltd

2. Royalty free stock from dreamstime.com

- Fig. 1.8, 1.10, 1.12, 1.14
- Fig. 2.3, 2.4, 2.6, 2.7, 2.8, 2.10, 2.11, 2.12, 2.17, 2.18, 2.21
- Fig. 3.7, 3.12, 3.13, 3.22, 3.25
- Fig. 5.10
- Fig. 6.12

3. AI generated images

- Fig. 1.1, 1.11
- Fig. 2.1, 2.5
- Fig. 5.1
- Fig. 6.1, 6.5, 6.10, 6.13

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